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Plasticizer-inclusive polymeric-inorganic hybrid layer for a lithium anode in a lithium-sulfur battery

Abstract

A lithium-sulfur battery including an anode structure, a cathode, a separator, and an electrolyte is provided. A protective layer may form within the anode structure responsive to operational discharge-charge cycling of the lithium-sulfur battery. The protective layer may include a polymeric backbone chain formed of interconnected carbon atoms collectively defining a segmental motion of the protective layer. Additional polymeric chains may be cross-linked to one another and at least some carbon atoms of the polymeric backbone chain. Each additional polymeric chain may be formed of interconnected monomer units. A plasticizer may be dispersed throughout the protective layer without covalently bonding to the polymeric backbone chain. The plasticizer may separate adjacent monomer units of at least some additional polymeric chains. Increasing separation of adjacent monomer units increases a cooperative segmental mobility of the additional polymeric chains and ionic conductivity of the protective layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Patent Application is a continuation-in-part application and claims priority to U.S. patent application Ser. No. 17/666,753 entitled “POLYMERIC-INORGANIC HYBRID LAYER FOR A LITHIUM ANODE” filed on Feb. 8, 2022, which is a continuation-in-part application and claims priority to U.S. patent application Ser. No. 17/584,666 entitled “SOLID-STATE ELECTROLYTE FOR LITHIUM-SULFUR BATTERIES” filed on Jan. 26, 2022, which is a continuation-in-part application and claims priority to U.S. patent application Ser. No. 17/578,240 entitled “LITHIUM-SULFUR BATTERY ELECTROLYTE COMPOSITIONS” filed on Jan. 18, 2022, which is a continuation-in-part application and claims priority to U.S. patent application Ser. No. 17/563,183 entitled “LITHIUM-SULFUR BATTERY CATHODE FORMED FROM MULTIPLE CARBONACEOUS REGIONS” filed on Dec. 28, 2021, which is a continuation-in-part of and claims priority to U.S. patent application Ser. No. 17/383,803 entitled "CARBONACEOUS MATERIALS FOR LITHIUM-SULFUR BATTERIES" filed on Jul. 23, 2021. This Patent Application also claims priority to Provisional Patent Application No. 63/149,894 entitled "CATHODE FOR A LITHIUM SULFUR BATTERY" filed on Feb. 16, 2021. The disclosures of all prior Applications are assigned to the assignee hereof, and are considered part of and are incorporated by reference in this Patent Application in their respective entireties.

TECHNICAL FIELD

(1) This disclosure relates generally to a lithium-sulfur battery, and, more particularly, to a protective layer, disposed on an anode of the lithium-sulfur battery, configured to prevent polysulfide species from contacting the anode.

DESCRIPTION OF RELATED ART

(2) Recent developments in batteries allow consumers to use electronic devices in many new applications. However, further improvements in battery technology are desirable.

SUMMARY

(3) This Summary is provided to introduce in a simplified form a selection of concepts that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to limit the scope of the claimed subject matter.

(4) One innovative aspect of the subject matter described in this disclosure may be implemented as a lithium-sulfur battery including a cathode, an anode structure positioned opposite to the cathode, a separator positioned between the anode structure and the cathode, and an electrolyte dispersed throughout the cathode and in contact with the anode structure. In one implementation, the anode structure may be formed as a single layer of solid lithium, which may output lithium cations (Li.sup.+) during operational discharge cycling of the lithium-sulfur battery. A solid-electrolyte interphase layer may be formed on the single layer of solid lithium. A protective layer may form on and at least partially within the solid-electrolyte interphase layer responsive to operational discharge-charge cycling of the lithium-sulfur battery.

(5) The protective layer may include a polymeric backbone chain formed of interconnected carbon atoms collectively defining a cooperative segmental mobility of the protective layer. Polymeric chains may be cross-linked to one another and to at least some carbon atoms of the polymeric backbone chain. Each polymeric chain is formed of interconnected monomer units. In some aspects, a plasticizer, e.g., any one or more of a polyethylene glycol (PEG or PEO) based oligomer, a nitrile, such as succinonitrile, glutaronitrile, adiponitrile, a solvent, including dimethoxyethane (DME), tetrahydrofuran (THF), diethyl ether, dioxolane (DOL), tetraethylene glycol dimethyl ether (TEGDME), toluene, bis(2,2-trifluoroethyl ether) (TEE), fluoroethylene carbonate (FEC), diethyl carbonate (DEC), dimethyl carbonate (DMC), propylene carbonate (PC), ethylene carbonate (EC), may be dispersed throughout the protective layer without covalently bonding to the polymeric backbone chain. In this way, the plasticizer may separate adjacent monomer units of at least some polymeric chains. Increasing separation of adjacent monomer units may be associated with an increased cooperative segmental mobility of polymeric chains and ionic conductivity of the protective layer. In addition, increasing the plasticizer in the protective layer may be associated with an increase in lithium cation (Li.sup.+) conductivity through the protective layer. In some other aspects, various linear polymeric and/or oligomeric chains may be chemically (e.g., by covalent bonding) attached to carbon atoms of the polymeric backbone chain to increase cooperative segmental mobility of the protective layer, and thereby also increase lithium cation (Li.sup.+) conductivity through the protective layer. For example, oligomeric substances such as polyoxypropylenediamine (e.g., JEFFAMINE® M-600) may be associated with an increase in the cooperative segmental mobility of the protective layer.

(6) In some aspects, the protective layer may melt at a glass transition temperature, such that an

increase in the glass transition temperature may be associated with a reduction in the cooperative segmental mobility of the plurality of additional polymeric chains. In some other aspects, a decrease in the cooperative segmental mobility of the plurality of additional polymeric chains may be associated with a decrease in lithium ion (Li.sup.+) conductivity through the protective layer. In this way, in some aspects, an increase in an amount of the plasticizer in the protective layer may be associated with an increase in the lithium ion (Li.sup.+) conductivity through the protective layer.
(7) Another innovative aspect of the subject matter described in this disclosure may be implemented as a lithium-sulfur battery including a cathode, an anode positioned opposite to the cathode, a separator positioned between the anode and the cathode, and an electrolyte dispersed throughout the cathode and in contact with the anode. In one implementation, a protective layer may be formed on the anode as a three-dimensional (3D) polymeric lattice including a first polymeric chain and a second polymeric chain positioned opposite one another. Each of the first and second polymeric chains may include carbon atoms at least temporarily chemically bonded to oxide ions (O.sup.2−), fluorine anions (F.sup.−), and/or nitrate anions (NO.sub.3.sup.−). Lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) may be dispersed throughout the 3D polymeric lattice to dissociate into lithium cations (Li.sup.+) and TFSI− anions. In this way, the first and second polymeric chains may form the 3D polymeric lattice with a cross-linking density sufficient to trap TFSI− anions by cross-linking with each other. For example, cross-linking may be initiated upon exposure to an energetic environment including ultraviolet (UV) energy and LiTFSI, such that LiTFSI may serve as a polymerization co-initiator compound.
(8) The first and second polymeric chains may form the 3D polymeric lattice through cross-linking polymerization reactions, one or more of which may include a ultraviolet (UV) curing that may progress at a curing rate. In some aspects, the polymerization co-initiator compound may increase the curing rate. In some instances, the lithium-sulfur battery may include additives dispersed uniformly throughout the 3D polymeric lattice. Each of the additives may include lithium nitrate (LiNO.sub.3), inorganic ionically-conductive ceramics including lithium lanthanum zirconium oxide (LLZO), NASICON-type oxide Li.sub.1+xAl.sub.xTi.sub.2-x(PO.sub.4).sub.3 (LATP) or lithium tin phosphorus sulfide (LSPS), or nitrogen-oxygen containing additives. In this way, inorganic ionically-conductive ceramics may be uniformly embedded in the 3D polymeric lattice and/or uniformly distributed throughout the protective layer. In addition, the protective layer may include desiccated solvents.
(9) Another innovative aspect of the subject matter described in this disclosure may be implemented as a lithium-sulfur battery including a cathode, an anode positioned opposite to the cathode, a protective layer formed on the anode, a separator positioned between the anode and the cathode, and an electrolyte dispersed throughout the cathode and in contact with the anode. The protective layer may trap various types of anions. In addition, the protective layer may be formed of multiple ingredients including relatively pliable oligomeric epoxy and/or polyol based compounds (e.g., one or more of which may eliminate formation of pinholes in the protective layer), a relatively rigid polymeric epoxy based compound, and/or photo-initiator molecules. Lithium-containing salts may be dispersed uniformly throughout the protective layer to dissociate into lithium (Li+) cations and at least one type of anion.
(10) In some instances, the protective layer may be formed on the anode responsive to ultraviolet (UV) curing of at least some of the multiple ingredients. In addition, in some aspects, the protective layer may include non-reactive diluents including 1,2-Dimethoxyethane (DME), tetrahydrofuran (THF), triethylene glycol dimethyl ether, or 2-Methyl-2-oxazoline (MOZ). In some other aspects, the protective layer may include reactive diluents including 1,3-Dioxolane (DOL), 3,3-Dimethyloxetane (DMO), 2-Ethyl-2-oxazoline (EOZ), or ε-Caprolactone (CL). In this way, a per-unit formulation weight of the protective layer may be based on a concentration level of non-reactive diluents or reactive diluents relative to ingredients of the protective layer. In addition, the reactive diluents may reduce mechanical stress of at least some of the cross-linking units within the

protective layer.
(11) In some aspects, reactive diluents may be removed from the protective layer. In some other aspects, the reactive diluents may remain in the protective layer after cross-linking of at least some ingredients. In this way, retention of reactive diluents in the protective layer after cross-linking of two or more ingredients improves lithium cation (Li+) diffusion through the electrolyte. In one implementation, the relatively rigid polymeric epoxy based compound is formed from repeating epoxy monomer units. For example, each of the repeating epoxy monomer units may include 3,4-epoxycyclohexylmethyl 3,4-epoxycyclohexanecarboxylate (ECC), which may cross-link with additional ECC monomer units and produce a network of polar groups. The network of polar groups may trap at least some anions produced upon dissociation of lithium-containing salts.
(12) Details of one or more implementations of the subject matter described in this disclosure are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages will become apparent from the description, the drawings, and the claims. Note that the relative dimensions of the following figures may not be drawn to scale.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows a diagram depicting an example battery, according to some implementations.
(2) FIG. **2** shows a diagram depicting another example battery, according to some implementations.
(3) FIG. **3** shows a diagram of an example electrode of a battery, according to some implementations.
(4) FIG. **4** shows a diagram a diagram of a portion of an example battery that includes a protective lattice, according to some implementations.
(5) FIG. **5** shows a diagram of an anode structure including a tin fluoride (SnF.sub.2) layer, according to some implementations.
(6) FIG. **6** shows a diagram of an enlarged portion of the anode structure of FIG. **5**, according to some implementations.
(7) FIG. **7** shows a diagram of a polymeric network of a battery, according to some implementations.
(8) FIG. **8**A shows a diagram of an example carbonaceous particle with graded porosity, according to some implementations.
(9) FIG. **8**B shows a diagram of an example of a tri-zone particle, according to some implementations.
(10) FIG. **8**C shows an example step function representative of the tri-zone particle of FIG. **8**B, according to some implementations.
(11) FIG. **8**D shows a graph depicting an example distribution of pore volume versus pore width of an example carbonaceous particle, according to some implementations.
(12) FIGS. **9**A and **9**B show electron micrographs of example carbonaceous particles, aggregates, and/or agglomerates depicted in FIG. **8**A and/or FIG. **8**B, according to some implementations.
(13) FIGS. **10**A and **10**B show transmission electron microscope (TEM) images of carbonaceous particles treated with carbon dioxide (CO.sub.2), according to some implementation.
(14) FIG. **11** shows a diagram depicting carbon porosity types prevalent in the anodes and/or the cathodes of the present disclosure, according to some implementations.
(15) FIG. **12** shows a graph depicting cumulative pore volume versus pore width for micropores and mesopores dispersed throughout the anode or cathode of a battery, according to some implementations.
(16) FIG. **13** shows graphs depicting battery performance per cycle number, according to some implementations.

(17) FIG. **14** shows a bar chart depicting capacity per cycle number, according to some implementations.
(18) FIG. **15** shows graphs depicting battery performance per cycle number, according to some implementations.
(19) FIG. **16** shows a graph depicting battery discharge capacity per cycle number, according to some implementations.
(20) FIG. **17** shows a graph depicting battery discharge capacity per cycle number, according to some implementations.
(21) FIG. **18** shows a graph depicting battery specific discharge capacity for various TBT-containing electrolyte mixtures, according to some implementations.
(22) FIG. **19** shows graphs depicting battery specific discharge capacity per cycle number for the battery of FIG. **1**, according to some implementations.
(23) FIG. **20** shows graphs depicting battery specific discharge capacity and discharge capacity retention per cycle number for the battery of FIG. **2**, according to other implementations.
(24) FIG. **21** shows graphs depicting battery specific discharge capacity and discharge capacity retention per cycle number for the battery of FIG. **2**, according to some other implementations.
(25) FIG. **22** shows a diagram of an example cathode of a battery, according to some implementations.
(26) FIGS. **23-25** show graphs depicting specific discharge capacity per cycle number, according to some implementations.
(27) FIG. **26**A shows a diagram depicting an example battery, according to some implementations.
(28) FIG. **26**B shows a diagram depicting an example battery, according to some implementations.
(29) FIG. **27** shows a graph depicting voltage drop per specific capacity, according to some implementations.
(30) FIG. **28** shows a diagram depicting an example battery, according to some implementations.
(31) FIG. **29** shows a diagram depicting an example cathode of the battery of FIG. **28**, according to some implementations.
(32) FIG. **30** shows a diagram depicting the protective layer of the battery of FIG. **28**, according to some implementations.
(33) FIG. **31**A shows a micrograph of an example baseline protective layer, according to some implementations.
(34) FIG. **31**B shows a micrograph of the protective layer of the battery of FIG. **28**, according to some implementations.
(35) FIG. **32** shows a micrograph of a cutaway of the protective layer of the battery of FIG. **28**, according to some implementations.
(36) FIG. **33**A shows an example cross-linking density of the protective layer of the battery of FIG. **28**, according to some implementations.
(37) FIG. **33**B shows another example cross-linking density of the protective layer of the battery of FIG. **28**, according to some implementations.
(38) FIG. **34** shows an example ring-opening (ROP) mechanism for triarylsulfonium salt (Ar.sub.3S.sup.+MtXn.sup.−), according to some implementations.
(39) FIG. **35** shows several example onium salts suitable for usage as cationic photo-initiators for the protective layer of the battery of FIG. **28**, according to some implementations.
(40) FIG. **36** shows a several example monomers of various cationic photo-polymerizable compositions suitable for forming the protective layer of the battery of FIG. **28**, according to some implementations.
(41) FIG. **37** shows ultraviolet (UV) curable monomers suitable for forming the protective layer of the battery of FIG. **28**, according to some implementations.
(42) FIG. **38** shows several example non-reactive diluents suitable for usage as additives to adjust dilution levels in UV-curable formulations prepared with the UV curable monomers of FIG. **37**,

according to some implementations.
(43) FIG. **39** shows several example reactive diluents suitable for usage as additives to adjust dilution levels in UV-curable formulations prepared with the UV curable monomers of FIG. **37**, according to some implementations.
(44) FIG. **40**A shows a graph of capacity (% of initial) against cycle number, according to some implementations.
(45) FIG. **40**B shows a graph of capacity (mAh/g) against cycle number, according to some implementations.
(46) FIG. **41**A shows another graph of capacity (% of initial) against cycle number, according to some implementations.
(47) FIG. **41**B shows another graph of capacity (mAh/g) against cycle number, according to some implementations.
(48) FIG. **42**A shows a diagram depicting an example battery, according to some implementations.
(49) FIG. **42**B shows an enlarged section of the battery of FIG. **42**A, according to some implementations.
(50) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(51) The following description is directed to some example implementations for the purposes of describing innovative aspects of this disclosure. However, a person having ordinary skill in the art will readily recognize that the teachings herein can be applied in a multitude of different ways. The described implementations can be implemented in any type of electrochemical cell, battery, or battery pack, and can be used to compensate for various performance related deficiencies. As such, the disclosed implementations are not to be limited by the examples provided herein, but rather encompass all implementations contemplated by the attached claims. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure.
(52) Batteries typically include several electrochemical cells that can be connected to each other to provide electric power to a wide variety of devices such as (but not limited to) mobile phones, laptops, electric vehicles (EVs), factories, and buildings. Certain types of batteries, such as lithium-ion or lithium-sulfur batteries, may be limited in performance by the type of electrolyte used or by uncontrolled battery side reactions. As a result, optimization of the electrolyte may improve the cyclability, the specific discharge capacity, the discharge capacity retention, the safety, and the lifespan of a respective battery. For example, in an unused or “fresh” battery, lithium cations (Li^+) are transported freely from the anode to the cathode upon activation and later during initial and subsequent discharge cycles. Then, during battery charge cycles, lithium cations (Li^+) may be forced to migrate back from their electrochemically favored positions in the cathode to the anode, where they are stored for subsequent use. This cyclical discharge-charge process associated with rechargeable batteries can result in the generation of undesirable chemical species that can interfere with the transport of lithium cations (Li^+) to and from the cathode during respective discharge and charge of the battery. Specifically, lithium-containing polysulfide intermediate species (referred to herein as “polysulfides”) are generated when lithium cations (Li^+) interact with elemental sulfur (or, in some configurations, lithium sulfide, Li_2S) present in the cathode. These polysulfides are soluble in the electrolyte and, as a result, diffuse throughout the battery during operational cycling, thereby resulting in loss of active material from cathode. Generation of excessive concentration levels of polysulfides can result in unwanted battery capacity decay and cell failure during operational cycling, potentially reducing the driving range for electric vehicles (EVs) and increasing the frequency with which such EVs need recharging.
(53) In some cases, polysulfides participate in the formation of inorganic layers in a solid electrolyte interphase (SEI) provided in the battery. In one example, the anode may be protected by

a stable inorganic layer formed in the electrolyte and containing 0.020 M Li_2S_5 (0.10 M sulfur) and 5.0 wt. % $LiNO_3$. The anode with a lithium fluoride and polysulfides (LiF—Li_2S_x) may enrich the SEI and result in a stable Coulombic efficiency of 95% after 233 cycles for Li—Cu half cells, while preventing formation of lithium dendrites or other uncontrolled lithium growths that can extend from the anode to the cathode and result in a failed or ruptured cell. However, when polysulfides are generated at certain concentrations (such as greater than 0.50 M sulfur), formation of the SEI may be hindered. As a result, lithium metal from the anode may be undesirably etched, creating a rough and imperfect surface exposed to the electrolyte. This unwanted deterioration (etching) of the anode due to a relatively high concentration of polysulfides may indicate that polysulfide dissolution and diffusion may be limiting battery performance.

(54) In some implementations, the porosity of a carbonaceous cathode may be adjusted to achieve a desired balance between maximizing energy density and inhibiting the migration of polysulfides into and/or throughout the battery's electrolyte. As used herein, carbonaceous may refer to materials containing or formed of one or more types or configuration of carbon. For example, cathode porosity may be higher in sulfur and carbon composite cathodes than in conventional lithium-ion battery electrodes. Denser electrodes with relatively low porosity may minimize electrolyte intake, parasitic weight, and cost. Sulfur utilization may be limited by the solubility of polysulfides and conversion from those polysulfides into lithium sulfide (Li_2S). The conversion of polysulfides into lithium sulfide may be based on the accessible surface area of the cathode. Aspects of the present disclosure recognize that cathode porosity may be adjusted based on electrolyte compositions to maximize battery volumetric energy density. In addition, or in the alternative, one or more protective layers or regions can be added to surfaces of the cathode and/or the anode exposed to the electrolyte to adjust cathode porosity levels. In some aspects, these protective layers or regions can inhibit the undesirable migration of polysulfides throughout the battery.

(55) Various aspects of the subject matter disclosed herein relate to a lithium-sulfur battery including a liquid-phase electrolyte, which may include a ternary solvent package and one or more additives. In some implementations, the lithium-sulfur battery may include a cathode, an anode positioned opposite to the cathode, and an electrolyte. The cathode may include several regions, where each region may be defined by two or more carbonaceous structures adjacent to and in contact with each other. In some instances, the electrolyte may be interspersed throughout the cathode and in contact with the anode. In some aspects, the electrolyte may include a ternary solvent package and 4,4′-thiobisbenzenethiol (TBT). In other instances, the electrolyte may include the ternary solvent package and 2-mercaptobenzothiazole (MBT).

(56) In various implementations, the ternary solvent package may include 1,2-Dimethoxyethane (DME), 1,3-Dioxolane (DOL), tetraethylene glycol dimethyl ether (TEGDME) and one or more additives, which may include a lithium nitrate ($LiNO_3$), all which may be in a liquid-phase. In some implementations, the ternary solvent package may be prepared by mixing approximately 5,800 microliters (μL) of DME, 2,900 microliters (μL) of DOL, and 1,300 microliters (μL) of TEGDME with one another to create a mixture. Approximately 0.01 mol of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) may be dissolved into the ternary solvent package to produce an approximate dilution level of 1 M LiTFSI in DME:DOL:TEGDME at a volume ratio of 2:1:1 including approximately 2 weight percent (wt. %) lithium nitrate. In other implementations, the ternary solvent package may be prepared with 2,000 microliters (μL) of DME, 8,000 microliters (μL) of DOL, and 2,000 microliters (μL) of TEGDME and include approximately 0.01 mol of dissolved lithium bis(trifluoromethanesulfonyl)imide (LiTFSI). In some aspects, the ternary solvent package may be prepared at a first approximate dilution level of 1 molar (M) LiTFSI in a mixture of DME:DOL:TEGDME. In other instances, the ternary solvent package may be prepared at a second approximate dilution level of approximately 1 M LiTFSI in DME:DOL:TEGDME at an approximate volume ratio of 1:4:1 and include either an addition of 5M TBT solution or an addition of 5M MBT solution, or an addition of other additives and/or chemical substances.

(57) In various implementations, each carbonaceous structure may include a relatively high-density outer shell region and a relatively low-density core region. In some aspects, the core region may be formed within an interior portion of the outer shell region. The outer shell region may have a carbon density between approximately 1.0 grams per cubic centimeter (g/cc) and 3.5 g/cc. The core region may have a carbon density of between approximately 0.0 g/cc and 1.0 g/cc or some other range lower than the first carbon density. In other implementations, each carbonaceous structure may include an outer shell region and core region having the same or similar densities, for example, such that the carbonaceous structure does not include a graded porosity.
(58) Various regions of the cathode may include microporous channels, mesoporous channels, and macroporous channels interconnected to each other to form a porous network extending from the outer shell region to the core region. For example, in some aspects, the porous network may include pores that each have a principal dimension of approximately 1.5 nm.
(59) In some implementations, one or more portions of the porous network may temporarily micro-confine electroactive materials such as (but not limited to) elemental sulfur within the cathode, which may increase battery specific capacity by complexing with lithium cations (Li.sup.+). In some aspects, the ternary solvent package may have a tunable polarity, a tunable solubility, and be capable of transporting lithium cations (Li.sup.+). In addition, the ternary solvent package may at least temporarily suspend polysulfides (PS) during charge-discharge cycles of the battery.
(60) Particular implementations of the subject matter described in this disclosure can be implemented to realize one or more potential advantages. In some implementations, the porous network formed by the interconnection of microporous, mesoporous, and macroporous channels within the cathode may include a plurality of pores having a multitude of different pore sizes. In some implementations, the plurality of pores may include micropores having a pore size less than approximately 2 nm, may include mesopores having a pore size between approximately 5 and 50 nm, and may include macropores having a pore size greater than approximately 50 nm. The micropores, mesopores, and macropores may collectively mitigate the undesirable migration or diffusion of polysulfides throughout the electrolyte. Since the poly sulfide shuttle effect may result in the loss of active material from the cathode, the ability to mitigate or reduce the polysulfide shuttle effect can increase battery performance.
(61) In one implementation, the micropores may have a pore size of approximately 1.5 nm selected to micro-confine elemental sulfur (S.sub.8, or smaller chains/fragments of sulfur, for example in the form of S.sub.2, S.sub.4 or S.sub.6) pre-loaded into the cathode. The micro-confinement of elemental sulfur within the cathode may allow TBT or MBT complexes generated during battery cycling to inhibit the migration of long-chain polysulfides within the mesopores of the cathode. Accumulation of these long-chain polysulfides within the mesopores of the cathode may cause the cathode to volumetrically expand to retain the polysulfides and thereby reduce the polysulfide shuttle effect. Accordingly, lithium cations (Li.sup.+) may continue to transport freely between the anode and the cathode via the electrolyte without being blocked or impeded by the polysulfides. The free movement of lithium cations (Li.sup.+) throughout the electrolyte without interference by polysulfides can increase battery performance.
(62) In addition, or the alternative, one or more protective layers, sheaths, films, and/or regions (collectively referred to herein as “protective layers”) may be disposed on the anode and/or the cathode and/or the separator and in contact with the electrolyte. The protective layers may include materials capable of binding with polysulfides to impede polysulfide migration and prevent lithium dendrite formation. In some aspects, the protective layers may be arranged in different configurations and used with any of the electrolyte chemistries and/or compositions disclosed herein, which in turn may result in complete tunability of the battery.
(63) In one implementation, carbonaceous materials may be grafted with fluorinated polymer chains and deposited on one or more exposed surfaces of the anode. The fluorinated polymer chains can be cross-linked into a polymeric network on contact with Lithium metal from the anode

surface via the Wurtz reaction. The cross-linked polymeric network formation may, in turn, suppress Lithium metal dendrite formation associated with the anode, and may also generate Lithium fluoride. Fluorinated polymers within the polymeric network may participate in chemical reactions during battery operational cycling to yield Lithium fluoride. Formation of the lithium fluoride may involve the chemical binding of lithium cations (Li.sup.+) from the electrolyte with fluorine ions.

(64) In addition, or the alternative, the polymeric network may be combined with any of the electrolyte chemistries and/or compositions disclosed herein and/or a protective sheath disposed on the cathode. In one implementation, the protective sheath can be formed by combining compounds containing di-functional, or higher functionality Epoxy and Amine or Amide compounds. Their intermolecular cross-linking would result in formation of 3D network with high chemical resistance to dissolution in electrolyte. Composition, for example, may include a tri-functional epoxy compound and a di-amine oligomer-based compound, which may react with each other to produce a protective lattice that can bind to polysulfides generated in the cathode and prevent their migration or diffusion into the electrolyte. In addition, the protective lattice may diffuse through one or more cracks that may form in the cathode due to battery cycling. The protective lattice, when diffused throughout such cracks formed in the cathode, may increase the structural integrity of the cathode, and reduce potential rupture of the cathode associated with volumetric expansion.

(65) In various implementations, one or more of the disclosed battery components may be combined with a conformal coating disposed on edges or surfaces of the anode exposed to the electrolyte. In some implementations, the conformal coating may include a graded interface layer that may replace the polymeric network. In some aspects, the graded interface layer may include a tin fluoride layer and a tin-lithium alloy region formed between the tin fluoride layer and the anode. The tin-lithium alloy region may form a layer of lithium fluoride uniformly dispersed between the anode and the tin-fluoride layer in response to operational cycling of the battery.

(66) In various implementations, a lithium-sulfur battery employing various aspects of the present disclosure may include an electroactive material extracted from an external source, e.g., a subterranean source and/or an extraterrestrial subterranean source. In such implementations, the cathode may be prepared as a sulfur-free cathode including functional pores that may micro-confine the electroactive material within the cathode. In some aspects, the cathode may include aggregates including a multitude of carbonaceous particles joined together, and may include agglomerates including a multitude of the aggregates joined together. In one implementation, the carbonaceous materials used to form the cathode (and/or the anode) may be tuned to define unique pore sizes, size ranges, and volumes. In some implementations, the carbonaceous particles may include non-tri-zone particles with and without tri-zone particles. In other implementations, the carbonaceous particles may not include tri-zone particles. Each tri-zone particle may include micropores, mesopores, and macropores, and both the non-tri-zone and tri-zone particles may each have a principal dimension in an approximate range of 20 nm to 300 nm. Each of the carbonaceous particles may include carbonaceous fragments nested within each other and separated from immediate adjacent carbonaceous fragments by mesopores. In some aspects, each of the carbonaceous particles may have a deformable perimeter that changes in shape and coalesces with adjacent materials.

(67) Some of the pores may be distributed throughout the plurality of carbonaceous fragments and/or the deformable perimeters of the carbonaceous particles. In various implementations, mesopores may be interspersed throughout the aggregates, and macropores may be interspersed throughout the plurality of agglomerates. In one implementation, each mesopore may have a principal dimension between 3.3 nanometers (nm) and 19.3 nm, each aggregate may have a principal dimension in an approximate range between 10 nm and 10 micrometers (μm), and each agglomerate may have a principal dimension in an approximate range between 0.1 μm and 1,000 μm. As further described below, specific combinations of pore sizes matched with unique

electrolyte formulations and protective layers can be used to reduce or mitigate the harmful effects of unwanted polysulfide diffusion, which may further increase battery performance.

(68) FIG. **1** shows an example battery **100**, according to some implementations. The battery **100** may be a lithium-sulfur electrochemical cell, a lithium-ion battery, or a lithium-sulfur battery. The battery **100** may have a body **105** that includes a first substrate **101**, a second substrate **102**, a cathode **110**, an anode **120** positioned opposite to the cathode **110**, and an electrolyte **130**. In some aspects, the first substrate **101** may function as a current collector for the anode **120**, and the second substrate **102** may function as a current collector for the cathode **110**. The cathode **110** may include a first thin film **111** deposited onto the second substrate **102**, and may include a second thin film **112** deposited onto the first thin film **111**. In some implementations, the electrolyte **130** may be a liquid-phase electrolyte including one or more additives such as lithium nitrate, tin fluoride, lithium iodide, lithium bis(oxalate)borate (LiBOB), cesium nitrate, cesium fluoride, ionic liquids, lithium fluoride, fluorinated ether, TBT, MBT, DPT and/or the like. Suitable solvent packages for these example additives may include various dilution ratios, including 1:1:1 of 1,3-dioxolane (DOL), 1,2-dimethoxyethane, (DME), tetraethylene glycol dimethyl ether (TEGDME), and/or the like.

(69) Although not shown for simplicity, in one implementation, a lithium layer may be electrodeposited on one or more exposed carbon surfaces of the anode **120**. In some instances, the lithium layer may include elemental lithium provided by the ex-situ electrodeposition of lithium onto exposed surfaces of the anode **120**. In some aspects, the lithium layer may include lithium, calcium, potassium, magnesium, sodium, and/or cesium, where each metal may be ex-situ deposited onto exposed carbon surfaces of the anode **120**. The lithium layer may provide lithium cations (Li.sup.+) available for transport to-and-from the cathode **110** during operational cycling of the battery **100**. As a result, the battery **100** may not need an additional lithium source for operation. Instead of using lithium sulfide, elemental sulfur (S.sub.8) may be pre-loaded in various pores or porous networks formed in the cathode **110**. During operational cycling of the battery, the elemental sulfur may form lithium-sulfur complexes that can micro-confine (at least temporarily) greater amounts of lithium than conventional cathode designs. As a result, the battery **100** may outperform batteries that rely on such conventional cathode designs.

(70) In various implementations, the lithium layer may dissociate and/or separate into lithium cations (Li.sup.+) **125** and electrons **174** during a discharge cycle of the battery **100**. The lithium cations (Li.sup.+) **125** may migrate from the anode **120** towards the cathode **110** through the electrolyte **130** to their electrochemically favored positions within the cathode **110**, as depicted in the example of FIG. **1**. As the lithium cations (Li.sup.+) **125** move through the electrolyte **130**, electrons **174** are released from lithium cations (Li.sup.+) **125** and become available to carry charge, and therefore conduct an electric current, between the anode **120** and cathode **110**. As a result, the electrons **174** may travel from the anode **120** to the cathode **110** through an external circuit to power an external load **172**. The external load **172** may be any suitable circuit, device, or system such as (but not limited to) a lightbulb, consumer electronics, or an electric vehicle (EV).

(71) In some implementations, the battery **100** may include a solid-electrolyte interphase layer **140**. The solid-electrolyte interphase layer **140** may, in some instances, be formed artificially on the anode **120** during operational cycling of the battery **100**. In such instances, the solid-electrolyte interphase layer **140** may also be referred to as an artificial solid-electrolyte interphase, or A-SEI. The solid-electrolyte interphase layer **140**, when formed as an A-SEI, may include tin, manganese, molybdenum, and/or fluorine compounds. Specifically, the molybdenum may provide cations, and the fluorine compounds may provide anions. The cations and anions may interact with each other to produce salts such as tin fluoride, manganese fluoride, silicon nitride, lithium nitride, lithium nitrate, lithium phosphate, manganese oxide, lithium lanthanum zirconium oxide (LLZO, Li.sub.7La.sub.3Zr.sub.2O.sub.12), etc. In some instances, the A-SEI may be formed in response to exposure of lithium cations (Li.sup.+) **125** to the electrolyte **130**, which may include solvent-based solutions including tin and/or fluorine.

(72) In various implementations, the solid-electrolyte interphase layer **140** may be artificially provided on the anode **120** prior to activation of the battery **100**. Alternatively, in one implementation, the solid-electrolyte interphase layer **140** may form naturally, e.g., during operational cycling of the battery **100**, on the anode **120**. In some instances, the solid-electrolyte interphase layer **140** may include an outer layer of shielding material that can be applied to the anode **120** as a micro-coating. In this way, formation of the solid-electrolyte interphase layer **140** on portions of the anode **120** facing the electrolyte **130** may result from electrochemical reduction of the electrolyte **130**, which in turn may reduce uncontrolled decomposition of the anode **120**.
(73) In some implementations, the battery **100** may include a barrier layer **142** that flanks the solid-electrolyte interphase layer **140**, for example, as shown in FIG. **1**. The barrier layer **142** may include a mechanical strength enhancer **144** coated and/or deposited on the anode **120**. In some aspects, the mechanical strength enhancer **144** may provide structural support for the battery **100**, may prevent lithium dendrite formation from the anode **120**, and/or may prevent protrusion of lithium dendrite throughout the battery **100**. In some implementations, the mechanical strength enhancer **144** may be formed as a protective coating over the anode **120**, and may include one or more carbon allotropes, carbon nano-onions (CNOs), nanotubes (CNTs), reduced graphene oxide, graphene oxide (GO), and/or carbon nano-diamonds. In some instances, the solid-electrolyte interphase layer **140** may be formed within the mechanical strength enhancer **144**.
(74) In some implementations, the first substrate **101** and/or the second substrate **102** may be a solid copper metal foil and may influence the energy capacity, rate capability, lifespan, and long-term stability of the battery **100**. For example, to control energy capacity and other performance attributes of the battery **100**, the first substrate **101** and/or the second substrate **102** may be subject to etching, carbon coating, or other suitable treatment to increase electrochemical stability and/or electrical conductivity of the battery **100**. In other implementations, the first substrate **101** and/or the second substrate **102** may include or may be formed from a selection of aluminum, copper, nickel, titanium, stainless steel and/or carbonaceous materials depending on end-use applications and/or performance requirements of the battery **100**. For example, the first substrate **101** and/or the second substrate **102** may be individually tuned or tailored such that the battery **100** meets one or more performance requirements or metrics.
(75) In some aspects, the first substrate **101** and/or the second substrate **102** may be at least partially foam-based or foam-derived, and can be selected from any one or more of metal foam, metal web, metal screen, perforated metal, or sheet-based three-dimensional (3D) structures. In other aspects, the first substrate **101** and/or the second substrate **102** may be a metal fiber mat, metal nanowire mat, conductive polymer nanofiber mat, conductive polymer foam, conductive polymer-coated fiber foam, carbon foam, graphite foam, or carbon aerogel. In some other aspects, the first substrate **101** and/or second substrate **102** may be carbon xerogel, graphene foam, graphene oxide foam, reduced graphene oxide foam, carbon fiber foam, graphite fiber foam, exfoliated graphite foam, or any combination thereof.
(76) FIG. **2** shows another example battery **200**, according to some implementations. The battery **200** may be similar to the battery **100** of FIG. **1** in many respects, such that description of like elements is not repeated herein. In some implementations, the battery **200** may be a next-generation battery, such as a lithium-metal battery and/or a solid-state battery featuring a solid-state electrolyte. In other implementations, the battery **200** may include an electrolyte **230** and may therefore include any of the protective layers and/or electrolyte chemistries or compositions disclosed herein.
(77) In some other implementations, the electrolyte **230** may be solid or substantially solid. For example, in some instances, the electrolyte **230** may begin in a gel phase and then later solidify upon activation of the battery **200**. The battery **200** may reduce specific capacity or energy losses associated with the polysulfide shuttle effect by replacing conventional carbon scaffolded anodes with a single solid metal layer of lithium deposited in an initially empty cavity. For example, while

the anode **120** of the battery **100** of FIG. **1** may include carbon scaffolds, the anode **220** of the battery **200** of FIG. **2** may be a lithium-metal anode devoid of any carbon material. In one implementation, the lithium-metal anode may be formed as a single solid lithium metal layer and referred to as a "lithium metal anode."

(78) Energy density gains associated with various cathode materials may be based on whether lithium metal is pre-loaded into the cathode **210** and/or is prevalent in the electrolyte **230**. Either the cathode **210** and/or the electrolyte **230** may provide lithium available for lithiation of the anode **220**. For example, batteries having high-capacity cathodes may need thicker or energetically denser anodes in order to supply the increased quantities of lithium needed for usage by the high-capacity cathodes. In some implementations, the anode **220** may include scaffolded carbonaceous structures capable of being incrementally filled with lithium deposited therein. These carbonaceous structures may be capable of retaining greater amounts of lithium within the anode **220** as compared to conventional graphitic anodes, which may be limited to solely hosting lithium intercalated between alternating graphene layers or may be electroplated with lithium. For example, conventional graphitic anodes may use six carbon atoms to hold a single lithium atom. In contrast, by using a pure lithium metal anode, such as the anode **220**, batteries disclosed herein may reduce or even eliminate carbon use in the anode **220**, which may allow the anode **220** to store greater amounts of lithium in a relatively smaller volume than conventional graphitic anodes. In this way, the energy density of the battery **200** may be greater than conventional batteries of a similar size.

(79) Lithium metal anodes, such as the anode **220**, may be prepared to function with a solid-state electrolyte designed to inhibit the formation and growth of lithium dendrites from the anode. In some aspects, a separator **250** may further limit dendrite formation and growth. The separator **250** may have a similar ionic conductivity as the electrolyte **130** of FIG. **1** yet still reduce lithium dendrite formation. In some aspects, the separator **250** may be formed from a ceramic containing material and may, as a result, fail to chemically react with metallic lithium. As a result, the separator **250** may be used to control lithium ion transport through pores dispersed across the separator **250** while concurrently preventing a short-circuit by impeding the flow or passage of electrons through the electrolyte **230**.

(80) In one implementation, a void space (not shown for simplicity) may be formed within the battery **200** at or near the anode **220**. Operational cycling of the battery **200** in this implementation may result in the deposition of lithium into the void space. As a result, the void space may become or transform into a lithium-containing region (such as a solid lithium metal layer) and function as the anode **220**. In some aspects, the void space may be created in response to chemical reactions between a metal-containing electrically inactive component and a graphene-containing component of the battery **200**. Specifically, the graphene-containing component may chemically react with lithium deposited into the void space during operational cycling and produce lithiated graphite (LiC.sub.6) or a patterned lithium metal. The lithiated graphite produced by the chemical reactions may generate or lead to the generation and/or liberation of lithium cations (Li.sup.+) and/or electrons that can be used to carry electric charge or a "current" between the anode **220** and the cathode **210** during discharge cycles of the battery **200**.

(81) And, in implementations for which the anode **220** is a solid lithium metal layer, the battery **200** may be able to hold more electroactive material and/or lithium per unit volume (as compared to batteries with scaffolded carbon and/or intercalated lithiated graphite anodes). In some aspects, the anode **220**, when prepared as a solid lithium metal layer, may result in the battery **200** having a higher energy density and/or specific capacity than batteries with scaffolded carbon and/or intercalated lithiated graphite anodes, thereby resulting in longer discharge cycle times and additional power output per unit time. In instances for which use of a solid-state electrolyte is not desired or not optimal, the electrolyte **230** of the battery **200** of FIG. **2** may be prepared with any of the liquid-phase electrolyte chemistries and/or compositions disclosed herein. In addition, or in the alternative, the electrolyte **230** may include lithium and/or lithium cations (Li.sup.+) available for

cyclical transport from the anode **220** to the cathode **210** and vice-versa during discharge and charge cycles, respectively.

(82) To reduce the migration of polysulfides **282** generated from elemental sulfur **281** pre-loaded in the cathode **210** into the electrolyte **230**, the battery **200** may include one or more unique polysulfide retention features. For example, given that polysulfides are soluble in the electrolyte **230**, some polysulfides may be expected to drift or migrate from the cathode **210** towards the anode **220** due to differences in electrochemical potential, chemical gradients, and/or other phenomena. The migration of polysulfides **282**, especially long-chain form polysulfides, may impede the transport of lithium cations (Li.sup.+) from the anode **220** to the cathode **210**, which in turn may reduce the number of electrons available to generate an electric current that can power a load **272**, such as an electric vehicle (EV). In some aspects, lithium cations (Li.sup.+) **225** may be transported from one or more start positions **226** in or near the anode **220** along transport pathways to one or more end positions **227** in or near the cathode **210**, as depicted in the example of FIG. **2**.

(83) In some implementations, a polymeric network **285** may be disposed on the anode **220** to reduce the uncontrolled migration of polysulfides **282** from the anode **220** to the cathode **210**. The polymeric network **285** may include one or more layers of carbonaceous materials grafted with fluorinated polymer chains cross-linked with each other via the Wurtz reaction upon exposure to Lithium anode surface. The carbonaceous materials in the polymeric network **285**, which may include (but are not limited to graphene, few layer graphene, FLG, many layer graphene, and MLG), may be chemically grafted with fluorinated polymer chains containing carbon-fluorine (C—F) bonds. These C—F bonds may chemically react with lithium metal from the surface of the anode **220** to produce highly ionic Carbon-Lithium bonds (C—Li). These formed C—Li bonds, in turn, may react with C—F bonds of polymer chains to form new Carbon-Carbon bonds that can also cross-link the polymer chains into (and thereby form) the polymeric network and generate lithium fluoride (LiF).

(84) The resulting lithium fluoride may be uniformly distributed along the entire perimeter of the polymeric network **285**, such that lithium cations (Li.sup.+) are uniformly consumed to produce an interface layer **283** that may form or otherwise include lithium fluoride during battery cycling. The interface layer **283** may extend along a surface or portion of the anode **220** facing the cathode **210**, as shown in FIG. **2**. As a result, the lithium cations (Li.sup.+) **225** are less likely to combine and/or react with each other and are more likely to combine and/or react with fluorine atoms made available by the fluorinated polymer chains in the polymeric network **285**. The resulting reduction of lithium-lithium chemical reactions decreases lithium-lithium bonding responsible for undesirable lithium-metal dendrite formation. In addition, in some implementations, the polymeric network **285** may replace the interphase layer **240** that either naturally or artificially develops between the anode **220** and the electrolyte **230**.

(85) In one implementation, the interface layer **283** of the polymeric network **285** is in contact with the anode **220**, and a protective layer **284** is disposed on top of the interface layer **283** (such as between the interface layer **283** and the interphase layer **240**). In some aspects, the interface layer **283** and the protective layer **284** may collectively define a gradient of cross-linked fluoropolymer chains of varying degrees of density, for example, as described with reference to FIG. **7**.

(86) In some other implementations, the battery **200** may include a protective lattice **280** disposed on the cathode **210**. The protective lattice **280** may include a tri-functional epoxy compound and a di-amine oligomer-based compound that may chemically react with each other to produce nitrogen and oxygen atoms. The nitrogen and oxygen atoms made available by the protective lattice **280** can bind with the polysulfides **282**, thereby confining the polysulfides **282** within the cathode **210** and/or the protective lattice **280**. Either of the cathode **210** and/or the protective lattice **280** may include carbon-carbon bonds and/or regions capable of flexing and/or volumetrically expanding during operational cycling of the battery **200**, which may confine polysulfides **282** generated during the operational cycling to the cathode **210**.

(87) The electrolyte **130** of FIG. **1** and the electrolyte **230** of FIG. **2** may be prepared according to one or more recipes disclosed herein. For example, a ternary solvent package used in the electrolyte **130** and/or the electrolyte **230** may include DME, DOL and TEGDME. In one implementation, a solvent mixture may be prepared by mixing 5800 μL DME, 2900 μL DOL and 1300 μL TEGDME and stirring at room temperature (77° F. or 25° C.). Next, 0.01 mol (2,850.75 mg) of LiTFSI may be weighed. Afterwards, the 0.01 mol of LiTFSI may be dissolved in solvent mixture by stirring at room temperature to prepare approximately 10 mL 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume 1:4:1). Finally, approximately 223 mg LiNO.sub.3 may be added to 10 mL solution to produce 10 mL 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=58:29:13) with approximately 2 wt. % LiNO.sub.3.
(88) In addition, or the alternative, a ternary solvent package used in the electrolyte **130** and/or the electrolyte **230** may include DME, DOL, TEGDME, and TBT or MBT. A solvent mixture may be prepared by mixing 2,000 μL DME, 8,000 μL DOL and 2,000 μL TEGDME and stirring at room temperature (68° F. or 25° C.). Next, 0.01 mol (2,850.75 mg) of LiTFSI may be weighed and dissolved in approximately 3 mL of the solvent mixture by stirring at room temperature. Next, the dissolved LiTFSI and an additional solvent mixture (˜8,056 mg) may be mixed in a 10 mL volumetric flask to produce approximately 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume 1:4:1). Finally, approximately 0.05 mmol (˜12.5 mg) TBT or MBT may be added to the 10 mL solution to produce 10 mL of 5M TBT or MBT solution.
(89) FIG. **3** shows an example electrode **300**, according to some implementations. In various implementations, the electrode **300** may be one example of the cathode **110** and/or the anode **120** of the battery **100** of FIG. **1**. In some other implementations, the electrode **300** may be one example of the cathode **210** of the battery **200** of FIG. **2**. When the electrode **300** is implemented as a cathode (such as the cathode **110** of the battery **100** of FIG. **1**), the electrode **300** may temporarily micro-confine an electroactive material, such as elemental sulfur, which may decrease the amount of sulfur available for reacting with lithium to produce polysulfides. In some aspects, the electrode **300** may provide an excess supply of lithium and/or lithium cations (Li.sup.+) that can compensate for first-cycle operational losses associated with lithium-based batteries.
(90) In some implementations, the electrode **300** may be porous and receptive of a liquid-phase electrolyte, such as the electrolyte **130** of FIG. **1**. Electroactive species, such as lithium cations (Li.sup.+) **125** suspended in the electrolyte **130**, may chemically react with elemental sulfur pre-loaded into pores of the electrode **300** to produce polysulfides, which in turn may be trapped in the electrode **300** during battery cycling. In some aspects, the electrode **300** may expand in volume along one or more flexure points to retain additional quantities of polysulfides created during battery cycling. By confining the polysulfides within the electrode **300**, aspects of the subject matter disclosed herein may allow the lithium cations (Li.sup.+) **125** to flow freely through the electrolyte **130** from the anode **120** to the cathode **110** during discharge cycles of the battery **100** (e.g., without being impeded by the polysulfides). For example, when lithium cations (Li.sup.+) **125** reach the cathode **110** and react with elemental sulfur contained in or associated with the cathode **110**, sulfur is reduced to lithium polysulfides (Li.sub.2S.sub.x) at decreasing chain lengths according to the order Li.sub.2S.sub.8.fwdarw.Li.sub.2S.sub.6.fwdarw.Li.sub.2S.sub.4.fwdarw.Li.sub.2S.sub.2.Math.Si.sub.2S, where 2≤x≤8). Higher order polysulfides may be soluble in various types of solvents and/or electrolytes, thereby interfering with the lithium ion transport necessary for healthy battery operation. Retention of such higher order polysulfides by the electrode **300** thereby allows the lithium cations (Li.sup.+) **125** to flow more freely through the electrolyte **130**, which in turn may increase the number of electrons available to carry charge from the anode **120** to the cathode **110**.
(91) The electrode **300** may include a body **301** defined by a width **305**, and may include a first thin film **310** and a second thin film **320**. The first thin film **310** may include a plurality of first aggregates **312** that join together to form a first porous structure **316** of the electrode **300**. In some

instances, the first porous structure **316** may have an electrical conductivity between approximately 0 and 500 S/m. In other instances, the first electrical conductivity may be between approximately 500 and 1,000 S/m. In some other instances, the first electrical conductivity may be greater than 1,000 S/m. In some aspects, the first aggregates **312** may include carbon nano-tubes (CNTs), carbon nano-onions (CNOs), flaky graphene, crinkled graphene, graphene grown on carbonaceous materials, and/or graphene grown on graphene.

(92) In some implementations, the first aggregates **312** may be decorated with a plurality of first nanoparticles **314**. In some instances, the first nanoparticles **314** may include tin, lithium alloy, iron, silver, cobalt, semiconducting materials and/or metals such as silicon and/or the like. In some aspects, CNTs, due to their ability to provide high exposed surface areas per unit volume and stability at relatively high temperatures (such as above 77° F. or 25° C.), may be used as a support material for the first nanoparticles **314**. For example, the first nanoparticles **314** may be immobilized (such as by decoration, deposition, surface modification or the like) onto exposed surfaces of CNTs and/or other carbonaceous materials. The first nanoparticles **314** may react with chemically available carbon on exposed surfaces of the CNTs and/or other carbonaceous materials.

(93) The second thin film **320** may include a plurality of second aggregates **322** that join together to form a second porous structure **326**. In some instances, the electrical conductivities of the first porous structure **316** and/or the second porous structure **326** may be between approximately 0 S/m and 250 S/m. In instances for which the first porous structure **316** includes a higher concentration of aggregates than the second porous structure **326**, the first porous structure **316** may have a higher electrical conductivity than the second porous structure **326**. In one implementation, the first electrical conductivity may be between approximately 250 S/m and 500 S/m, while the second electrical conductivity may be between approximately 100 S/m and 250 S/m. In another implementation, the second electrical conductivity may be between approximately 250 S/m and 500 S/m. In yet another implementation, the second electrical conductivity may be greater than 500 S/m. In some aspects, the second aggregates **322** may include CNTs, CNOs, flaky graphene, crinkled graphene, graphene grown on carbonaceous materials, and/or graphene grown on graphene.

(94) The second aggregates **322** may be decorated with a plurality of second nanoparticles **324**. In some implementations, the second nanoparticles **324** may include iron, silver, cobalt, semiconducting materials and/or metals such as silicon and/or the like. In some instances, CNTs may also be used as a support material for the second nanoparticles **324**. For example, the second nanoparticles **324** may be immobilized (such as by decoration, deposition, surface modification or the like) onto exposed surfaces of CNTs and/or other carbonaceous materials. The second nanoparticles **324** may react with chemically available carbon on exposed surfaces of the CNTs and/or other carbonaceous materials.

(95) In some aspects, the first thin film **310** and/or the second thin film **320** (as well as any additional thin films disposed on their respective immediately preceding thin film) may be created as a layer or region of material and/or aggregates. The layer or region may range from fractions of a nanometer to several microns in thickness, such as between approximately 0 and 5 microns, between approximately 5 and 10 microns, between approximately 10 and 15 microns, or greater than 15 microns. Any of the materials and/or aggregates disclosed herein, such as CNOs, may be incorporated into the first thin film **310** and/or the second thin film **320** to result in the described thickness levels.

(96) In some implementations, the first thin film **310** may be deposited onto the second substrate **102** of FIG. **1** by chemical deposition, physical deposition, or grown layer-by-layer through techniques such as Frank-van der Merwe growth, Stranski-Krastonov growth, Volmer-Weber growth and/or the like. In other implementations, the first thin film **310** may be deposited onto the second substrate **102** by epitaxy or other suitable thin-film deposition process involving the epitaxial growth of materials. The second thin film **320** and/or subsequent thin films may be

deposited onto their respective immediately preceding thin film in a manner similar to that described with reference to the first thin film **310**.

(97) In various implementations, each of the first aggregates **312** and/or the second aggregates **322** may be a relatively large particle formed by many relatively small particles bonded or fused together. As a result, the external surface area of the relatively large particle may be significantly smaller than combined surface areas of the many relatively small particles. The forces holding an aggregate together may be, for example, covalent, ionic bonds, or other types of chemical bonds resulting from the sintering or complex physical entanglement of former primary particles.

(98) As discussed above, the first aggregates **312** may join together to form the first porous structure **316**, and the second aggregates **322** may join together to form the second porous structure **326**. The electrical conductivity of the first porous structure **316** may be based on the concentration level of the first aggregates **312** within the first porous structure **316**, and the electrical conductivity of the second porous structure **326** may be based on the concentration level of the second aggregates **322** within the second porous structure **326**. In some aspects, the concentration level of the first aggregates **312** may cause the first porous structure **316** to have a relatively high electrical conductivity, and the concentration level of the second aggregates **322** may cause the second porous structure **326** to have a relatively low electrical conductivity (such that the first porous structure **316** has a greater electrical conductivity than the second porous structure **326**). The resulting differences in electrical conductivities of the first porous structure **316** and the second porous structure **326** may create an electrical conductivity gradient across the electrode **300**. In some implementations, the electrical conductivity gradient may be used to control or adjust electrical conduction throughout the electrode **300** and/or one or more operations of the battery **100** of FIG. **1**.

(99) As used herein, the relatively small source particles may be referred to as “primary particles,” and the relatively large aggregates formed by the primary particles may be referred to as “secondary particles.” As shown in FIG. **1**, FIGS. **8** to **10**, and elsewhere throughout the present disclosure, the primary particles may be or include multiple graphene sheets, layers, regions, and/or nanoplatelets fused and/or joined together. Thus, in some instances, carbon nano-onions (CNOs), carbon nano-tubes (CNTs), and/or other tunable carbon materials may be used to form the primary particles. In some aspects, some aggregates may have a principal dimension (such as a length, a width, and/or a diameter) between approximately 500 nm and 25 μm. Also, some aggregates may include innately-formed smaller collections of primary particles, referred to as “innate particles,” of graphene sheets, layers, regions, and/or nanoplatelets joined together at orthogonal angles. In some instances, these innate particles may each have a respective dimension between approximately 50 nm and 250 nm.

(100) The surface area and/or porosity of these innate particles may be imparted by secondary processes, such as carbon-activation by a thermal, plasma, or combined thermal-plasma process using one or more of steam, hydrogen gas, carbon dioxide, oxygen, ozone, KOH, ZnCl.sub.2, H.sub.3PO.sub.4, or other similar chemical agents alone or in combination. In some implementations, the first porous structure **316** and/or the second porous structure **326** may be produced from a carbonaceous gaseous species that can be controlled by gas-solid reactions under non-equilibrium conditions. Producing the first porous structure **316** and/or the second porous structure **326** in this manner may involve recombination of carbon-containing radicals formed from the controlled cooling of carbon-containing plasma species (which can be generated by excitement or compaction of feedstock carbon-containing gaseous and/or plasma species in a suitable chemical reactor).

(101) In some implementations, the first aggregates **312** and/or the second aggregates **322** may have a percentage of carbon to other elements, except hydrogen, within each respective aggregate of greater than 99%. In some instances, a median size of each aggregate may be between approximately 0.1 microns and 50 microns. The first aggregates **312** and/or the second aggregates

322 may also include metal organic frameworks (MOFs).
(102) In some implementations, the first porous structure **316** and second porous structure **326** may collectively define a host structure **328**, for example, as shown in FIG. **3**. In some instances, the host structure **328** may be based on a carbon scaffold and/or may include decorated carbons, for example, as shown in FIG. **8**. The host structure **328** may provide structural definition to the electrode **300**. In some instances, the host structure **328** may be fabricated as a positive electrode and used in the cathode **110** of FIG. **1**. In other implementations, the host structure **328** may be fabricated as a negative electrode and used in the anode **120** of FIG. **1**. In some other implementations, the host structure **328** may include pores having different sizes, such as micropores, mesopores, and/or macropores defined by the IUPAC. In some instances, at least some of the micropores may have a width of approximately 1.5 nm, which may be large enough to allow sulfur to be pre-loaded into the electrode **300** and yet small enough to confine polysulfides within the electrode **300**.
(103) The host structure **328**, when provided within the electrode **300** as shown in FIG. **3**, may include microporous, mesoporous, and/or macroporous pathways created by exposed surfaces and/or contours of the first porous structure **316** and/or the second porous structure **326**. These pathways may allow the host structure **328** to receive an electrolyte, for example, by transporting lithium cations (Li.sup.+) towards the cathode **110** of the battery **100**. Specifically, the electrolyte **130** may infiltrate the various porous pathways of the host structure **328** and uniformly disperse throughout the electrode **300** and/or other portions of the battery **100**. Infiltration of the electrolyte **130** into such regions of the host structure **328** may allow the lithium cations (Li.sup.+) **125** migrating from the anode **120** towards the cathode **110** to react with elemental sulfur associated with the cathode **110** to form lithium-sulfur complexes. As a result, the elemental sulfur may retain additional quantities of lithium cations (Li.sup.+) that would otherwise be achievable using non-sulfur chemistries such as lithium cobalt oxide (LiCoO) or other lithium-ion cells.
(104) In some aspects, each of the first porous structure **316** and/or the second porous structure **326** may have a porosity based on one or more of a thermal, plasma, or combined thermal-plasma process using one or more of steam, hydrogen gas, carbon dioxide, oxygen, ozone, KOH, ZnCl2, H3PO4, or other similar chemical agents alone or in combination. For example, in one implementation, the macroporous pathways may have a principal dimension greater than 50 nm, the mesoporous pathways may have a principal dimension between approximately 20 nm and 50 nm, and the microporous pathways may have a principal dimension less than 4 nm. As such, the macroporous pathways and mesoporous pathways can provide tunable conduits for transporting lithium cations (Li.sup.+) **125**, and the microporous pathways may confine active materials within the electrode **300**.
(105) In some implementations, the electrode **300** may include one or more additional thin films (not shown for simplicity). Each of the one or more additional thin films may include individual aggregates interconnected with each other across different thin films, with at least some of the thin films having different concentration levels of aggregates. As a result, the concentration levels of any thin film may be varied (such as by gradation) to achieve particular electrical resistance (or conductance) values. For example, in some implementations, the concentration levels of aggregates may progressively decline between the first thin film **310** and the last thin film (such as in a direction **195** depicted in FIG. **1**), and/or the individual thin films may have an average thickness between approximately 10 microns and approximately 200 microns. In addition, or in the alternative, the first thin film **310** may have a relatively high concentration of carbonaceous aggregates, and the second thin film **320** may have a relatively low concentration of carbonaceous aggregates. In some aspects, the relatively high concentration of aggregates corresponds to a relatively low electrical resistance, and the relatively low concentration of aggregates corresponds to a relatively high electrical resistance.
(106) The host structure **328** may be prepared with multiple active sites on exposed surfaces of the

first aggregates **312** and/or the second aggregates **322**. These active sites, as well as the exposed surfaces of the first aggregates **312** and/or the second aggregates **322**, may facilitate ex-situ electrodeposition prior to incorporation of the electrode **300** into the battery **100**. Electroplating is a process that can create a lithium layer **330** (including lithium on exposed surfaces of the host structure **328**) through chemical reduction of metal cations by application and/or modulation of an electric current. In implementations for which the electrode **300** serves as the anode **120** of the battery **100** in FIG. **1**, the host structure **328** may be electroplated such that the lithium layer **330** has a thickness between approximately 1 and 5 micrometers (μm), 5 μm and 20 μm, or greater than 20 μm. In some instances, ex-situ electrodeposition may be performed at a location separate from the battery **100** prior to the assembly of the battery **100**.
(107) In various implementations, excess lithium provided by the lithium layer **330** may increase the number of lithium cations (Li.sup.+) **125** available for transport in the battery **100**, thereby increasing the storage capacity, longevity, and performance of the battery **100** (as compared with traditional lithium-ion and/or lithium-sulfur batteries).
(108) In some aspects, the lithium layer **330** may produce lithium-intercalated graphite (LiC.sub.6) and/or lithiated graphite based on chemical reactions with the first aggregates **312** and/or the second aggregates **322**. Lithium intercalated between alternating graphene layers may migrate or be transported within the electrode **300** due to differences in electrochemical gradients during operational cycling of the battery **100**, which in turn may increase the energy storage and power delivery of the battery **100**.
(109) FIG. **4** shows a diagram of a portion of an example battery **400** that includes a protective lattice **402**, according to some implementations. In some implementations, the protective lattice **402** may be disposed on the anode **220** of the battery **200**. In other implementations, the protective lattice **402** may be disposed on the cathode **210** of the battery **200** (or other suitable batteries). In some aspects, the protective lattice **402** may be one example of the protective lattice **280** of FIG. **2**. The protective lattice **402** may function with many components (e.g., anode, cathode, current collectors associated, carbonaceous materials, electrolyte, and separator) in a manner similar to the battery **100** of FIG. **1** and/or the battery **200** of FIG. **2**.
(110) The protective lattice **402** may include a tri-functional epoxy compound and a di-amine oligomer-based compound that can chemically react with each other to produce a 3D lattice structure (e.g., as shown in FIG. **6** and FIG. **8**). In some aspects, the protective lattice **402** may prevent polysulfide migration within the battery **400** by providing nitrogen and oxygen atoms that can chemically bind with lithium present in the polysulfides, thereby impeding the migration of polysulfides through the electrolyte **130**. As a result, lithium cations (Li.sup.+) **125** can be more freely transported from the anode **120** and the cathode **110** of FIG. **1**, thereby increasing battery performance metrics.
(111) Cyclical usage of the cathode **110** may cause the formation of cracks **404** that at least partially extend into the cathode **110**. In one implementation, the protective lattice **402** may disperse throughout the cracks **404**, thereby reducing susceptibility of the cathode **110** to rupture during volumetric expansion of the cathode **110** caused by the retention of polysulfides within the cathode **110** during cyclical usage. In one implementation, the protective lattice **402** of FIG. **4** may have a cross-linked, 3D structure based on chemical reactions between di-functional, or higher functionality Epoxy and Amine or Amide compounds. For example, the di-functional, or higher functionality Epoxy compound may be trimethylolpropane triglycidyl ether (TMPTE), tris(4-hydroxyphenyl)methane triglycidyl ether, or tris(2,3-epoxypropyl) isocyanurate, and di-functional, or higher functionality Amine compound may be dihydrazide sulfoxide (DHSO) or one of polyetheramines, for example JEFFAMINE® D-230 characterized by repeating oxypropylene units in the backbone.
(112) In various implementations, the chemical compounds may be combined and reacted with each other in any number of quantities, amounts, ratios and/or compositions to achieve different

performance capabilities relating to binding with polysulfides generated during operation of the battery **400**. For example, in one implementation, 113 mg of TMPTE and 134 mg of JEFFAMINE® D-230 polyetheramine may be mixed together and diluted with 1 mL to 10 mL of tetrahydrofuran (THF), or any other solvent. Additional amounts of TMPTE and/or JEFFAMINE may be mixed together and diluted in THF, or any other solvent, at an example ratio of 113 mg of TMPTE for every 134 mg of JEFFAMINE® D-230 polyetheramine. For this implementation, proof-of-concept (POC) data shows that the protective lattice **402** of FIG. **4** has a defined weight of approximately 2.6 wt. % of the cathode **110** of FIG. **1** or the cathode **210** of FIG. **2**. In other implementations, the protective lattice **402** may have a weight of approximately 2 wt. % to 21 wt. % of the cathode **110** and/or the cathode **210**, where an impedance increases of the cathode **110** and/or the cathode **210** may be expected at a weight level of approximately 10 wt. % or more for the protective lattice **402**.
(113) In various implementations, the protective lattice **402** may be fabricated based on a mole and/or molar ratio of —NH.sub.2 group and epoxy groups and may further accommodate various forms of cross-linking between di-functional, or higher functionality Epoxy and Amine or Amide compound. In some aspects, such forms of cross-linking may include a fully cross-linked stage, e.g., where one —NH.sub.2 group is chemically bonded with two epoxy groups and may further extend to configurations including one NH.sub.2 group chemically bonded with only one epoxy group. Still further, in one or more implementations, mixtures including excess quantities (above the ratios presented here) of —NH.sub.2 groups may be prepared to provide additional polysulfide binding capability for the protective lattice **402**.
(114) In some other implementations, the protective lattice **402** may be prepared by mixing 201 g of TMPTE with between 109 g and 283 g of JEFFAMINE® D-230 polyetheramine. The resulting mixture may be then diluted with 1 L to 20 L of a selected solvent (such as THF). The resultant diluted solution may be deposited and/or otherwise disposed on the cathode **110** to achieve a crosslinker content between 1 wt. % to 10 wt. %. Additional TMPTE and/or JEFFAMINE may be mixed together and diluted in THF, or another suitable solvent, at an example ratio of 201 g of TMPTE for every 109 g to 283 g of JEFFAMINE® D-230 polyetheramine.
(115) In still other implementations, the protective lattice **402** may be prepared by mixing 201 g of TMPTE with between 74 g and 278 g DHSO. The resulting mixture may be then diluted with 1 L to 20 L of a selected solvent (such as THF). The resultant diluted solution may be deposited and/or otherwise disposed on the cathode **110** to achieve a crosslinker content between 1 wt. % to 10 wt. %. Additional TMPTE and/or JEFFAMINE may be mixed together and diluted in THF, or another suitable solvent, at an example ratio of 201 g of TMPTE for every 201 g to 278 g of JEFFAMINE® D-230 polyetheramine.
(116) In one implementation, di-functional, or higher functionality Epoxy compound may chemically react with di-functional, or higher functionality amine compound to produce the protective lattice **402** in a 3D cross-linked form, which may include both functional epoxy compounds and amine containing molecules. In some aspects, the protective lattice **402**, when deposited on the cathode **110** of FIG. **1** or the cathode **210** of FIG. **2**, may have a thickness between approximately 1 nm and 5 μm.
(117) In some implementations, the protective lattice **402** may increase the structural integrity of the cathode **110** or the cathode **210**, may reduce surface roughness, and may retain polysulfides in the cathode. For example, in one implementation, the protective lattice **402** may serve as sheath on exposed surfaces of the cathode and bind with polysulfides to prevent their migration and diffusion into the electrolyte **130**. In this way, aspects of the subject matter disclosed herein may prevent (or at least reduce) battery capacity decay by suppressing the polysulfide shuttle effect. In some aspects, the protective lattice **402** may also fill the cracks **404** formed in the cathode of FIG. **4** to improve cathode coating integrity. In various implementations, the protective lattice **402** may be prepared by drop casting processes in the presence of a solvent, where the resultant solution can

penetrate in cracks **404** of the cathode **110** and bind with polysulfides in the cathode **110** to prevent their migration and/or diffusion throughout the electrolyte **130**.

(118) In various implementations, the protective lattice **402** may provide nitrogen atoms and/or oxygen atoms that can chemically bond with lithium in the polysulfides generated during operational battery cycling. In one example, the polysulfides may bond with available nitrogen atoms provided by, for example, DHSO. In another example, the polysulfides may bond with available oxygen atoms provided by, for example, DHSO. In yet another example, the polysulfides may bond with other available oxygen atoms.

(119) In some other implementations, the recipes described above may be altered by replacing TMPTE with a tris(4-hydroxyphenyl)methane triglycidyl ether **910** and/or a tris(2,3-epoxypropyl) isocyanurate. In various implementations, the di-amine oligomer-based compound may be (or may include) a JEFFAMINE® D-230, or other polyetheramines containing polyether backbone normally based on either propylene oxide (PO), ethylene oxide (EO), or mixed PO/EO structure, for example JEFFAMINE® D-400, JEFFAMINE® T-403. The protective lattice **402** may also include various concentration levels of inert molecules, e.g., polyethylene glycol chains of various lengths, which may allow to fine-tune mechanical properties of protective lattice and the chemical bonding of various atoms to lithium present in the polysulfides.

(120) FIG. **5** shows a diagram of an anode structure **500** that includes a tin fluoride (SnF_2) layer, according to some implementations. Specifically, the diagram depicts a cut-away schematic view of the anode structure **500** in which all of the components associated with a first region A have identical counterparts in a second region B, where the first and second regions A and B have opposite orientations around a current collector **520**. As such, the description below with reference to the components of first region A is equally applicable to the components of second region B. In some aspects, the anode **502** may be one example of the anode **120** of FIG. **1** and/or the anode **220** of FIG. **2**.

(121) As discussed, lithium-sulfur batteries, such as the battery **100** of FIG. **1** and the battery **200** of FIG. **2**, operate as conversion-chemistry type electrochemical cells in that sulfur pre-loaded into the cathode may dissolve rapidly into the electrolyte prior to and during operation. Lithium, which may be provided by lithiated anodes and/or may be prevalent in the electrolyte, dissociates into lithium cations (Li^+) suitable for transport from the anode to the cathode through the electrolyte. The production of lithium cations (Li^+) is associated with a corresponding release of electrons, which may flow through an external circuit to power a load, as described with reference to FIG. **1**. However, when lithium dissociates into lithium cations (Li^+) and electrons (e^-), some of the lithium cations (Li^+) may undesirably react with polysulfides produced in the cathode, and therefore may no longer be available to generate an output current or voltage. This consumption of lithium cations (Li^+) by polysulfides reduces the overall capacity of the host cell or battery, and may also facilitate corrosion of the anode, which can result in cell failure.

(122) In some implementations, the protective layer **516** may be provided as passivation coating that can reduce the chemical reactivity of the anode **502** during cell assembly or formation. In some aspects, the protective layer **516** may be permeable to lithium cations (Li^+) while concurrently protecting the anode **502** from corrosion caused by chemical reactions between lithium cations (Li^+) and polysulfides. In other implementations, the protective layer **516** may be an artificial solid-electrolyte interphase (A-SEI) that can replace naturally occurring SEIs and/or other types of conventional A-SEIs. In various implementations, the protective layer **516** may be deposited as a liner on top of one or more films disposed on the anode **502**. In some aspects, the protective layer **516** may be a self-generating layer that forms during electrochemical reactions associated with operational cycling of the battery. In some aspects, the protective layer **516** may have a thickness that is less than 5 microns. In other aspects, the protective layer **516** may have a thickness between 0.1 and 1.0 microns.

(123) In various implementations, one or more engineered additives that may facilitate the

formation and/or deposition of the protective layer **516** on the anode **502** may be provided within the electrolyte of the battery. In other implementations, the engineered additives may be an active ingredient of the protective layer **516**. In some aspects, the protective layer **516** may provide tin ions and/or fluoride anions that can prevent undesirable lithium growths from a first edge **5181** and a second edge **5182** of the anode.

(124) A graded layer **514** may be formed and/or deposited onto the anode **502** beneath the protective layer **516**. In various implementations, the graded layer **514** may prevent lithium contained in or associated with the anode **502** from participating in undesirable chemical interactions and/or reactions with the electrolyte **540** that can lead to the growth of lithium-containing dendrites from the anode **502**. The graded layer **514** may also facilitate the production of lithium fluoride based on chemical reactions between dissociated lithium cations (Li.sup.+) and fluoride ions. As discussed, the presence of lithium fluoride in or near the anode **502** can decrease the polysulfide shuttle effect. For example, formation of lithium fluoride (e.g., form available lithium cations (Li.sup.+) and fluorine ions) may occur uniformly across the entirety of the first edge **5181** and/or the second edge **5182** of the anode. In this way, localized regions of high lithium concentration in the electrolyte **540** near the anode **502** are substantially inhibited. As a result, lithium-lithium bonds contributing to the formation of lithium containing dendritic structures extending length-wise from the anode are correspondingly inhibited, thereby yielding free passage of lithium cations (Li.sup.+) from the anode **502** into the electrolyte (e.g., as encountered during battery operational cycling). In some aspects, the uniform distribution of lithium throughout the graded layer **514** can increase a uniformity of a lithium-ion flux during battery operational cycling. In some aspects, the graded layer **514** may be approximately 5 nanometers (nm) in thickness.

(125) In one or more implementations, the graded layer **514** may structurally reinforce the host battery in a manner that not only decreases or prevents lithium-containing dendritic growth from the anode **502** but also increases the ability of the anode **502** to expand and contract during operational cycling of the host battery without rupturing. In some aspects, the graded layer **514** has a 3D architecture with a graded concentration gradient (e.g., of one or more formative materials and/or ingredients including carbon, tin, and/or fluorine), which facilitates rapid lithium-ion transport. As a result, the graded layer **514** markedly improves overall battery efficiency and performance.

(126) In some implementations, the graded layer **514** may provide an electrochemically desirable surface upon which the protective layer **516** may be grown or deposited. For example, in some aspects, the graded layer **514** may include compounds and/or organometallic compounds including (but not limited to) aluminum, gallium, indium, nickel, zinc, chromium, vanadium, titanium, and/or other metals. In other aspects, the graded layer **514** may include oxides, carbides and/or nitrides of aluminum, gallium, indium, nickel, zinc, chromium, vanadium, titanium, and/or other metals.

(127) In some implementations, the graded layer **514** may include carbonaceous materials including (but not limited to) flaky graphene, few layer graphene (FLG), carbon nano onions (CNOs), graphene nanoplatelets, or carbon nanotubes (CNTs). In other implementations, the graded layer **514** may include carbon, oxygen, hydrogen, tin, fluorine and/or other suitable chemical compounds and/or molecules derived from tin fluoride and one or more carbonaceous materials. The graded layer **514** may be prepared and/or deposited either directly or indirectly on the anode **502** at a different concentration levels. For example, the graded layer **514** may include 5 wt. % carbonaceous materials with a balance of 95 wt. % tin fluoride, which may result in a relatively uniform disassociation of fluorine atoms and/or fluoride anions from the tin fluoride.

(128) Other suitable ratios include: 5% carbonaceous materials with 95% tin fluoride; 10% carbonaceous materials with 90% tin fluoride, 15% carbonaceous materials with 85% tin fluoride, 20% carbonaceous materials with 80% tin fluoride, 25% carbonaceous materials with 75% tin fluoride, 30% carbonaceous materials with 70% tin fluoride, 35% carbonaceous materials with 65% tin fluoride, 40% carbonaceous materials with 60% tin fluoride, 45% carbonaceous materials

with 55% tin fluoride, 50% carbonaceous materials with 50% tin fluoride, 55% carbonaceous materials with 45% tin fluoride, 55% carbonaceous materials with 45% tin fluoride, 60% carbonaceous materials with 40% tin fluoride, 65% carbonaceous materials with 35% tin fluoride, 70% carbonaceous materials with 30% tin fluoride, 75% carbonaceous materials with 25% tin fluoride, 80% carbonaceous materials with 20% tin fluoride, 85% carbonaceous materials with 15% tin fluoride, 90% carbonaceous materials with 10% tin fluoride, 95% carbonaceous materials with 5% tin fluoride. The fluorine atoms and/or fluoride anions may then uniformly react and combine with lithium cations (Li.sup.+) to form lithium fluoride, as further discussed below.
(129) In some implementations, lithium cations (Li.sup.+) cycling between the anode **502** and the cathode (not shown in FIG. **5**) may produce a tin-lithium alloy region **512** within the graded layer **514**. In some aspects, operational cycling of the host battery may result in a uniform dispersion of lithium fluoride within the tin-lithium alloy region **512**. The uniform dispersion of lithium fluoride may facilitate a defluorination reaction of at least some of tin (II) fluoride (SnF.sub.2) within the tin fluoride layer **510** (and additional tin fluoride which may have dispersed into the graded layer **514** and/or the protective layer). The fluorine atoms and/or fluoride anions made available by the defluorination reaction may chemically bond with at least some of the lithium cations (Li.sup.+) present in or near the anode **502**, to create lithium fluoride (LiF) and correspondingly thereby prevent at least some of the lithium cations (Li.sup.+) from bonding with each other and creating a lithium dendritic growth from the anode **502**.
(130) For example, at least a portion of the fluorine atoms and/or fluoride anions present in the tin fluoride may dissociate from the protective layer **516** and produce tin cations (Sn.sup.2+) and fluorine anions (2F.sup.−) via one or more chemical reactions. The fluorine atoms and/or fluoride anions dissociated from the protective layer **516** may chemically bond to at least some of the lithium cations (Li.sup.+) present in the electrolyte **540** and/or dispersed throughout the protective layer **516** or the graded layer **514**. In some aspects, the dissociated fluorine atoms may form Li—F bonds or Li—F compounds in the tin-lithium alloy region **512**. In other aspects, the dissociated fluorine atoms may form a tin fluoride layer **510** within the graded layer **514**.
(131) In addition, in one implementation, at least some of the defluorinated tin fluoride may disperse uniformly throughout the graded layer **514** to produce lithium fluoride (LiF) crystals. The lithium fluoride crystals may act as an electrical insulator and prevent the flow of electrons from the anode **502** into the electrolyte **540** through the first edge **5181** and/or the second edge **5182** of the anode **502**.
(132) In various implementations, the graded layer **514** may be deposited on the anode **502** by one or more of atomic layer deposition (ALD), chemical vapor deposition (CVD), or physical vapor deposition (PVD). For example, ALD may be used to deposit protective films on the anode **502** such as, for example, an ALD film that at least partially reacts with the electrolyte **540** during high-pressure bonding processes. Accordingly, the ALD film may be used to produce the protective layer **516** or the graded layer **514** using an atomic plane available for lithium transfer. Such lithium transfer may be similar in principle to that observed for few layer graphene (FLG) or graphite, where alternating graphene layers in FLG or graphite intercalate lithium cations (Li.sup.+) in various forms including as lithium titanium oxide (LTO), lithium iron phosphate (PO.sub.3) (LFP). The described forms of intercalated lithium, e.g., LTO and/or LFP, may be oriented to facilitate rapid lithium atom and/or lithium ion transport and/or diffusion, which may be conducive for the formation and/or synthesis of lithium fluoride (e.g., in the tin fluoride layer **510** and/or elsewhere), as described earlier. Additional forms of intercalated lithium, e.g., perovskite lithium lanthanum titanate (LLTO), may also function to store lithium within the anode **502**.
(133) In some implementations, the graded layer **514** may include various distinct types and/or forms of carbon and/or carbonaceous materials, each having one or more physical attributes that can be selected or configured to adjust the reactivity of carbon with contaminants (such as polysulfides) present in the electrolyte **540** and/or the anode **502**. In some aspects, the selectable

physical attributes may include (but are not limited to) porosity, surface area, surface functionalization, or electric conductivity. In addition, the graded layer **514** may include binders or other additives that can be used to adjust one or more physical attributes of the carbonaceous materials to achieve a desired reactivity of carbon supplied by the carbonaceous materials with polysulfides present in the electrolyte **540** and/or the anode **502**.

(134) In one implementation, carbonaceous materials within the graded layer **514** may capture unwanted contaminants and thereby prevent the contaminants from chemically reacting with lithium available at exposed surfaces of the anode **502**. Instead, the unwanted contaminants (e.g., polysulfides) may chemically react with various exposed surfaces of the carbonaceous materials within the graded layer **514** (e.g., through carbon-lithium interactions). In some implementations, the carbonaceous materials within the graded layer **514** may cohere to the available lithium. The degree of cohesion between the carbonaceous materials and the lithium cations (Li.sup.+) may be selected or modified via chemical reactions induced during preparation of the graded layer **514**.

(135) In some implementations, various carbon allotropes may be incorporated within the graded layer **514** (such as in one or more portions of the tin-lithium alloy region **512** and/or the tin fluoride layer **510**). These carbon allotropes may be functionalized with one or more reactants and used to form a sealant layer and/or region at an interface of carbon nanodiamonds within the graded layer **514** and the electrolyte **540**. In some aspects, the carbon nanodiamonds may increase the mechanical robustness of the anode **502** and/or the graded layer **514**. In other aspects, the carbon nanodiamonds may also provide exposed carbonaceous surfaces that may be used to decrease the polysulfide shuttle effect by micro-confining and/or bonding with polysulfides present in the electrolyte **540** in a manner that retains the polysulfides within defined regions of the battery external to the anode **502**.

(136) Alternatively, in other implementations, the carbon nanodiamonds within the graded layer **514** may be replaced with carbons and/or carbonaceous materials including surfaces and/or regions having a specific LA dimensions (e.g., sp.sup.2 hybridized carbon), reduced graphene oxide (rGO), and/or graphene. In some aspects, employing the carbonaceous materials disclosed herein within a battery may increase carbon stacking and layer formation within the graded layer **514**. Exfoliated and oxidized carbonaceous materials may also yield more uniform layered structures within the graded layer **514** (as compared to carbonaceous materials that have not been exfoliated and oxidized). In some aspects, solvents such as tetrabutylammonium hydroxide (TBA) and/or dimethyl formamide (DMF) treatments may be applied to the carbonaceous materials disclosed herein to increase the wetting of exposed carbonaceous surfaces within the graded layer **514**.

(137) In some implementations, slurries used to form the graded layer **514** may be doped to improve or otherwise influence the crystalline structure of carbonaceous materials within the graded layer **514**. For example, addition of certain dopants may influence the crystalline structure of the carbonaceous materials in a certain corresponding way, and functional groups may be added (e.g., via grafting onto exposed carbon atoms within the carbonaceous materials) within the graded layer **514**.

(138) In some implementations, carbonaceous materials having exposed surfaces functionalized with one or more of fluorine-containing or silicon-containing functional groups may be included within the graded layer **514**. In other implementations, carbonaceous materials having exposed surfaces functionalized with one or more of fluorine-containing or silicon-containing functional groups may be deposited beneath the graded layer **514** to form a stable SEI on at an interface between the graded layer **514** and the anode **502**. In one implementation, the stable SEI may replace the protective layer **516**. In some implementations, the graded layer **514** may be slurry cast and/or deposited using other techniques onto the anode **502** with lithium and carbon interphases, any of which may be functionalized with silicon and/or nitrogen to inhibit the diffusion and migration of polysulfides towards exposed surfaces of the anode **502**. In addition, specific polymers and/or crosslinkers may be incorporated within the graded layer **514** to mechanically

strengthen the graded layer **514**, to improve lithium ion transport across the graded layer **514**, or to increase the uniformity of lithium ion flux across the graded layer **514**. Example polymers and/or polymeric materials suitable for incorporation within the graded layer **514** may include poly(ethylene oxide) and poly(ethyleneimine). Example crosslinkers suitable for incorporation within the graded layer **514** may include inorganic linkers (e.g., borate, aluminate, silicate), multifunctional organic molecules (e.g., diamines, diols), polyurea, or high molecular weight (MW) (e.g., >10,000 daltons) carboxymethyl cellulose (CMC).
(139) Various fabrication methods may be employed to produce the graded layer **514**. In one implementation, direct coating of the interface between the anode **502** and the electrolyte **540** prior to the deposition and/or formation of the graded layer **514** may be performed with a dispersion of carbonaceous materials and other chemicals dissolved in a carrier (e.g., a solvent, binder, polymer). In another implementation, deposition of the graded layer **514** may be performed as a separate operation, or may be added to various other active ingredients (e.g., metals, carbonaceous materials, tin fluoride and/or the like) into a slurry that can be cast onto the anode **502**. Alternatively, in another implementation, the protective layer **516** may be transferred directly onto the anode **502** by a calendar roll lamination processes. The protective layer **516** and/or the graded layer **514** may also incorporate partially-cured lithium ion conductive epoxies to, for example, increase adhesion with lithium better during the calendar roll lamination processes.
(140) In one implementation, a carbon-inclusive layered structure (not shown in FIG. **5**) may be disposed on the anode **502** as a replacement for the graded layer **514**. The carbon-inclusive layered structure may include an atomic plane available for lithium transfer, and may uniformly transport lithium cations (Li.sup.+) provided by the electrolyte **540** throughout the protective layer **516** in a manner that can guide the formation of lithium fluoride in various portions of the battery. In various implementations, the carbon-inclusive layered structure may include one or more arrangements of few layer graphene (FLG) or graphite and/or may intercalate with lithium and produce one or more reaction products including lithium tin oxide (LTO), lithium iron phosphate (LFP), or perovskite lithium lanthanum titanate (LLTO).
(141) In some implementations, the tin fluoride layer **510** may function as a protection layer against corrosion, including corrosion of copper-inclusive surfaces and/or regions of the protective layer **516**, the graded layer **514**, or the anode **502**. In some aspects, the tin fluoride layer **510** may also provide a uniform seed layer suitable for lithium deposition, and thereby inhibiting dendrite formation. In addition, in some implementations, the tin fluoride layer **510** may include one or more lithium ion intercalating compounds, any one or more having a low voltage penalty. Suitable lithium ion intercalating compounds may include graphitic carbon (e.g., graphite, graphene, reduced graphene oxide, rGO). In one implementation, during fabrication of the anode **502**, lithium cations (Li.sup.+) may tend to intercalate prior to plating onto exposed carbonaceous surfaces within the tin fluoride layer **510**. In this way, the tin fluoride layer **510** will have a uniform Li distribution ready to act as a seed layer prior to initiation of lithium plating and/or electroplating operations.
(142) In one implementation, one or more conformal coatings may be applied over portions of the anode **502** such that the resulting conformal coating contacts and conforms to the first edge **5181** and/or the second edge **5182** of the anode **502**. In some aspects, the conformal coating may begin as a first spacer edge protection region **5301** and a second spacer edge protection region **5302** that react or otherwise combine with one or more of the protective layer **516**, the tin-lithium alloy region **512**, and/or the tin fluoride layer **510** to form a conformal coating **544** that at least partially seals and protects surfaces and/or interfaces between lithium in the anode **502** and various substances suspended in the electrolyte, e.g., copper (Cu). In some aspects, the dissociation of fluorine atoms from tin fluoride present in the conformal coating **544** may react with lithium in the anode **502** to form lithium fluoride, rather than form or grow into lithium dendrites. In this way, the conformal coating **544** may decrease lithium dendrite formation or growth from the anode **502**.

(143) The conformal coating **544** may be deposited or disposed over the anode **502** at any number of different thicknesses. In some aspects, the conformal coating **544** may be less than 5 μm thick. In other aspects, the conformal coating **544** may be less than 2 μm thick. In some other aspects, the conformal coating **544** may be less than 1 μm thick. These thickness levels may impede the migration of polysulfides towards the anode **502** during battery cycling, thereby preventing at least some of the lithium cations (Li.sup.+) from reacting with the polysulfides. Lithium cations (Li.sup.+) that do not react with the polysulfides are available for transport from the anode to the cathode during discharge cycles of the battery.
(144) The conformal coating **544** (as well as the protective layer **516** and the graded layer **514**) can uniquely regulate lithium ion flux toward the first edge **5181** and/or the second edge **5182** of the anode **502**, and thereby prevent corrosion of the anode **502**. Such regulation may function in a similar manner to gate spacers used during the fabrication of polysilicon (poly-Si) gates. Specifically, gate spacer or gate sidewall constructs may be used to protect and mechanically support polysilicon gates during the fabrication of integrated circuits (ICs). Similarly, edge protection provided by the conformal coating **544** for the anode **502** of FIG. **5** regulates lithium ion flux toward the first edge **5181** and/or the second edge **5182** of the anode **502**, and thereby prevents corrosion of the anode **502**. This type of edge protection provided by the conformal coating **544** for the anode **502** may equally apply to other battery and/or electrical cell formats and/or configurations such as (but not limited to) cylindrical cells, stacked cells, and/or the like, with various constructs engineered specifically to fit within the parameters of each of these designs.
(145) In some implementations, fabrication and/or deposition of the conformal coating **544**, the protective layer **516**, and/or the graded layer **514** on the anode **502** may depend on the type of battery or cell construct in which the anode **502** is incorporated, e.g., cylindrical cells compared to pouch cells and/or prismatic cells. In one implementation, for cylindrical cells, metal anodes may be constructed from an electroactive material, typically metallic lithium, and/or lithium-containing alloys, such as graphitic and/or other carbonaceous composited including lithium, as well as any plenary uniform or multi-layer sheet of material. In one example, a solid metal lithium foil used as the anode **502** may be attached to a copper substrate used as the current collector **520** to facilitate electron transfer through a tab **546** to an external load, as depicted in the example of FIG. **5**. In other implementations, the anode structure **500** may include the anode **502** without the current collector **520**, where carbonaceous materials contained within the anode **502** may provide an electrically conductive medium coupled to a circuit.
(146) In some implementations, the anode structure **500** may be incorporated into electrochemical cells and/or batteries by winding around a mandrel. Cylindrical cell layouts typically use double-sided anodes, such as the anode structure **500**. In some implementations, cylindrical cell constructions employing the anode structure **500** may use the conformal coating **544** to protect the first edge **5181** and/or the second edge **5182** of the anode **502**. The uniform protection provided by the conformal coating **544** may be referred to herein as “edge protection.” In one implementation, edge protection can be incorporated into a cell employing the anode structure **500** by extending the size and/or area of the protective layer **516** to overlap beyond any geometrically induced edge effects, e.g., surface roughness, of the anode.
(147) In other implementations, the anode structure **500** may be incorporated into pouch cells and/or prismatic cells. Generally, two constructs of pouch and/or prismatic cells may be manufactured, including (1): jelly-roll type cells (e.g., seen in industry as lithium-polymer batteries), two mandrel wound electrodes may be produced in a manner similar to cylindrical cells as discussed earlier; and (2): stacked plate type cells, which may be cut from a sheet of a pre-cast and/or pre-laminated prepared anode, leaving an unprotected edge of, for example, the anode **502** (when prepared in a stacked-plate type configuration) exposed and vulnerable to corrosion, fast ion fluxes and exposure within the cells. The conformal coating **544**, in a stacked-plate type configuration, may protect the anode **502** and prevent lithium over-saturation in the electrolyte **540**.

In this way, the conformal coating **544** can control lithium plating on the anode **502** during operational cycling of the battery.

(148) In some implementations, one or more chemical reactions may occur between the electrolyte **540** and the anode **502** (involving solvent decomposition and/or additive reactions) during cell assembly or cell rest period. These chemical reactions may assist in the production of the conformal coating **544**. In some aspects, elevated and/or reduced temperatures (e.g., relative to room temperature and/or 20° C.) may be used as a stimulus for lithium-induced polymerization of the conformal coating **544**. For example, the lithium-induced polymerization may occur in the presence of one or more catalysts and/or by using lithium metal, and its associated chemical reactivity, as an inducing agent to initiate free-radical based polymerization of component species within any one or more layers of the anode structure **500** and/or the conformal coating **544**. In addition, electrochemical reactions under electrical bias in either the forward or reverse direction may be used to fabricate and/or deposit the conformal coating **544** onto the anode **502**, as well as usage of secondary metals and/or salts as additives that may decompose to form an alloy on the first edge **5181** and/or the second edge **5182** of metallic lithium in the anode **502** exposed to the electrolyte **540**. For example, suitable additives may contain one or more metallic species, e.g., desired for co-alloying with lithium or to be used as a blocking layer to reduce lithium transfer to the first edge **5181** and/or the second edge **5182** of the anode **502**.

(149) FIG. **6** shows a schematic diagram of an enlarged portion **600** of the anode structure **500** of FIG. **5**, according to some implementations. The enlarged portion **600** illustrates placement of the first spacer edge protection region **5301** and the second spacer edge protection region **5302** (collectively referred to as the edge protection region **530** in FIG. **6**) in a direction orthogonal to the first edge **5181** and/or the second edge **5182**, as shown in FIG. **5**. As a result, the edge protection region **530**, which may include the carbonaceous materials **610** organized into structures and/or lattices, may block lithium cations (Li.sup.+) from undesirably escaping the anode **502** across the edge protection region **530**. In this way, lithium ion dissociation, flux, transport, and/or other movement may be channeled effectively throughout the enlarged portion **600** of FIG. **6** (as well as the anode structure **500** of FIG. **5**), thereby yielding optimal battery operational cycling. In some implementations, carbonaceous materials **610** used to produce the edge protection region may include few layer graphene (FLG), multi-layer graphene (MLG), graphite, carbon nano-tubes (CNTs), carbon nano-onions (CNOs) and/or the like. The carbonaceous materials **610** (e.g., shown in FIG. **8**A, FIG. **8**B, FIG. **9**A, FIG. **9**B, FIG. **10**A and/or FIG. **10**B) may be synthesized, self-nucleated, or otherwise joined together at varying concentration levels to provide for complete tunability of the edge protection region **530**. For example, the density, thickness, and/or compositions of may be designed to reduce lithium ion permeation more than the protective layer **516** or the graded layer **514** to direct lithium ion permeation accordingly. In some implementations, the edge protection region **530** may be less than 5 μm thick. In other aspects, the edge protection region **530** may be less than 2 μm thick. In some other aspects, the edge protection region **530** may be less than 1 μm thick. In some implementations, a conductive additive **640** may be added to the carbonaceous materials **610**, as well as a binder **620**.

(150) FIG. **7** shows a diagram of a polymeric network **710**, according to some implementations. In some aspects, the polymeric network **710** may be one example of the polymeric network **285** of FIG. **2**. The polymeric network **710** may be disposed on an anode **702**. The anode **702** may be formed as an alkali metal layer having one or more exposed surfaces that include any number of alkali metal-containing nanostructures or microstructures. The alkali metal may include (but is not limited to) lithium, sodium, zinc, indium and/or gallium. The anode **702** may release alkali cations during operational cycling of the battery.

(151) A layer **714** of carbonaceous materials may be grafted with fluorinated polymer chains and deposited over one or more exposed surfaces of the anode **702**. The grafting may be based on (e.g., initiated by) activation of carbonaceous material with one or more radical initiators, for example,

benzoyl peroxide (BPO) or azobisisobutyronitrile (AIBN), followed by reaction with monomer molecules. The polymeric network **710** may be based on the fluorinated polymer chains cross-linked with one another and carbonaceous materials of the layer **714** such that the layer **714** is consumed during generation of the polymeric network **710**. In some implementations, the polymeric network **710** may have a thickness approximately between 0.001 μm and 5 μm and include between approximately 0.001 wt. % to 2 wt. % of the fluorinated polymer chains. In some other implementations, the polymeric network **710** may include between approximately 5 wt. % to 100 wt. % of the plurality of carbonaceous materials grafted with fluorinated polymer chains and a balance of fluorinated polymers, or one or more non-fluorinated polymers, or one or more cross-linkable monomers, or combinations thereof. In one implementation, carbonaceous materials grafted with fluorinated polymer chains may include 5 wt. % to 50 wt. % of fluorinated polymer chains and a balance of carbonaceous material.
(152) During battery cycling, carbon-fluorine bonds within the polymeric network **710** may chemically react with newly forming Lithium metal and convert into carbon-Lithium bonds (C—Li). These C—Li bonds may, in turn, react with carbon-fluorine bonds within the polymeric network **710** via a Wurtz reaction **750**, to further cross-link polymeric network by newly formed C—C bonds and to form an alkali-metal containing fluoride (such as lithium fluoride (LiF)). Additional polymeric network cross-linking leading to uniform formation of the alkali-metal containing fluoride may thereby suppress alkali metal dendrite formation **740** associated with the anode **702**, thereby improving battery performance and longevity. In one implementation, grafting of fluorinated m/acrylate (FMA) to one or more exposed graphene surfaces of carbonaceous materials in the layer **714** may be performed in an organic solution, e.g., leading to the formation of graphene-graft-poly-FMA and/or the like. Incorporation of carbon-fluorine bonds on exposed graphene surfaces may enable the Wurtz reaction **750** to occur between carbon-fluorine bonds and metallic surface of an alkali metal (e.g., lithium) provided by the anode **702**. In this way, completion of the Wurtz reaction **750** may result in the formation of the polymeric network **710**. In some aspects, the polymeric network **710** may include a density gradient **716** pursuant to completion of the Wurtz reaction **750**. The density gradient **716** may include interconnected graphene flakes and may be infused with one or more metal-fluoride salts formed in-situ. In addition, layer porosity and/or mechanical properties may be tuned by carbon loading and/or a combination of functionalized carbons, each having a unique and/or distinct physical structure.
(153) In some implementations, carbonaceous materials within the density gradient **716** may include one or more of flat graphene, wrinkled graphene, a plurality of carbon nano-tubes (CNTs), or a plurality of carbon nano-onions (CNOs) (e.g., as depicted in FIG. **8**A and/FIG. **8**B and as shown in the micrographs of FIGS. **9**A-**9**B and FIGS. **10**A-**10**B). In one implementation, graphene nanoplatelets may be dispersed throughout and isolated from each other within the polymeric network **710**. The dispersion of the graphene nanoplatelets includes one or more different concentration levels. In one implementation, the dispersion of the graphene nanoplatelets may include at least some of the carbonaceous materials functionalized with at least some of the fluorinated polymer chains.
(154) For example, the fluorinated polymer chains may include one or more acrylate or methacrylate monomers including 2,2,3,3,4,4,5,5,6,6,7,7-Dodecafluoroheptyl acrylate (DFHA), 3,3,4,4,5,5,6,6,7,7,8,8,9,9,10,10,10-Heptadecafluorodecyl methacrylate (HDFDMA), 2,2,3,3,4,4,5,5-Octafluoropentyl methacrylate (OFPMA), Tetrafluoropropyl methacrylate (TFPM), 3-[3,3,3-Trifluoro-2-hydroxy-2-(trifluoromethyl)propyl]bicyclo[2.2.1]hept-2-yl methacrylate (HFA monomer), or vinyl-based monomers including 2,3,4,5,6-Pentafluorostyrene (PFSt).
(155) In some implementations, fluorinated polymer chains may be grafted to a surface of the layer of carbonaceous materials and may thereby chemically interact with the one or more surfaces of the alkali metal of the anode via the Wurtz reaction **750**. In organic chemistry, organometallic chemistry, and inorganic main-group polymers, the Wurtz reaction is a coupling reaction, whereby

two alkyl halides are reacted with sodium metal (or some other metal) in dry ether solution to form a higher alkane. In this reaction alkyl halides are treated with alkali metal, for example, sodium metal in dry ethereal (free from moisture) solution to produce higher alkanes. In case of Sodium intermediate product of the Wurtz reaction are highly polar and highly reactive Carbon-Sodium metal bonds, which in turn are chemically reacting with Carbon-Halide bonds to yield newly formed C—C bonds and Sodium Halide. A formation of new Carbon-Carbon bonds allows to use the Wurtz reaction for the preparation of higher alkanes containing even number of carbon atoms, for example:

(156) $2R\text{-}X + 2\mathrm{Na}$ .fwdarw. $R\text{-}R + 2\mathrm{Na}^{+}X^{-}$ (Eq. 1)

(157) Other metals have also been used to influence Wurtz coupling, among them silver, zinc, iron, activated copper, indium and a mixture of manganese and copper chloride. The related reaction dealing with aryl halides is called the Wurtz-Fittig reaction. This can be explained by the formation of free radical intermediate and its subsequent disproportionation to give alkene. The Wurtz reaction **750** occurs through a free-radical mechanism that makes possible side reactions producing alkene products. In some implementations, chemical interactions associated with the Wurtz reaction described above may form an alkali metal fluoride, e.g., lithium fluoride.

(158) In one implementation, the polymeric network **710** may include an interface layer **718** in contact with the anode **702**. A protective layer **720** may be disposed on top of the interface layer **718**, which may be based on the Wurtz reaction **750** at an interface between the anode **702** and the polymeric network **710**. The interface layer **718** may have a relatively high cross-linking density (e.g., of fluorinated polymers and/or the like), a high metal-fluoride concentration, and a relatively low carbon-fluorine bond concentration. In contrast to the interface layer **718**, the protective layer **720** may have a relatively low cross-linking density, a low metal-fluoride concentration, and a high carbon-fluorine bond concentration.

(159) In some implementations, the interface layer **718** may include cross-linkable monomers such as methacrylate (MA), acrylate, vinyl functional groups, or a combination of epoxy and amine functional groups. In one implementation, the protective layer **720** may be characterized by the density gradient **716**. In this way, the density gradient **716** may be associated with one or more self-healing properties of the protective layer **720** and/or may strengthen the polymeric network **710**. In some implementations, the protective layer **720** may further suppress alkali metal dendrite formation **740** from the anode **702** during battery cycling.

(160) Operationally, the interface layer **718** may suppress alkali metal dendrite formation **740** associated with the anode **702** by uniformly producing metal-fluorides, e.g., lithium fluoride, at an interface across the length of the anode **702**. The uniform production of metal fluorides causes dendrite surface dissolution, e.g., via conversion into metal-fluorides, ultimately suppressing alkali metal dendrite formation **740**. In addition, cross-linking of fluorinated polymer chains over remaining dendrites may further suppress alkali metal dendrite formation **740**. In some implementations, the density gradient **716** may be tuned to control the degree of cross-linking between the fluorinated polymer chains.

(161) FIG. **8**A shows a simplified cutaway view of an example carbonaceous particle **800** with graded porosity, according to some implementations. The carbonaceous particle **800** may be synthesized in a reactor, and output in a controlled manner to produce the cathode **110** and/or anode **120** of FIG. **1**, the cathode **210** and/or anode **220** of FIG. **2**, or the electrode **300** of FIG. **3**. The carbonaceous particle **800**, which may also be referred to as a composition of matter, includes a plurality of regions nested within each other. Each region may include at least a first porosity region **811** and a second porosity region **812**. The first porosity region **811** may include a plurality of first pores **801**, and the second porosity region **812** may include a plurality of second pores **802**. In some aspects, each region may be separated from immediate adjacent regions by at least some of the first pores **801**. The first pores **801** may be dispersed throughout the first porosity region **811** of

the carbonaceous particle **800**, and the second pores **802** may be dispersed throughout the second porosity region **812** of the carbonaceous particle **800**. In this way, the first pores **801** may be associated with a first pore density, and the second pores **802** may be associated with a second pore density that is different than the first pore density. In some aspects, the first pore density may be between approximately 0.0 cubic centimeters (cc)/g and 2.0 cc/g, and the second pore density may be between approximately 1.5 and 5.0 cc/g. In some aspects, the first pores **801** may be configured to retain polysulfides **820**, and the second pores **802** may provide exit pathways from the carbonaceous particle **800**.
(162) A group of carbonaceous particles **800** may be joined together to form a carbonaceous aggregate (not shown for simplicity), and a group of carbonaceous aggregates may be joined together to form a carbonaceous agglomerate (not shown for simplicity). In some implementations, the first pores **801** and second pores **802** may be dispersed throughout aggregates formed by respective groups of the carbonaceous particles **800**. In some aspects, the first porosity region **811** may be at least partially encapsulated by the second porosity region **812** such that a respective agglomerate may include some of the first pores **801** and/or some of the second pores **802**.
(163) In some implementations, the carbonaceous particle **800** may have a principal dimension “A” in an approximate range between 20 nm and 150 nm, an aggregate formed by a group of the carbonaceous particle **800** may have a principal dimension in an approximate range between 20 nm and 10 μm, and an agglomerate formed by a group of aggregates may have a principal dimension in an approximate range between 0.1 μm and 1,000 μm. In some aspects, at least some of the first pores **801** and the second pores **802** has a principal dimension in an approximate range between 1.3 nm and 32.3 nm. In one implementation, each of the first pores **801** has a principal dimension in an approximate range between 0 nm and 100 nm.
(164) The carbonaceous particle **800** may also include a plurality of deformable regions **813** distributed along a perimeter **810** of the carbonaceous particle **800**. The carbonaceous particle **800** may conduct electricity along joined boundaries with (such as the perimeter **810**) one or more other carbonaceous particles. The carbonaceous particle **800** may also confine polysulfides **820** within the first pores **801** and/or at one or more blocking regions **822**, thereby inhibiting the migration of polysulfides **820** towards the anode and increasing the rate at which lithium cations (Li.sup.+) can be transported from the anode to the cathode of a host battery.
(165) In some implementations, the carbonaceous particle **800** may have a surface area of exposed carbon surfaces in an approximate range between 10 m.sup.2/g to 3,000 m.sup.2/g. In other implementations, the carbonaceous particle **800** may have a composite surface area including sulfur **824** micro-confined within a number of the first pores **801** and/or a number of the second pores **802**. As used herein, the first pores **801** and/or the second pores **802** that micro-confine the polysulfides **820** may be referred to as “functional pores.” In some aspects, one or more of the carbonaceous particles, the aggregates formed by corresponding groups of carbonaceous particles, or the agglomerates formed by corresponding groups of aggregates may include one or more exposed carbon surfaces configured to nucleate the sulfur **824**. The composite surface area may be in an approximate range between 10 m.sup.2/g to 3,000 m.sup.2/g, and the carbonaceous particle **800** may have a sulfur to carbon weight ratio between approximately 1:5 to 10:1. In some aspects, the carbonaceous particle **800** may have an electrical conductivity in an approximate range between 100 S/m to 20,000 S/m at a pressure of 12,000 pounds per square in (psi).
(166) In some implementations, the carbonaceous particle **800** may include a surfactant or a polymer that includes one or more of styrene butadiene rubber, polyvinylidene fluoride, poly acrylic acid, carboxyl methyl cellulose, polyvinylpyrrolidone, and/or polyvinyl acetate that can act as a binder to join a group of the carbonaceous particles **800** together. In other implementations, the carbonaceous particle **800** may include a gel-phase electrolyte or a solid-phase electrolyte disposed within at least some of the first pores **801** or second pores **802**.
(167) FIG. **8**B shows a diagram of an example of a tri-zone particle **850**, according to some

implementations. In various implementations, the tri-zone particle **850** may be one example of the carbonaceous particle **800** of FIG. **8**A. The tri-zone particle **850** may include three discrete zones such as (but not limited to) a first zone **851**, a second zone **852**, and a third zone **853**. In some aspects, each of the zones **851-853** surrounds and/or encapsulates a preceding zone. For example, the first zone **851** may be surrounded by or encapsulated by the second zone **852**, and the second zone **852** may be surrounded by or encapsulated by the third zone **853**. The first zone **851** may correspond to an inner region of the tri-zone particle **850**, the second zone **852** may correspond to an intermediate transition region of the tri-zone particle **850**, and the third zone **853** may correspond to an outer region of the tri-zone particle **850**. In some aspects, the tri-zone particle **850** may include a permeable shell **855** that deforms in response to contact with one or more adjacent non-tri-zone particles and/or tri-zone particles **850**.

(168) In some implementations, the first zone **851** may have a relatively low density, a relatively low electrical conductivity, and a relatively high porosity, the second zone **852** may have an intermediate density, an intermediate electrical conductivity, and an intermediate porosity, and the third zone **853** may have a relatively high density, a relatively high electrical conductivity, and a relatively low porosity. In some aspects, the first zone **851** may have a density of carbonaceous material between approximately 1.5 g/cc and 5.0 g/cc, the second zone **852** may have a density of carbonaceous material between approximately 0.5 g/cc and 3.0 g/cc, and the third zone **853** may have a density of carbonaceous material between approximately 0.0 and 1.5 g/cc. In other aspects, the first zone **851** may include pores having a width between approximately 0 and 40 nm, the second zone **852** may include pores having a width between approximately 0 and 35 nm, and the third zone **853** may include pores having a width between approximately 0 and 30 nm. In some other implementations, the second zone **852** may not be defined for the tri-zone particle **850**. In one implementation, the first zone **851** may have a principal dimension D.sub.1 between approximately 0 nm and 100 nm, the second zone **852** may have a principal dimension D.sub.2 between approximately 20 nm and 150 nm, and the third zone **853** may have a principal dimension D.sub.3 of approximately 200 nm.

(169) Aspects of the present disclosure recognize that the unique layout of the tri-zone particle **850** and the relative dimensions, porosities, and electrical conductivities of the first zone **851**, the second zone **852**, and the third zone **853** can be selected and/or modified achieve a desired balance between minimizing the polysulfide shuttle effect and maximizing the specific capacity of a host battery. Specifically, in some aspects, the pores may decrease in size and volume from one zone to other. In some implementations, the tri-zone particle may consist entirely of one zone with a range of pore sizes and pores distributions (e.g., pore density). For the example of FIG. **8**B, the Pores **861** associated with the first zone **851** or the first porosity region have relatively large widths and may be defined as macropores, the pores **862** associated with the second zone **852** or the second porosity region have intermediate-sized widths and may be defined as mesopores, and the pores **863** associated with the third zone **853** or the third porosity region have relatively small widths and may be defined as micropores.

(170) A group of tri-zone particles **850** may be joined together to form an aggregate (not shown for simplicity), and a group of the aggregates may be joined together to form an agglomerate (not shown for simplicity). In some implementations, a plurality of mesopores may be interspersed throughout the aggregates formed by respective groups of the carbonaceous particles **800**. In some aspects, the first porosity region **811** may be at least partially encapsulated by the second porosity region **812** such that a respective aggregate may include one or more mesopores and one or more macropores. In one implementation, each mesopore may have a principal dimension between 3.3 nanometers (nm) and 19.3 nm, and each macropore may have a principal dimension between 0.1 μm and 1,000 μm. In some instances, the tri-zone particle **850** may include carbon fragments intertwined with each other and separated from one another by at least some of the mesopores.

(171) In some implementations, the tri-zone particle **850** may include a surfactant or a polymer that

includes one or more of styrene butadiene rubber, polyvinylidene fluoride, poly acrylic acid, carboxyl methyl cellulose, polyvinylpyrrolidone, and/or polyvinyl acetate that can act as a binder to join a group of the carbonaceous materials together. In other implementations, the tri-zone particle **850** may include a gel-phase electrolyte or a solid-phase electrolyte disposed within at least some of the pores.
(172) In some implementations, the tri-zone particle **850** may have a surface area of exposed carbonaceous surfaces in an approximate range between 10 m.sup.2/g to 3,000 m.sup.2/g and/or a composite surface area (including sulfur micro-confined within pores) in an approximate range between 10 m.sup.2/g to 3,000 m.sup.2/g. In one implementation, a composition of matter including a multitude of tri-zone particles **850** may have an electrical conductivity in an approximate range between 100 S/m to 20,000 S/m at a pressure of 12,000 pounds per square in (psi) and a sulfur to carbon weight ratio between approximately 1:5 to 10:1.
(173) FIG. **8**C shows an example step function **800**C representative of the average pore volumes in each of the regions of the tri-zone particle **850** of FIG. **8**B, according to some implementations. As discussed, the pores distributed throughout the tri-zone particle **850** may have different sizes, volumes, or distributions. In some implementations, the average pore volume may decrease based on a distance between a center of the tri-zone particle **850** and an adjacent zonae, for example, such that pores associated with the first zone **851** or the first porosity region have a relatively large volume or pore size, pores associated with the second zone **852** or the second porosity region have an intermediate volume, and pores associated with the third zone **853** or the third porosity region have a relatively small volume. The interior region has a higher pore volume than the regions near the periphery. The region with higher pore volume provides for high sulfur loading whereas the lower pore volume outer regions mitigate the migration of polysulfides during cell cycling. In the example of FIG. **8**C, the average pore volume in the inner region is approximately 3 cc/g, the average pore volume in the outermost region is −0.5 cc/g and the average pore volume in the intermediate region is between 0.5 cc/g and 3 cc/g.
(174) FIG. **8**D shows a graph **800**D depicting an example distribution of pore volume versus pore width of carbonaceous particles described herein. As depicted in the graph **800**D, pores associated with a relatively high pore volume may have a relatively low pore width, for example, such that the pore width generally increases as the pore volume decreases. In some aspects, pores having a pore width less than approximately 1.0 nm may be referred to as micropores, pores having a pore width between approximately 3 and 11 nm may be referred to as mesopores, and pores having a pore width greater than approximately 24 nm may be referred to as macropores.
(175) FIG. **9**A shows a micrograph **900** of a plurality of carbonaceous structures **902**, according to some implementations. In some implementations, each of the carbonaceous structures **902** may have a substantially hollow a core region surrounded by various monolithic carbon growths and/or layering. In some aspects, the monolithic carbon growths and/or layering may be examples of the monolithic carbon growths and/or layering described with reference to FIGS. **8**A and **8**B. In some instances, the carbonaceous structures **902** may include several concentric multi-layered fullerenes and/or similarly shaped carbonaceous structures organized at varying levels of density and/or concentration. For example, the actual final shape, size, and graphene configuration of each of the carbonaceous structures **902** may depend on various manufacturing processes. The carbonaceous structures **902** may, in some aspects, demonstrate poor water solubility. As such, in some implementations, non-covalent functionalization may be utilized to alter one or more dispersibility properties of the carbonaceous structures **902** without affecting the intrinsic properties of the underlying carbon nanomaterial. In some aspects, the underlying carbon nanomaterial may be formative a sp.sup.2 carbon nanomaterial. In some implementations, each of the carbonaceous structures **902** may have a diameter between approximately 20 and 500 nm. In various implementations, groups of the carbonaceous structures **902** may coalesce and/or join together to form the aggregates **904**. In addition, groups of the aggregates **904** may coalesce and/or join

together to form the agglomerates **906**. In some aspects, one or more of the carbonaceous structures **902**, the aggregates **904**, and/or the agglomerates **906** may be used to form the anode and/or the cathode of the battery **100** of FIG. **1**, the battery **200** of FIG. **2**, or the electrode **300** of FIG. **3**.

(176) FIG. **9**B shows a micrograph **950** of an aggregate formed of carbonaceous material, according to some implementations. In some implementations, the aggregate **960** may be an example of one of the aggregates **904** of FIG. **9**A. In one implementation, exterior carbonaceous shell-type structures **952** may fuse together with carbons provided by other carbonaceous shell-type structures **954** to form a carbonaceous structure **956**. A group of the carbonaceous structures **956** may coalesce and/or join with one another to form the aggregate **1010**. In some aspects, a core region **958** of each of the carbonaceous structures **956** may be tunable, for example, in that the core region **958** may include various defined concentration levels of interconnected graphene structures, as described with reference to FIG. **8**A and/or FIG. **8**B. In some implementations, some of the carbonaceous structures **956** may have a first concentration of interconnected carbons approximately between 0.1 g/cc and 2.3 g/cc at or near the exterior carbonaceous shell-type structure **952**. Each of the carbonaceous structures **956** may have pores to transport lithium cations (Li.sup.+) extending inwardly from toward the core region **1008**.

(177) In some implementations, the pores in each of the carbonaceous structures **956** may have a width or dimension between approximately 0.0 nm and 0.5 nm, between approximately 0.0 and 0.1 nm, between approximately 0.0 and 6.0 nm, or between approximately 0.0 and 35 nm. Each carbonaceous structures **956** may also have a second concentration at or near the core region **958** that is different than the first concentration. For example, the second concentration may include several relatively lower-density carbonaceous regions arranged concentrically. In one implementation, the second concentration may be lower than the first concentration at between approximately 0.0 g/cc and 1.0 g/cc or between approximately 1.0 g/cc and 1.5 g/cc. In some aspects, the relationship between the first concentration and the second concentration may be used to achieve a balance between confining sulfur or polysulfides within a respective electrode and maximizing the transport of lithium cations (Li.sup.+). For example, sulfur and/or polysulfides may travel through the first concentration and be at least temporarily confined within and/or interspersed throughout the second concentration during operational cycling of a lithium-sulfur battery.

(178) In some implementations, at least some of the carbonaceous structures **956** may include CNO oxides organized as a monolithic and/or interconnected growths and be produced in a thermal reactor. For example, the carbonaceous structures **956** may be decorated with cobalt nanoparticles according to the following example recipe: cobalt(II) acetate (C.sub.4H.sub.6CoO.sub.4), the cobalt salt of acetic acid (often found as tetrahydrate Co(CH.sub.3CO.sub.2).sub.2. 4 H.sub.2O, which may be abbreviated as Co(Oac).sub.2. 4 H.sub.2O, may be flowed into the thermal reactor at a ratio of approximately 59.60 wt % corresponding to 40.40 wt % carbon (referring to carbon in CNO form), resulting in the functionalization of active sites on the CNO oxides with cobalt, showing cobalt-decorated CNOs at a 15,000× level, respectively. In some implementations, suitable gas mixtures used to produce Carbon #29 and/or the cobalt-decorated CNOs may include the following steps: Ar purge 0.75 standard cubic feet per minute (scfm) for 30 min; Ar purge changed to 0.25 scfm for run; temperature increase: 25° C. to 300° C. 20 mins; and temperature increase: 300°-500° C. 15 mins.

(179) Carbonaceous materials described with reference to FIGS. **9**A and **9**B may include or otherwise be formed from one or more instances of graphene, which may include a single layer of carbon atoms with each atom bound to three neighbors in a honeycomb structure. The single layer may be a discrete material restricted in one dimension, such as within or at a surface of a condensed phase. For example, graphene may grow outwardly only in the x and y planes (and not in the z plane). In this way, graphene may be a two-dimensional (2D) material, including one or several layers with the atoms in each layer strongly bonded (such as by a plurality of carbon-carbon

bonds) to neighboring atoms in the same layer.
(180) In some implementations, graphene nanoplatelets (e.g., formative structures included in each of the carbonaceous structures **956**) may include multiple instances of graphene, such as a first graphene layer, a second graphene layer, and a third graphene layer, all stacked on top of each other in a vertical direction. Each of the graphene nanoplatelets, which may be referred to as a GNP, may have a thickness between 1 nm and 3 nm, and may have lateral dimensions ranging from approximately 100 nm to 100 μm. In some implementations, graphene nanoplatelets may be produced by multiple plasma spray torches arranged sequentially by roll-to-roll (R2R) production. In some aspects, R2R production may include deposition upon a continuous substrate that is processed as a rolled sheet, including transfer of 2D material(s) to a separate substrate. In some instances, the R2R production may be used to form the first thin film **310** and/or the second thin film **320** of the electrode **300** of FIG. **3**, for example, such that the concentration level of the first aggregates **312** within the first thin film **310** is different than the concentration level of the second aggregates **322** within the second thin film **320**. That is, the plasma spray torches used in the R2R processes may spray carbonaceous materials at different concentration levels to create the first thin film **310** and/or the second thin film **320** using specific concentration levels of graphene nanoplatelets. Therefore, R2R processes may provide a fine level of tunability for the battery **100** of FIG. **1** and/or the battery **200** or FIG. **2**.
(181) FIGS. **10**A and **10**B show transmission electron microscope (TEM) images **1000** and **1050**, respectively, of carbonaceous particles treated with carbon dioxide (CO.sub.2), according to some implementations. The carbonaceous particles shown in FIGS. **10**A and **10**B may include or otherwise be formed from one or more instances of graphene, which may include a single layer of carbon atoms with each atom bound to three neighbors in a honeycomb structure.
(182) FIG. **11** shows a diagram **1100** depicting carbon porosity types of various carbonaceous aggregates, according to some implementations. In various implementations, the carbonaceous aggregates described with reference to FIG. **11** may be examples of the aggregates **904** of FIG. **9**A and/or the carbonaceous structures **956** of FIG. **9**B. In some aspects, the carbonaceous aggregates described with reference to FIG. **11** may be used to form the electrode **300** of FIG. **3**. As discussed, the aggregates may be formed from or may include a group of carbonaceous structures such as the carbonaceous structure **902** of FIG. **9**A or the carbonaceous structures **956** of FIG. **9**B. In some aspects, the carbonaceous structures may be CNOs.
(183) The carbonaceous structures may be used to form an electrode (such as the electrode **300** of FIG. **3**) having any of the porosity types shown in the diagram **1100**. For example, the electrode may include any of a porosity type 1 **1110**, a porosity type II **1120**, and a porosity type III **1130**. In some implementations, the porosity type 1 **1110** may include a first pore **1111**, a second pore **1112**, and a third pore **1113**, all sized with a principal dimension of less than 5 nm to retain polysulfides within the electrode. Some polysulfides may grow in size upon forming larger complexes and become immovably lodged within pores of the porosity type I **1110**. In some implementations, aggregates may be joined together to create pores of the porosity type II **1120** and/or porosity type III **1130** that can retain larger polysulfides and/or polysulfide complexes.
(184) FIG. **12** shows a graph **1200** depicting pore size versus pore distribution of an example electrode, according to some implementations. As used herein, “Carbon **1**” refers to structured carbonaceous materials including mostly micropores (such as less than 5 nm in principal dimension), and “Carbon **2**” refers to structured carbonaceous materials including mostly mesopores (such as between approximately 20 nm to 50 nm in principal dimension). In some implementations, an electrode suitable for use in one of the batteries disclosed herein may be prepared to have the pore size versus pore distribution depicted in the graph **1200**.
(185) FIG. **13** shows a first graph **1300** and a second graph **1310** depicting battery performance per cycle number, according to some implementations. Specifically, the first graph **1300** shows the specific discharge capacity of an example battery employing an electrolyte **1302** disclosed herein

relative to the specific discharge capacity of a conventional battery employing a conventional electrolyte. The second graph shows the capacity retention of the battery employing the electrolyte **1302** relative to the capacity retention of the battery employing the conventional electrolyte. In some aspects, the electrolyte **1302** may be one example of the electrolyte **130** of FIG. **1** or the electrolyte **230** of FIG. **2**. In the first graph **1300** and the second graph **1310**, the conventional electrolyte is prepared as 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=1:1:1) with 2 wt. % LiNO.sub.3.

(186) FIG. **14** shows a bar chart **1400** depicting battery performance per cycle number, according to some implementations. Specifically, the bar chart **1400** depicts the specific discharge capacity per cycle number of an example battery employing an electrolyte **1402** disclosed herein relative to the specific discharge capacity per cycle number of a conventional battery employing a conventional electrolyte. In some aspects, the electrolyte **1402** may be one example of the electrolyte **130** of FIG. **1** or the electrolyte **230** of FIG. **2**. In the bar chart **1400**, the conventional electrolyte is prepared as 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=1:1:1). The bar chart **1400** shows that employing the electrolyte **1402** in an example battery (such as the battery **100** of FIG. **1** or the battery **200** of FIG. **2**) may increase the specific discharge capacity of the battery by approximately 28% at the 3.sup.rd cycle number, by approximately 30% at the 50.sup.th cycle number, and by approximately 39% at the 60.sup.th as compared to a battery employing the conventional electrolyte.

(187) FIG. **15** shows a first graph **1500** and a second graph **1510** depicting battery performance per cycle number, according to some implementations. Specifically, the first graph **1500** shows the electrode discharge capacity per cycle number of an example lithium-sulfur coin cell employing an electrolyte **1502** disclosed herein relative to the electrode discharge capacity per cycle number of an example lithium-sulfur coin cell battery employing a conventional electrolyte, and the second graph **1510** shows the capacity retention per cycle number of the lithium-sulfur coin cell battery employing the electrolyte **1502** relative to the electrode discharge capacity per cycle number of the lithium-sulfur coin cell battery employing the conventional electrolyte. In some aspects, the electrolyte **1502** may be one example of the electrolyte **130** of FIG. **1** or the electrolyte **230** of FIG. **2**. The lithium-sulfur coin cell battery is cycled at a discharge rate of 1 C (such as fully discharged within one hour), at 100% depth-of-discharge (DOD) and is kept at approximately at room temperature (68° F. or 20° C.). The conventional electrolyte is prepared as 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=1:1:1) with 2 wt. % LiNO.sub.3.

(188) FIG. **16** shows a graph **1600** depicting electrode discharge capacity per cycle number, according to some implementations. Specifically, the graph **1600** depicts the electrode discharge capacity per cycle number of an example battery employing an electrolyte **1602** disclosed herein relative to the electrode discharge capacity of a conventional battery employing a conventional electrolyte. In some aspects, the electrolyte **1602** may be one example of the electrolyte **130** of FIG. **1** or the electrolyte **230** of FIG. **2**. The conventional electrolyte is prepared as 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=1:1:1) with 2 wt. % LiNO.sub.3, and the electrolyte **1602** is prepared as 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=58:29:13) with approximately 2 wt. % LiNO.sub.3.

(189) FIG. **17** shows another graph **1700** depicting electrode discharge capacity per cycle number, according to some implementations. Specifically, the graph **1700** depicts the electrode discharge capacity per cycle number of an example battery employing an electrolyte **1702** and solvent package **1704** disclosed herein relative to the electrode discharge capacity of a conventional battery employing a conventional electrolyte and solvent package. The conventional electrolyte is prepared as 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=1:1:1) with approximately 2 wt. % LiNO.sub.3, and the electrolyte **1702** is prepared as 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=58:29:13) with 2 wt. % LiNO.sub.3.

(190) The conventional solvent package is prepared as 1 M LiTFSI in DME:DOL:TEGDME

(volume:volume:volume=1:1:1), and the solvent package **1704** is prepared as 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=58:29:13).

(191) FIG. **18** shows a graph **1800** depicting specific discharge capacity per cycle number for various TBT-containing electrolyte mixtures, according to some implementations. As shown in the graph **1800**, “181” indicates an electrolyte without any TBT additions, resulting in a 0 M TBT concentration level, “181-25TBT” indicates an electrolyte prepared at a 25 M TBT concentration level and so on and so forth. In some implementations, a 5M TBT concentration level may result in an approximate 70 mAh/g discharge capacity increase relative to the electrolyte without any TBT additions.

(192) FIG. **19** shows a first graph **1900** depicting electrode discharge capacity per cycle number and a second graph **1910** depicting electrode capacity retention per cycle number, according to some implementations. Specifically, the first graph **1900** depicts the electrode discharge capacity per cycle number of an example battery that includes a protective lattice disclosed herein relative to the electrode discharge capacity of an example battery that does not include the protective lattice disclosed herein. The second graph **1910** depicts the electrode capacity retention per cycle number of an example battery that includes the protective lattice disclosed herein relative to the electrode capacity retention of an example battery that does not include the protective lattice disclosed herein. In some aspects, the protective lattice may be one example of the protective lattice **402** of FIG. **4**. Performance results for both the first graph **1900** and the second graph **1910** include usage of an electrolyte prepared with 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=58:29:13) with 2 wt. % LiNO.sub.3.

(193) FIG. **20** shows a first graph **2000** depicting electrode discharge capacity per cycle number and a second graph **2010** depicting electrode capacity retention per cycle number, according to other implementations. Specifically, the first graph **2000** depicts the electrode discharge capacity per cycle number of an example battery that includes the polymeric network of FIG. **7**. The second graph **2010** depicts the discharge capacity retention per cycle number of an example battery that includes the polymeric network of FIG. **7**. The battery may be one example of the battery **100** of FIG. **1** or the battery **200** of FIG. **2**. Performance results for both the first graph **2000** and the second graph **2010** include usage of an electrolyte prepared with 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=58:29:13) with 2 wt. % LiNO.sub.3.

(194) FIG. **21** shows a first graph **2100** depicting electrode discharge capacity per cycle number and a second graph **2110** depicting electrode capacity retention per cycle number, according to some other implementations. Specifically, the first graph **2100** depicts the electrode discharge capacity per cycle number of an example battery that includes the protective layer **516** of FIG. **5**. The second graph **2110** depicts the discharge capacity retention per cycle number of an example battery that includes the protective layer **516** of FIG. **5**. The battery may be one example of the battery **100** of FIG. **1** or the battery **200** of FIG. **2**. Performance results for both the first graph **1900** and the second graph **1910** include usage of an electrolyte prepared with 1 M LiTFSI in DME:DOL:TEGDME (volume:volume:volume=58:29:13) with 2 wt. % LiNO.sub.3.

(195) FIG. **22** shows an example cathode **2200** having a body **2201** and a width **2205**, according to some implementations. In some implementations, the cathode **2200** may be one example of the electrode **300** of FIG. **3**. The cathode **2200** may be similar to the electrode **300** of FIG. **3** in many respects, such that description of like elements is not repeated herein. In one implementation, the cathode **2200** includes a first porous carbonaceous region **2210** and a second porous carbonaceous region **2220** positioned adjacent to the first porous carbonaceous region **2210**. The first porous carbonaceous region **2210** may be formed of a first concentration level of carbonaceous materials, and the second porous carbonaceous region **2220** formed of a second concentration level of carbonaceous materials dissimilar to the first concentration level of carbonaceous materials. For example, the second porous carbonaceous region **2220** may have a lower concentration level of carbonaceous materials than the first porous carbonaceous region **2210** as shown in FIG. **22**. In

some aspects, additional porous carbonaceous regions (not shown in FIG. **22** for simplicity) maybe coupled with at least the second porous carbonaceous region.

(196) Specifically, these additional porous carbonaceous regions may be arranged in order of incrementally decreasing concentration levels of carbonaceous materials in a direction away from the first porous carbonaceous region **2210** to provide for complete ionic transport and electrical current tunability. That is, in one implementation, the second porous carbonaceous region **2220** may face a bulk electrolyte (e.g., provided in the liquid phase) and the first porous carbonaceous region **2210** of the cathode **2200** may be coupled with a current collector (not shown in FIG. **22** for simplicity). In this way, denser carbonaceous regions, such as the first porous carbonaceous region **2210**, may facilitate higher levels of electrical conduction (shown in FIG. **22** as "e.sup.−") between adjacent contact points of carbonaceous materials, while sparser carbonaceous regions, such as the second porous carbonaceous region **2220**, may facilitate higher levels of lithium ion transport associated with improved lithium-sulfur battery discharge-charge cycling relative to conventional lithium ion batteries. In some implementations, additional carbonaceous regions coupled with and positioned adjacent to the second porous carbonaceous region **2220** may have a lower density of carbonaceous materials than the second porous carbonaceous region **2220**. In this way, the additional carbonaceous regions of lower density may accommodate higher levels of lithium ion transport to, for example, permit for tuning of various performance characteristics of the electrode **300**.

(197) In one implementation, the first porous carbonaceous region **2210** may include first non-tri-zone particles **2211**. The configuration of the first non-tri-zone particles **2211** within the first porous carbonaceous region is one example configuration. Other placements, orientations, alignments and/or the like are possible for the non-tri-zone particles. In some aspects, each non-tri-zone particle may be an example of one or more carbonaceous materials disclosed elsewhere in the present disclosure. The first porous carbonaceous region **2210** may also include first tri-zone particles **2212** interspersed throughout the first non-tri-zone particles **2211** as shown in FIG. **22**, or positioned in any other placement, orientation, or configuration. Each first tri-zone particle **2212** may be one example of the tri-zone particle **850** of FIG. **8**B. In addition, or the alternative, each first tri-zone-particle **2212** may include first carbon fragments **2213** intertwined with each other and separated from one another by mesopores **2214**. Each tri-zone-particle may have a first deformable perimeter **2215** configured to coalesce with adjacent first non-tri-zone particles **2211** and/or first tri-zone particles **2212**.

(198) The first porous carbonaceous region **2210** may also include first aggregates **2216**, where each aggregate includes a multitude of the first tri-zone particles **2212** joined together. In one or more particular examples, each first aggregate may have a principal dimension in a range between 10 nanometers (nm) and 10 micrometers (μm). The mesopores **2214** may be interspersed throughout the first plurality of aggregates, where each mesopore has a principal dimension between 3.3 nanometers (nm) and 19.3 nm. In addition, the first porous carbonaceous region **2210** may include first agglomerates **2217**, where each agglomerate includes a multitude of the first aggregates **2216** joined to each other. In some aspects, each first agglomerate **2217** may have a principal dimension in an approximate range between 0.1 μm and 1,000 μm. Macropores **2218** may be interspersed throughout the first aggregates **2216**, where each macropore may have a principal dimension between 0.1 μm and 1,000 μm. In some implementations, one or more of the above-discussed carbonaceous materials, allotropes and/or structures may be one or more examples of that shown in FIGS. **9**A and **9**B.

(199) The second porous carbonaceous may include second non-tri-zone particles **2221**, which may be one example of the first non-tri-zone particles **2211**. The second porous carbonaceous region **2220** may include second tri-zone particles **2222**, which may each be one example of each of the first tri-zone particles **2212** and/or may be one example of the tri-zone particle **850** of FIG. **8**B. In addition, or the alternative, each second tri-zone particle **2222** may include second carbon

fragments **2223** intertwined with each other and separated from one another by the mesopores **2214**. Each second tri-zone particle **2222** may have a second deformable perimeter **2225** configured to coalesce with one or more adjacent second non-tri-zone particles **2221** or second tri-zone particles **2222**.

(200) In addition, the second porous carbonaceous region **2220** may include second aggregates **2226**, where each second aggregate **2226** may include a multitude of the second tri-zone particles **2222** joined together. In one or more particular examples, each second aggregate **2226** may have a principal dimension in a range between 10 nanometers (nm) and 10 micrometers (μm). The mesopores **2214** may be interspersed throughout the second aggregates **2226**, each mesopore may have a principal dimension between 3.3 nanometers (nm) and 19.3 nm. Further, the second porous carbonaceous region **2220** may include second agglomerates **2227**, each second agglomerate **2227** may include a multitude of the second aggregates **2226** joined to each other, where each agglomerate may have a principal dimension in an approximate range between 0.1 μm and 1,000 μm. The macropores **2218** may be interspersed throughout the second plurality of aggregates, where each macropore having a principal dimension between 0.1 μm and 1,000 μm. In some implementations, one or more of the above-discussed carbonaceous materials, allotropes and/or structures may be one or more examples of that shown in FIGS. **9**A and **9**B.

(201) In one implementation, the first porous carbonaceous region **2210** and/or the second porous carbonaceous region **2220** may include a selectively permeable shell (not shown in FIG. **22** for simplicity), which may form a separated liquid phase on the first porous carbonaceous region **2210** or the second porous carbonaceous region **2220**, respectively. An electrolyte, such as any of the electrolytes disclosed in the present disclosure, may be dispersed within the first porous carbonaceous region and/or the second porous carbonaceous region for lithium ion transport associated with lithium-sulfur battery discharge-charge operational cycling.

(202) In one or more particular examples, the first porous carbonaceous region **2210** may have an electrical conductivity in an approximate range between 500 S/m to 20,000 S/m at a pressure of 12,000 pounds per square in (psi). The second porous carbonaceous region **2220** may have an electrical conductivity in an approximate range between 0 S/m to 500 S/m at a pressure of 12,000 pounds per square in (psi). The first agglomerates **2217** and/or second agglomerates **2227** may include aggregates connected to each other with one or more polymer-based binders.

(203) In some aspects, each first tri-zone particle **2212** may have a first porosity region (not shown in FIG. **22** for simplicity) located around a center of the first tri-zone particle **2212**. Similarly, each second tri-zone particle **2222** may have a first porosity region (not shown in FIG. **22** for simplicity) located around a center of the second tri-zone particle **2222**. The first porosity region may include first pores. A second porosity region (not shown in FIG. **22** for simplicity) may surround the first porosity region. The second porosity region may include second pores. In one implementation, the first pores may define a first pore density, and the second pores may define a second pore density that is different the first pore density.

(204) In some aspects, the mesopores **2214** may be grouped into first mesopores and second mesopores (both not shown in FIG. **22** for simplicity). In one or more particular examples, the first mesopores may have a first mesopore density, and the second mesopores may have a second mesopore density that is different than the first mesopore density. In addition, the macropores **2218** may be grouped into first macropores that may have a first pore density, and second macropores (both not shown in FIG. **22** for simplicity) that may have a second pore density different than the first pore density.

(205) In one implementation, the first porous carbonaceous region **2210** and/or the second porous carbonaceous region **2220** may nucleate sulfur, such as that necessary to facilitate operational discharge-charge cycling of any of the lithium-sulfur batteries disclosed by the present disclosure. For example, the cathode **2200** may have a sulfur to carbon weight ratio between approximately 1:5 to 10:1. In some aspects, one or more electrically conductive additives may be dispersed within the

first porous carbonaceous region **2210** and/or the second porous carbonaceous region **2220** to, for example, correspondingly influence discharge-charge cycling performance of the cathode **2200**. In addition, a protective sheath, such as the protective lattice **402** of FIG. **4**, may be disposed on the cathode.

(206) In one implementation, the example cathode **2200** of FIG. **22** and/or any of the battery configurations presented in the present disclosure (such as the battery **100** and/or **200**), may be prepared with an electrolyte (such as the electrolyte **130** and/or **230**) dispersed throughout the respective battery configuration. In addition, or the alternative, the electrolyte **130**, the electrolyte **230** and/or the like may be formulated according to the following numbered examples:

(207) TABLE-US-00001 Example 1 A 0.4 molar (M) solution of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture. The liquid solvent mixture (alternatively referred to as a “ternary solvent package”) has a 58:28:13 volume ratio of dimethoxyethane (DME), 1,3-dioxolane (DOL), and tetraethylene glycol dimethyl ether (TEGDME). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 2 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and tetrahydrofuran (THF). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 3 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and toluene. An additive including 26 grams of lithium nitrate (LiNO3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO3). Example 4 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and dimethyl sulfoxide (DMSO). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 5 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and tetramethyl urea (TMU). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 6 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and dimethyl formamide (DMF). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 7 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and methoxyperfluorobutane (MPB). An additive including 26 grams of lithium nitrate (LiNO.sub.3) may be added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 8 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and trifluoro ethyl ether (TFE). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 9 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and triethylene glycol dimethyl

ether (TrigDME). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 10 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and methyl tert-butyl ether (MTBE). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 11 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and dimethyl trisulfide (DMTS). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 12 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and acetonitrile (can). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 13 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and 1,1,2,2-tetrafluoro-1- 1(2,2,2-trifluoroethoxy)ethane (TFETFE). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 14 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and DAP. An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 15 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and TTE. An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 16 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and 2-Methyltetrahydrofuran (MeTHF). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 17 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and bis(2-methoxyethyl) ether (DEGDME). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.4 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 18 A 0.1 molar (M) solution of LiTFSI is prepared from approximately 28.71 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 95:5 volume ratio of DME and DOL. No additional lithium nitrate (LiNO.sub.3) is added to the 0.1 molar (M) solution of LiTFSI. Example 19 A 0.1 molar (M) solution of LiTFSI is prepared from approximately 28.71 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 50:25:25 volume ratio of DME, DOL, and bis(2-methoxyethyl) ether (DEGDME). An additive including 26 grams of lithium nitrate (LiNO.sub.3) is added to the 0.1 molar (M) solution of LiTFSI to achieve a dilution level of 2 percent by weight of lithium nitrate (LiNO.sub.3). Example 20 A 0.4 molar (M) solution of LiTFSI is prepared from approximately 114.83 grams of powdered LiTFSI dissolved in 1 liter of a liquid solvent mixture having a 58:29:13 volume ratio of DME, DOL, and tetraethylene glycol dimethyl ether (TEGDME). No additional lithium nitrate (LiNO.sub.3) is added to the 0.1 molar (M) solution of LiTFSI.

(208) In some aspects, ionic conductivity of the lithium cations (Li.sup.+) **125** transported throughout the electrolyte **130**, such as when prepared according to any one or more of the above-presented examples, may depend on the molecular structure of various component substances of the electrolyte **130**. For example, substances that are hydrophobic and less polar may have lower ionic conductivity values. Substances that are hydrophilic and more polar may have higher ionic conductivity values. In this way, component materials used in the above electrolyte formulation examples of the electrolyte **130** may be ranked from lowest ionic conductivity to highest ionic conductivity according to the following order: DMTS, TOL, TFETFE, MPB, MTBE, TrigDME, THF, TEE, TMU, DMSO, DMF, ACN.

(209) Of the electrolyte components disclosed above in examples 1-20, lithium nitrate (LiNO.sub.3) may dissociate into lithium cations (Li.sup.+) (Li.sup.+) and nitrate anions (NO.sub.3.sup.−). In this way, the lithium nitrate (LiNO.sub.3) may produce nitrogen-oxygen containing compounds (not shown in the Figures for simplicity), which may be derived from and/or based on nitrate anions (NO.sub.3.sup.−). The electrolyte **130**, such as when prepared according to any one or more of the examples presented above, may prevent diffusion of nitrogen-oxygen containing compounds generated during operational discharge-charge cycling of the battery **100**. In addition, some nitrate anions (NO.sub.3.sup.−) may form a solvation sheath (not shown in the Figures for simplicity) on the anode **120**. In this way, the electrolyte **130** may be prepared to permit nitrogen-oxygen additives to coat the anode **120** at least partially and thereby prevent the extension of dendrites from the anode **120** toward the cathode **110** through the electrolyte **130**. The solvation sheath may form coordination complexes between LiTFSI in the electrolyte **130** and the lithium cations (Li.sup.+) **125**. The coordination complexes may include a central atom or ion (such as the lithium cations (Li.sup.+) **125**), which may be metallic and may be referred to as “the coordination center,” and a surrounding array of bound molecules or ions, which may be referred to ligands or complexing agents.

(210) Protection against dendrite formation provided by the solvation sheath may be at least partially compromised due to continued reduction of nitrogen compounds prevalent in the nitrogen-oxygen additives that form, for example, nitrite (NO.sub.2.sup.−), which may produce gas resulting in pockets of gas bubbles in the electrolyte **130**. These gas bubbles may interfere with transport of the lithium cations (Li.sup.+) **125**, may cause expansion of the battery **100** and may also lead to undesirable hindrance of lithium ionic transport. To address these limitations, Example 18 of the above-presented electrolyte formulations may be prepared without the addition of lithium nitrate (LiNO.sub.3) and/or other types of nitrogen-oxygen containing additives.

(211) FIGS. **23-25** show graphs depicting specific discharge capacity per cycle number for one or more of the Examples 1-20 presented earlier, according to some implementations. FIG. **23** shows a graph **2300** of specific discharge capacity (mAh/g) per cycle number depicting performance improvements of the battery **100** of FIG. **1**A and/or other battery configurations disclosed herein. Regarding the graph **2300**, “control” refers to the electrolyte **130** prepared according to Example 1, “Toluene” refers to the electrolyte **130** prepared according to Example 3, and “TMU” refers to the electrolyte **130** prepared according to Example 5. In some aspects, toluene in the electrolyte **130** prepared according to Example 3 may provide favorable unexpected results based on its non-polar nature and correspondingly poor interactions with, for example, the polysulfides **128** of FIG. **1**B. That is, toluene has a unique chemical structure that may impede facile transport of the polysulfides **128** through the electrolyte **130** when prepared according to Example 3. Benefits associated with usage of varying concentrations or dilution levels of toluene within, for example, the liquid solvent mixture (also referred to as the ternary solvent package) may increase in significance proportionate to cycling rate. That is, usage of toluene may be even more beneficial in terms of cathode capacity retention than that shown in the graph **2300** when observing a lithium-sulfur battery discharged at a C/3 rate (corresponding to complete battery discharge over a time period of 3 hours) relative to a

traditional discharge rate of 1 C (corresponding to complete battery discharge over a time period of 1 hour). In this way, toluene may be particularly well suited for end-use applications that may involve longer battery life or discharge times, such as electric vehicles (EVs).
(212) Toluene may be uniquely suited to out-perform other solvents based on its chemical structure and non-polar nature, having an approximate ionic conductivity value (mS/cm) of 5.128026. In this way, toluene in the electrolyte **130**, such as when prepared according to Example 3, may contribute to higher specific capacity and improved capacity retention during battery cycling by impeding movement of the polysulfides **128** within the electrolyte **130**, thereby freeing up volume in the electrolyte **130** available to transport the lithium cations (Li.sup.+) **125**. In addition, or the alternative, toluene may act as a favorable solvent for the elemental sulfur **126**, when pre-loaded (such as by capillary infusion or some other suitable technique) into the cathode **110** or the cathode **2200**. In one implementation, toluene may assist in the de-passivation of the cathode **110** or the cathode **2200** to pre-condition the battery **100** to prevent dropping beneath the minimum designed voltage (of the battery **100**) once the external load **172** is applied.
(213) By impeding movement of the polysulfides **128** in the electrolyte **130**, toluene may improve sulfur retention within the cathode **110** and overall sulfur related kinetics, that is sulfur utilization in forming coordination complexes with the lithium cations (Li.sup.+) **125**. Toluene also may improve interfacial regions between the anode **120** and the electrolyte **130** by preventing movement of the polysulfides **128** from contacting the anode **120**. In addition, toluene may increase the boiling point and/or decrease the volatility of the electrolyte **130**, which may improve safety and reliability of the electrolyte **130**. Further, toluene may lower the freezing point of the electrolyte **130**, which may assist in low temperature performance of the battery **100**. Toluene may also lower the density of the electrolyte **130** as well, which may improve specific energy, since the mass of the electrolyte **130** may impact the performance and/or efficiency of the battery **100**.
(214) In some implementations, the ability of toluene to improve the performance of the electrolyte **130** may depend at least in part on the ability of toluene to solubilize certain forms of elemental sulfur, such as cyclooctasulfur (S.sub.8). In some aspects, toluene may have a normalized first cycle discharge capacity (Ah/g) of approximately between 0.6 (Ah/g) to 0.8 (Ah/g) at a S.sub.8 solubility level of approximately 0.0275 (mol/L). These values present a marked improvement when compared to, for example, ACN, which has a normalized first cycle discharge capacity (Ah/g) of approximately between 0.37 (Ah/g) to 0.41 (Ah/g) at a S.sub.8 solubility level of approximately 0.0075 (mol/L), indicating that toluene tends to solvate S.sub.8 better and provides correspondingly improve discharge capacity.
(215) FIG. **24** shows a graph **2400** of specific discharge capacity (mAh/g) per cycle number depicting performance improvements of the battery **100** of FIG. **1**A and/or other battery configurations disclosed herein. Regarding the graph **2400**, “MPB” refers to the electrolyte **130** prepared according to Example 7, “TrigDME” refers to the electrolyte **130** prepared according to Example 9, and “TEE” refers to the electrolyte **130** prepared according to Example 8.
(216) FIG. **25** shows a graph **2500** of specific discharge capacity (mAh/g) per cycle number depicting performance improvements of the battery **100** of FIG. **1**A and/or other battery configurations. Regarding the graph **2500**, “TFETFE” refers to the electrolyte **130** prepared according to Example 13, “DMTS” refers to the electrolyte **130** prepared according to Example 11, “MTBE” refers to the electrolyte **130** prepared according to Example 10, and “ACN” refers to the electrolyte **130** prepared according to Example 12.
(217) In some aspects, a cathode (such as the cathode **2200** of FIG. **22**) may be positioned opposite to an anode (such as the anode **120** of FIG. **1**) and have an overall porosity between 40% and 70%. In one example, the cathode **2200** may include non-hollow carbon spherical (NHCS) particles joined together. Each NHCS particle may be one example of the first tri-zone particles **2212** of FIG. **22**, the second tri-zone particles **2222** of FIG. **22**, the carbonaceous particle **800** of FIG. **8**A, and/or the like. At least some NHCS particles may coalesce together and thereby collectively form

tubular NHCS particle agglomerates, which may be one example of the aggregate **960** of FIG. **9**B. Each NCHS particle may have a diameter between 30 nanometers (nm) and 60 nm, and may include a first region and a second region. In one implementation, the first region may be defined by the first pores **801** of FIG. **8**A, and the second region may be defined by the second pores **802** of FIG. **8**A. In this way, the first region may be adjacent to a center of a respective NHCS particle and may have a first density of carbonaceous materials, and the second region may be adjacent to a surface a respective NHCS particle. The second region may encapsulate the first region and have a second density of carbonaceous materials that is lower than the first density of carbonaceous materials. The first region and the second region may be in fluid communication with each other.

(218) In addition, the cathode **2200** may include interconnected channels (not shown in FIG. **22** for simplicity) defined in shape by adjacent NHCS particles. Some interconnected channels may be pre-loaded with an elemental sulfur and retain polysulfides (PS) based on one or more of the first density of carbonaceous materials or the second density of carbonaceous materials. An electrolyte, which may be prepared by any of the formulations presented in Examples 1-20, may be interspersed throughout the cathode and in contact with the anode. A separator may be positioned between the anode and the cathode.

(219) FIG. **26**A shows an example battery **2600**A, according to some other implementations. The battery **2600**A may be an example of other battery configurations disclosed herein. In one implementation, the battery **2600**A may be implemented as a lithium-sulfur battery, and may include an anode **2620** (e.g., an anode active material comprising a foil of lithium), a cathode **2610**, and a solid-state electrolyte **2630**. In some instances, the solid-state electrolyte **2630** may replace one or more electrolyte solution compositions presented in Examples 1-20. The cathode **2610** may be one example of other cathode configurations disclosed herein, such as the cathode **110** of FIG. **1**, the cathode **210** of FIG. **2**, and/or the cathode **2200** of FIG. **22**. In some aspects, the cathode may be loaded with elemental sulfur of 3 milligrams (mg) per cubic centimeter (cm.sup.3). In other aspects, the cathode **2610** may be loaded with other concentrations of elemental sulfur of 3 milligrams (mg) per cubic centimeter (cm.sup.3) suitable for maximizing the efficiency of discharge-charge cycling of the battery **2600**A. In some aspects, the cathode **2610** may be porous and formed from a composition of matter (not shown in FIG. **26**A for simplicity) including a plurality of pores **2612**. The composition of matter may be one example of various carbonaceous materials and/or structures disclosed herein, such as the first tri-zone particles **2212** of FIG. **22**, the second tri-zone particles **2222** of FIG. **22**, the carbonaceous particle **800** of FIG. **8**A, one or more instances of the aggregate **960** of FIG. **9**B and/or the like.

(220) The solid-state electrolyte **2630** may be dispersed throughout at least the pores **2612** of the cathode **2610**, and may also be in contact with the anode **2620**. In some aspects, the solid-state electrolyte **2630** may be formed as a membrane and thereby provide ionic conduction capabilities associated with a separator, such as the separator **150** of FIG. **1**. In one implementation, the solid-state electrolyte **2630** may be formed from and/or include a polymer matrix **2631**, which may be formed of glass fibers **2633** interconnected with each other. In some aspects, the polymer matrix **2631** may have an ionic conductivity (e.g., conducting lithium cations (Li.sup.+)) and may include between 8 weight percent (wt. %) and 12 wt. % of polyethylene oxide (PEO) **2632**, between 13 wt. % and 17 wt. % of polyvinylidene difluoride (PVDF) **2634**, between 3 wt. % and 7 wt. % of polyetheramine **2535** having repeated oxypropylene units (not shown in FIG. **26**A for simplicity) in its backbone, and between 5 wt. % and 10 wt. % of one or more lithium-containing salts including one or more of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) or lithium iodide (LiI) (not shown in FIG. **26**A for simplicity). In some implementations, at least some of the lithium-containing salts may dissociate into lithium cations (Li.sup.+) and thereby assist in lithium ionic transport between the anode **2620** and the cathode **2610** during operational discharge-charge cycling of the battery **2600**A.

(221) Similar to the various other lithium-sulfur battery configurations disclosed herein, the battery

2600A may generate undesirable lithium-containing polysulfide species (not shown in FIG. **26**A for simplicity) during operational discharge-charge cycling of the battery **2600**A. In some instances, the cathode **2610** may at least partially trap and/or retain the lithium-containing polysulfide species, thereby preventing blockage of lithium transport pathways (e.g., as shown as a charge cycle flow **2625**) within the solid-state electrolyte **2630**. In some aspects, the anode **2620** may be formed as a layer of lithium that provides lithium cations (Li.sup.+) upon activation of the battery **2600**A. In this way, the layer of lithium may provide lithium cations (Li.sup.+) during operational discharge-charge cycling of the battery **2600**A. In other aspects, the cavity **2622** may receive lithium deposits from the cathode **2610** during operational charge cycling of the battery **2600**A. That is, lithium cations (Li.sup.+) may travel along the charge cycling flow **2624** from the cathode **2610** to the cavity **2622** as may be associated with the return of the electrons **2674** back towards the battery **2600**A as may be encountered in or associated with battery charge and/or recharge cycling. In this way, the cavity **2622** may transform into the anode **2620**, which may be capable of again providing lithium cations (Li.sup.+) to their respective electrochemically favored positions in the cathode **2610** during operational discharge cycling of the battery **2600**A.

(222) In one implementation, the composition of matter used to form the cathode **2610** may be formed from and/or include one or more non-tri-zone particles, tri-zone particles, aggregates, or agglomerates as disclosed herein. In some aspects, the cathode **2610** may be one example of the cathode **2200** of FIG. **22**. That is, each tri-zone particle used in the cathode **2610** may include carbon fragments intertwined with each other. At least some carbon fragments may be separated from one another by mesopores. A deformable perimeter may be defined upon coalescence of one or more adjacent non-tri-zone particles or tri-zone particles. Each aggregate may include a multitude of the tri-zone particles joined together and have a principal dimension in a range between 10 nanometers (nm) and 10 micrometers (μm). Mesopores may be interspersed throughout the aggregates. Each mesopore may have a principal dimension between 3.3 nanometers (nm) and 19.3 nm. Each agglomerate may be formed from a multitude of the aggregates joined to each other and have a principal dimension in an approximate range between 0.1 μm and 1,000 μm. Macropores may be interspersed throughout the aggregates, where each macropore having a principal dimension between 0.1 μm and 1,000 μm.

(223) In some implementations, the ionic conductivity of the solid-state electrolyte **2630**, when formed and/or deposited as a membrane (not shown in FIG. **26**A for simplicity) on the anode **2620**, may be based on relative concentration levels of one or more lithium-containing salts doped into the polymer matrix **2631**. In this way, the ionic conductivity of the solid-state electrolyte **2630** may be between 0.97×10.sup.−3 siemens per meter (S/m) and 1.03×10.sup.−3 S/m at a temperature between 18 degrees Celsius (° C.) and 22° C. In other implementations, the membrane may be coated onto the anode **2620**, such that the ionic conductivity of the membrane is between 3.97×10.sup.−6 siemens per meter (S/m) and 4.03×10.sup.−6 S/m at a temperature between 18 degrees Celsius (° C.) and 22° C. In some aspects, higher quantities of one or more lithium-containing salts may be associated with an increase in the ionic conductivity of the polymer matrix **2631**.

(224) In one implementation, the solid-state electrolyte **2630**, when formed as a membrane, has a thickness between 10 micrometers (μm) and 50 μm, and may have a uniform density throughout its thickness. For example, in some instances, the solid-state electrolyte **2630** may have a density between 2 grams per cubic centimeter (g/cm.sup.3) and 3 g/cm.sup.3. In some aspects, the membrane may be coated onto a sacrificial polymer (not shown in FIG. **26**A for simplicity), which may be disposed on the anode **2620** facing the solid-state electrolyte **2630**. In this way, the solid-state electrolyte **2630** may prevent electrons from traveling from the anode **2620** to the cathode **2610** through the solid-state electrolyte **2630**. In addition, contact points between the solid-state electrolyte **2630** and the anode **2620** may prevent impedance growth of the battery **2600**A.

(225) In one implementation, the cathode **2610** has a thickness between 50 micrometers (μm) and

150 μm, and a density between 5 grams per cubic centimeter (g/cm.sup.3) and 15 g/cm.sup.3. In some aspects, the solid-state electrolyte **2360** may be prepared without a liquid-phase electrolyte, such as Examples 1-20 disclosed herein. In addition, or the alternative, the solid-state electrolyte **2630** may localize lithium-containing polysulfide species within the cathode **2610** and/or prevent growth of lithium-containing dendritic structures from the anode **2620**. In some aspects, the anode **2620** may volumetrically expand between 5% and 20% of its initial size during operational discharge-charge cycling of the battery **2600**A. In some instances, the solid-state electrolyte **2630** may provide interfacial stability between the anode **2620** and the solid-state electrolyte **2630** during operational discharge-charge cycling of the battery **2600**A, for example, to reduce or limit volumetric expansion of the anode **2620**.
(226) FIG. **26**B shows another example battery **2600**B, according to some other implementations. In one implementation, the battery **2600**B may be another example of the battery **2600**A. In some aspects, the cavity **2622** may replace the anode **2620**, and may be incrementally filled with lithium provided from the cathode **2610** during operational discharge-charge cycling of the battery **2600**B. In this way, once the cavity **2622** of FIG. **26**B is filled with lithium to become the anode **2620** of FIG. **26**A, the battery **2600**B may function in a manner similar or identical to the battery **2600**A.
(227) FIG. **27** shows a graph **2700** depicting voltage drop per specific capacity for an example configuration of the battery **2600**A of FIG. **26**A, according to some implementations. Regarding the graph **2700**, the battery **2600**A was prepared with a sulfur loading level of 3 milligrams (mg) per cubic centimeter (cm.sup.3) and in a coil cell format. In addition, the battery **2600**A was prepared with the solid-state electrolyte **2630** at a thickness level of 18 micrometers (μm). Region **2702** may be representative of unique voltage drop behavior associated with formation and/or dissociation of lithium-containing polysulfide intermediates generated during discharge-charge operational cycling of the battery **2600**A.
(228) FIG. **28** shows another example battery **2800**, according to some other implementations. The battery **2800** may be an example of other battery configurations disclosed herein. In one implementation, the battery **2800** may be implemented as a lithium-sulfur battery, and may include a cathode **2810**, an anode structure **2822** including an anode **2820** (e.g., an anode active material comprising a foil of lithium) positioned opposite to the cathode, a separator **2850** positioned between the anode **2820** and the cathode **2810**, and an electrolyte **2830**. In some aspects, the electrolyte **2830** may be formulated by mixing at least two or more solvents, such as those disclosed in Examples 1-20 presented earlier. The electrolyte **2830** may be dispersed throughout the cathode **2810** and in contact with the anode **2820**. In some aspects, the anode **2820** may be a single foil of solid metallic lithium. In this way, at least some lithium cations (Li.sup.+) **2825** output by the anode **2820** may participate in dissociation reactions and/or combination reactions during operational discharge-charge cycling of the battery **2800**. That is, lithium cations (Li.sup.+) **2825** output from the anode **2820** may be transported through the electrolyte **2830** and retained in their electrochemically favored positions (not shown in FIG. **28** for simplicity) within the cathode **2810** during discharge cycles of the battery **2800**. Then, during charge cycles of the battery **2800**, the lithium cations (Li.sup.+) **2825** may be forced to return to the anode **2820** upon exposure to an outside current source.
(229) In addition, a solid-electrolyte interphase **2840** may form on the anode **2820**. In this way, a protective layer **2860** may form at least partially within and/or on the solid-electrolyte interphase **2840** and face the cathode **2810**. In some aspects, the solid-electrolyte interphase **2840** may form one or more compounds on the anode **2820** based on one or more oxidation-reduction reactions involving lithium cations (Li.sup.+) and one or more solvents of the electrolyte **2830**. In some implementations, the protective layer **2860** may be at least partially formed from carbonaceous materials including one or more of flat graphene, wrinkled graphene, carbon nano-tubes (CNTs), carbon nano-onions (CNOs), or non-hollow carbon spherical particles (NHCS), one or more of which may be one example of the carbonaceous structure **956** of FIG. **9**B.

(230) FIG. **29** shows a diagram depicting an example cathode **2900**, according to some implementations. The cathode **2900** may be one example of other cathode configurations disclosed herein, such as the cathode **2200** of FIG. **22** and/or the cathode **2810** of FIG. **28**. In some implementations, the cathode **2900** may include a porous structure **2915** with interconnected channels **2916** defined by adjacent and interconnected non-hollow carbon spherical particles (NHCS) particles **2917**, each of which may be one example of the tri-zone particle **850** of FIG. **8**B, the aggregate **960** of FIG. **9**B, and/or other carbonaceous materials described in the present disclosure. In this way, at least some of the interconnected channels **2916** may be loaded with elemental sulfur **2926** in the cathode **2900** prior to activation and discharge-charge cycling of, for example, the battery **2800** of FIG. **28**. The elemental sulfur **2926** may form coordination complexes with at least some of the lithium cations (Li.sup.+) **2825** to increase specific capacity of the cathode **2900**, for example, compared to conventional lithium ion chemistries. In some aspects, the cathode **2900** may have a width **2910** formed of a first region **2911** disposed on a substrate **2902** (e.g., a copper or other metal current collector), and may have multiple additional regions, such as a second region **2912** positioned adjacent to the first region **2911**. Each of the regions including the first region **2911**, the second region **2912**, and/or additional subsequent regions positioned adjacent to the second region **2912** (not shown for simplicity in FIG. **29**) may be defined in shape, size, and orientation by a respective concentration level of NHCS particles **2917**. That is, in some aspects, the first region **2911** may be prepared with a relatively higher concentration of NHCS particles **2917**, thereby resulting in increased electrical conduction between adjacent graphene sheets within each NHCS particle, as may be desirable near or at the substrate **2902**. In contrast, the second region **2912** may be prepared with a relatively lower concentration of NHCS particles **2917**, thereby permitting for additional transport of lithium ions (Li.sup.+) **2925** into the width **2910** of the cathode **2900** for complexation with elemental sulfur **2926** contained within the interconnected channels **2916**.
(231) FIG. **30** shows a diagram **3000** depicting region "A" of the protective layer **2860** of the battery of FIG. **28**, according to some implementations. In some aspects, region "A" may be one example of various anode protective layers disclosed herein, including the protective layer **2860**. In one implementation, region "A" may be at least partially formed from polymeric materials, such as a first polymeric chain **3010** with carbon atoms **3014** provided by at least some of the carbonaceous materials. Oxide anions (O.sup.2−) **3022**, fluorine anions (F.sup.−) **3012**, and nitrate anions (NO.sup.3−) **3024** may be uniformly dispersed throughout region "A" and grafted onto one or more of the carbon atoms **3014**. Region "A" may also include a second polymeric chain **3020** including at least some of the carbon atoms **3014**, which may be also provided by carbonaceous materials. Similar to the first polymeric chain **3010**, the carbon atoms **3014** of the second polymeric chain **3020** may also be grafted to one or more of oxide anions (O.sup.2−) **3022**, fluorine anions (F.sup.−) **3012**, and nitrate anions (NO.sup.3−) **3024**, one or more of which may be uniformly dispersed throughout the protective layer **2860**. In some aspects, the second polymeric chain **3020** may be positioned opposite to the first polymeric chain **3010**. In addition, in one implementation, the first polymeric chain **3010** and the second polymeric chain **3020** may be configured to cross-link with each other based on exposure to one or more nitrogen-containing groups, such as the nitrate anions (NO.sup.3−) **3024**.
(232) In some aspects, inorganic and/or ionic conductor **3018**, such as lithium-containing salts including lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), may be uniformly dispersed throughout the electrolyte **2830** and/or the protective layer **2860**, and thereafter dissociate into lithium cations (Li.sup.+) **2825** and TFSI.sup.− anions **2826**. In addition, the first polymeric chain **3010** and the second polymeric chain **3020** may at least partially cross-link with each other and form carbon-carbon bonds **3026** upon exposure to an environment **3050**. In some instances, the environment **3050** may include a polymerization initiator and/or ultraviolet (UV) radiation.
(233) Specifically, exposure of the first and second polymeric chains **3010** and **3020** to the

polymerization initiator and/or the UV radiation may cause region “A” to be formed as a three-dimensional (3D) lattice having a cross-linking density defined by a number of cross-link points per-unit volume. In some aspects, the cross-link points may be configured to at least partially trap the TFSI.sup.− anions **2826** produced upon dissociation of LiTFSI. For example, the number of cross-link points per-unit volume may restrict re-dissolution of lithium-containing additives in region “A” toward the electrolyte **2830**. In some aspects, the cross-linking density of the protective layer **2860** may inhibit swelling above 10% of an initial volume of the protective layer **2860** by, for example, preventing absorption of at least one of the solvents in the electrolyte **2830**. In some other aspects, the cross-linking density of the protective layer **2860** may control swelling between 10%-50% of an original volume of the protective layer **2860** by controlling absorption of at least one solvent. In this way, the cross-linking density of the protective layer **2860** may improve and/or may be associated with an improvement of lithium ion (Li+) transport throughout the anode structure **2822** and the electrolyte **2830**.

(234) In one implementation, the protective layer **2860** may have a modulus of elasticity between 3 gigapascals (GPa) and 100 GPa, a glass transition temperature above 60° Celsius (C) and may cure at a temperature of less than 81° Celsius (C). Upon activation of the battery **2800**, lithium fluoride (LiF) may form based on one or more chemical reactions (e.g., the Wurtz reaction of FIG. **7**). For example, lithium fluoride (LiF) may be configured to form based on a combination of fluorine anions (F−) and lithium cations (Li.sup.+). In some aspects, the combination of fluorine anions (F−) and lithium cations (Li.sup.+) may generate lithium oxide (Li.sub.2O), lithium nitrate (LiNO.sub.3) and/or nitrogen-oxygen containing compounds. In one implementation, lithium fluoride (LiF) may be formed based on a combination of lithium cations (Li.sup.+) **2825** output from the anode **2820** and fluorine anions (F−) **3012** (of FIG. **30**) grafted onto the first polymeric chain **3010** and/or the second polymeric chain **3020**. In some aspects, the combination of lithium cations (Li.sup.+) **2825** and fluorine anions (F.sup.−) **3012** may consume at least some of the lithium cations (Li.sup.+) **2825** output from the anode **2820**, thereby reducing lithium-containing dendritic growth (not shown in FIG. **28** for simplicity) from the anode **2820**. Reducing lithium-containing dendritic growth from the anode may, in turn, increase the charge rate, the discharge rate, the energy density, the cycle life of the battery **2800**, or any combination thereof. In addition, lithium fluoride (LiF), lithium oxide (Li.sub.2O), lithium nitrate (LiNO.sub.3) or nitrogen-oxygen containing additives may form one or more regions across one or more of the anode **2820** or the solid-electrolyte interphase **2840**.

(235) In some aspects, the inorganic and/or ionic conductor **3018** of the protective layer **2860** may include additives, such as lithium salts including lithium nitrate (LiNO.sub.3) and/or inorganic ionically conductive ceramics including one or more of lithium lanthanum zirconium oxide (LLZO), NASICON-type oxide Li.sub.1+xAl.sub.xTi.sub.2-x(PO.sub.4).sub.3 (LATP) and/or lithium tin phosphorus sulfide (LSPS), and/or nitrogen-oxygen containing additives. At least some of these additives dispersed throughout region “A” and/or the protective layer **2860** may dissociate to produce the lithium cations (Li.sup.+) **2825**. In this way, the presence of at least some additives within the protective layer **2860** may increase the charge rate, the discharge rate, and/or the energy density of the lithium-sulfur battery **2800**.

(236) In one implementation, the first polymeric chain **3010** may be formed from a first plurality of interconnected monomer units, “C”, (e.g., of FIGS. **36** and **37**), and the second polymeric chain **3020** may be formed from a second plurality of interconnected monomer units, “D”. In some aspects, the first plurality of interconnected monomer units, “C”, and the second plurality of interconnected monomer units, “D”, may be identical. In other aspects, the first plurality of interconnected monomer units, “C”, and the second plurality of interconnected monomer units, “D”, may be distinct from each other.

(237) In some aspects, the first polymeric chain **3010** and the second polymeric chain **3020** may cross-link with each other based on exposure to nitrogen-containing groups (e.g., nitrate ions

NO.sub.3.sup.−), some of which may cure in an epoxy and/or include an amine-containing group. In addition, or the alternative, the first polymeric chain **3010** and/or the second polymeric chain **3020** may be prepared to include liquid bisphenol A epichlorohydrin-based epoxy resin, polyoxyethylene bis(glycidyl ether) having an average M.sub.n of 500 (PEG-DEG-500), and polyoxypropylenediamine. For example, in one implementation, the protective layer **2860** may be prepared to include between 2 wt. %-5 wt. %, of difunctional bisphenol A/epichlorohydrin derived liquid epoxy resin, between 15 wt. %-25 wt. % of polyoxyethylene bis(glycidyl ether) (PEG-DEG-500) having an average M.sub.n of 500, between 20 wt. %-25 wt. % of diaminopolypropylene glycol, between 5 wt. %-15 wt. % of poly(propylene glycol) bis(2-aminopropyl ether), between 5 wt. %-15 wt. % of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), and between 40 wt. %-60 wt. % of lithium lanthanum zirconium oxide (LLZO). In some other aspects, the protective layer **2860** may be prepared to include between 2 wt. %-5 wt. % of difunctional bisphenol A/epichlorohydrin derived liquid epoxy resin, between 15 wt. %-25 wt. % of polyoxyethylene bis(glycidyl ether) having an average M.sub.n of 500, between 5 wt. %-15 wt. % of 3,4-epoxy cyclohexyl methyl-3,4-epoxy cyclohexane carboxylate (ECC), between 15 wt. %-20 wt. % of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), between 40 wt. %-60 wt. % of lithium lanthanum zirconium oxide (LLZO), and between 1 wt. %-5 wt. % of diphenyliodonium hexafluorophosphate (DPIHFP). In addition, the protective layer **2860** may be coated and/or deposited onto the anode **2820** by a roll-to-roll apparatus. For example, the protective layer is one or more of spray coated, gravure coated, micro gravure coated, slot-die coated, doctor-blade coated, and/or Mayer's rod spiral-coated onto the anode.

(238) In some implementations, the battery **2800** may be arranged in one or more additional configurations. For example, the first polymeric chain **3010** and the second polymeric chain **3020** may participate in one or more cross-linking polymerization reactions with each other and form the protective layer **2860** based on exposure to one or more ultraviolet (UV) curing accelerators, including one or more cationic photo initiators. As discussed, the UV curing accelerators may be provided by the environment **3050**, which may include an ultraviolet (UV) radiation source.

(239) In some aspects, one or more cross-linking polymerization reactions may include ring-opening polymerization (ROP). In addition, the one or more ultraviolet (UV) curing accelerators includes a plurality of onium salts, which may include one or more triphenylsulfonium salts, one or more diazonium salts, one or more diaryliodonium salts, one or more ferrocenium salts, and/or one or more metallocene compounds. In some aspects, the one or more ultraviolet (UV) curing accelerators may include one or more antimony salts and/or polyols, which may serve as reactive diluents.

(240) In some aspects, the protective layer **2860** may be formed from multiple polymers uniformly mixed together and/or cross-linked (e.g., via cationic polymerization and/or ultraviolet (UV curing)) to form a three-dimensional (3D) lattice. The protective layer **2860** may be formulated according to one or more recipes disclosed in formulation recipe Examples 21-22 below:

Example 21: For Amine-Curable Epoxy-Based Membrane Compositions

(241) TABLE-US-00002 Weight Percent Components (wt. %) of Total Polyoxyethylene bis(glycidyl ether) (PEG- 19.64 DEG-500) having an average M.sub.n of 500 Undiluted clear difunctional bisphenol 3.62 A/epichlorohydrin derived liquid epoxy resin (e.g., EPON ™ Resin 828) Polyoxypropylenediamine (e.g., 10.04 JEFFAMINE ® D-230) Lithium bis(trifluoromethanesulfonyl)imide 16.7 (LiTFSI) Lithium Lanthanum Zirconium Oxide 50 (LLZO) - 500 nm particle size

Example 22: For Amine-Curable Epoxy-Based Membrane Compositions

(242) TABLE-US-00003 Weight Percent Components (wt. %) of Total Polyoxyethylene bis(glycidyl ether) (PEG- 11.7 DEG-500) having an average M.sub.n of 500 4,4′-Methylenebis(N,N-diglycidylaniline) 11.7 Polyoxypropylenediamine (e.g., 9.94 JEFFAMINE ® D-230) Lithium bis(trifluoromethanesulfonyl)imide 16.7 (LiTFSI) Lithium Lanthanum Zirconium

Oxide 50 (LLZO) - 500 nm particle size

(243) In some aspects, the protective layer **2860** may be prepared following the steps of Example 23, an example procedure for preparing a 30 milliliter (mL) sample-scale dispersion of any one or more of the formulation recipes provided by Examples 21-22 and/or 24-28 disclosed herein:

Example 23: Procedure for Preparing the Protective Layer

(244) TABLE-US-00004 Phase Steps 1. Dispersion Preparation 1. Weigh components listed in one of the Examples 21-22 or 24- 28 outside of a glovebox; pour and/or disperse the listed components into a 4 ounce (oz) glass bottle. 2. Transfer the bottle to inside a glovebox, weigh and optionally add LLZO (e.g., in the amount listed by the respective Example) and optionally add DME (e.g., depending on the level of dilution sought) as a non-reactive diluent. 3. Wrap bottle lid with paraffin, and sonicate the wrapped for a duration of 60 minutes (min.) outside the glovebox in a fume hood, or inside glovebox if a sonicator is available; vortex the wrapped bottle for 1 min., then sonicate the wrapped for another 60 min. 4. Transfer the bottle back to glovebox to prepare for spray coating the anode. 2. Spray coating 1. Clean a spray nozzle with acetone prior to spraying the liquid dispersion prepared in Phase 1. 2. Spray all contents of the bottle (e.g., the entire dispersion) onto 6 centimeter (cm) by 60 cm lithium foil (e.g., the anode) within glovebox using the spray nozzle. 3. Spray for a total duration of approximately 30-45 min. 3. (Optional) UV Curing 1. Expose the spray-coated dispersion (e.g., protective layer 2860) disposed on the lithium anode foil to UV light (e.g., having a wavelength of 254 nanometers (nm)) for 10-20 seconds. 2. Repeat if desired to increase cross-linking density of the protective layer 2860. 4. Drying 1. Dry (e.g., cure) the spray-coated dispersion (e.g., protective layer 2860) onto the lithium anode foil at room temperature (18° C.-22° C.) for more than 24 consecutive hours within the glovebox.

(245) When prepared according to Example 21, the protective layer **2860** has an on-set curing temperature of 68° C. and a glass-transition temperature (T.sub.g) of −16° C. In some aspects, the protective layer **2860** may have one or more openings, also referred to as “pinholes,” shown by a pinhole **3102**A in FIG. **31**A formed in the protective layer **2860** that permit for undesirable pass-through of polysulfides **2882** towards the anode **2820** in a direction “B” of FIG. **28**. The polysulfides **2882** may at least partially coat the anode **2820** and thereby impede free movement and/or transport of the lithium cations (Li.sup.+) **2825** between the anode **2820** and the cathode **2810**, which in turn may facilitate the discharge-charge cycling operation of the battery **2800**. To address the undesirable formation of pinholes in the protective layer **2860** when prepared according to Example 21, the protective layer **2860** may be prepared according to Example 22. In this configuration, the protective layer **2860** has an on-set curing temperature of 81° C., a curing peak temperature of 124° C., a curing enthalpy of 104 J/g, and a T.sub.g of 63° C. Variations of either Example 21 or Example 22 are possible where component loading levels are adjusted +/−3% from that listed. However, any formulation of the protective layer **2860** may be prepared to result in T.sub.g greater than 60° C. and an on-set curing temperature of less than 81° C., or a curing peak of less than 124° C., for the protective layer **2860** to cure at room temperature (e.g., 18° C.-22° C.). In addition, the protective layer **2860** may be disposed onto the anode **2820** according to Example 23 with a thickness between 100 nm to 3 μm.

(246) The protective layer **2860** may be formulated according to either of Example 21 or Example 22 by following the procedure set forth in Example 23 to address commonly encountered operational challenges facing conventional lithium-sulfur batteries. For example, in conventional lithium-sulfur batteries using liquid-phase electrolyte solutions, unwanted and uncontrolled dissolution of polysulfides due to polysulfide shuttle may contribute to battery failure. In one or more particular examples, polysulfide attack on lithium can cause rapid battery capacity reduction due to continuous solid-electrolyte interphase (SEI) growth. This growth consumes lithium cations, thereby resulting in fewer lithium being available for transport through to the cathode for healthy discharge-charge operational cycling in conventional lithium-sulfur batteries. In addition, remaining lithium cations may adhere to other lithium cations due to, for example, gradients in

electrochemical potential conducive for lithium-lithium metallic bonding, thereby producing lithium-containing dendritic structures, which grow and extend from the anode towards the cathode and may thereby cause short-circuiting of conventional lithium-sulfur batteries.

(247) To protect lithium contained in the anode **2820** from the effects of poly sulfide migration from the cathode **2810** to the anode **2820**, and to suppress lithium-containing dendrite formation, the protective layer **2860** may be directly coated onto the anode **2820** prior to activation and operational discharge-charge cycling of the battery **2800**. As such, the protective layer **2860** may be prepared to include the first polymeric chain **3010** and the second polymeric chain **3020**. Each of the first and second polymeric chains **3010** and **3020** may have repeating monomer units (e.g., of FIGS. **36** and **37**), and may be identical or dissimilar to each other. In addition, in some aspects, when the protective layer **2860** is formulated according to Example 21 or Example 22, at least some of the carbon atoms **3014** on each polymeric chain may form carbon-carbon bonds with each other via ring-opening (ROP) cationic polymerization (e.g., optionally catalyst-based), and may thereby form an amine-curable epoxy-based membrane. In some other aspects, when the protective layer **2860** is formulated according one of Examples 24 to 28 (presented below), at least some of the carbon atoms **3014** on each polymeric chain may form carbon-carbon bonds **3026** with each other via ultraviolet (UV) curable ROP cationic polymerization.

(248) In addition, catalyst-based or UV-curing of di- and/or multi-functional components may initiate and/or facilitate cross-linking of the first polymeric chain **3010** to the second polymeric chain **3020** by the carbon-carbon bonds **3026** for efficient protection of the anode **2820**. Functional groups (not shown in FIG. **28** or FIG. **30** for simplicity) may be attached to at least some of the inorganic and/or ionic conductors **3018** for additional tunability of the protective layer **2860**. In this way, the protective layer **2860**, when formulated by any recipe disclosed by Example 21 through Example 28, may provide several advantages relative to conventional anode protective layers, which typically have only one component.

(249) In some implementations, the protective layer **2860** may be formulated to include multiple polymeric components, such as where the first polymeric chain **3010** and the second polymeric chain **3020** are different from each other to make the protective layer **2860** relatively more flexible for curing on and over the anode **2820** after spray-coating. This flexibility may prevent the protective layer **2860** from disintegrating during the fabrication process. In addition, directly coating the protective layer **2860** onto the anode **2820** may uniformly strip and plate at least some of the lithium cations (Li+) **2825** during operational discharge-charge cycling of the battery **2800**, thereby minimizing lithium dendrite formation. In some aspects, inorganic and/or ionic conductors **3018**, such as LLZO, dispersed throughout the protective layer **2860** may be ionically conductive and designed to minimize impedance growth of the battery **2800**. In addition, UV curing may be used to replace conventional heat-drying processes to accelerate curing of the protective layer **2860** and minimize potential adverse effects of processing time of the protective layer **2860** on the anode **2820**.

(250) FIG. **31**A shows a micrograph of an example baseline protective layer **3100**A, according to some implementations. The baseline protective layer **3100**A may be one example of the protective layer **2860** of FIG. **28** and prepared according to Example 21 presented earlier, thereby resulting in formation of the pinhole **3102**A. In some aspects, the pinhole **3102**A may be sized as shown in FIG. **31**A. In some other aspects, the pinhole **3102**A may be smaller or larger than as shown in FIG. **31**A. In addition, the baseline protective layer **3100**A may have multiple instances of the pinhole **3102**A, which collectively may negatively interfere with healthy operational discharge-charge cycling of the battery **2800**. For example, the pinhole **3102**A may permit passage of at least some of the polysulfides **2882** through the pinhole **3102** to contact the anode **2820** and/or otherwise interfere with formation of the solid-electrolyte interphase **2840**, thereby reducing cycling efficiency of the battery **2800**.

(251) FIG. **31**B shows a micrograph of an example protective layer **3100**B, according to some

implementations. The protective layer **3100**B may be one example of the protective layer **2860** of the battery of FIG. **28** when prepared according to Example 22 by the process disclosed in Example 23. The protective layer **3100**B may minimize pinhole formation to have no pinholes or one or more smaller pinholes **3102**B, thereby reducing risk of at least some of the polysulfides **2882** contacting the anode **2820** and correspondingly improving operational discharge-charge performance of the battery **2800**.

(252) FIG. **32** shows a micrograph of a cutaway **3200** of the protective layer **2860** of the battery **2800** of FIG. **28**, according to some implementations. The protective layer **2860** may be prepared according to Example 21 by the process disclosed in Example 23. In some other aspects, the protective layer **2860** may be prepared according to other recipes disclosed in the Examples, including Example 22.

(253) FIG. **33**A shows an example cross-linking density **3300**A of the protective layer **2860** of the battery **2800** of FIG. **28**, according to some implementations. The cross-linking density **3300**A may be one example of a cross-linking density of the protective layer **2860** when prepared according to Example 21 by the process disclosed in Example 23, and thereby have a certain number (e.g., 9) of cross-link points **3306**A per unit area **3302**A. In this way, each cross-link point **3306**A may be separated from adjacent cross-link points **3306**A by a dimension **3304**A.

(254) FIG. **33**B shows another example cross-linking density **3300**B of the protective layer **2860** of the battery **2800** of FIG. **28**, according to some implementations. The cross-linking density **3300**B may be one example of a cross-linking density of the protective layer **2860** when prepared according to Example 22 by the process disclosed in Example 23, and thereby have a certain number (e.g., 25) of cross-link points **3306**B per unit area **3302**B. In this way, each cross-link point **3306**B may be separated from adjacent cross-link points **3306**B by a dimension **3304**B, which may be smaller than the dimension **3304**A and thereby configured to trap at least some of the TFSI.sup.− anions **2826** subsequent to dissociation of LiTFSI in the electrolyte **2830** and/or included in the protective layer **2860**. By trapping at least some of the TFSI.sup.− anions **2826** within the cross-linking density **3300**B, the protective layer **2860** may function to prevent the trapped TFSI.sup.− anions **2826** from blocking passage of the lithium cations (L+) **2825** and/or contacting the anode **2820**, thereby improving operational discharge-charge cycling performance of the battery **2800**.

(255) FIG. **34** shows an example ring-opening (ROP) mechanism **3400** for triarylsulfonium salt (Ar.sub.3S.sup.+MtXn.sup.−), according to some implementations. Generally, usage of UV initiators to initiate ROP polymerization via cross-linking of the first polymeric chain **3010** with the second polymeric chain **3020** may accelerate curing of the protective layer **2860** from a maximum of 24 hours to 1-3 seconds, which may be desirable for large-scale manufacturing processes. In some aspects, usage of UV initiators may accelerate epoxy cross-linking for existing epoxy systems (e.g., Examples 21-22, and/or Examples 24-28 to be disclosed herein), without requiring the additional introduction of new chemistries. In addition, UV initiators selected to enable epoxy cross-linking may be cationic UV initiators. In this way, when various epoxies are exposed to UV radiation, they may produce a relatively strong Lewis acid and/or Brønsted-Lowry acid, which may then correspondingly initiate ROP of epoxy groups.

(256) The ROP mechanism **3400** is illustrated for triarylsulfonium salt (Ar.sub.3S.sup.+MtXn.sup.−), which may be representative of the inorganic and/or ionic conductor **3018** incorporated in the protective layer **2860**. In some aspects, HMtXn is a Lewis acid (e.g., HBF.sub.4, HPF.sub.6, HAsF.sub.6, HSbF.sub.6) and M is a monomer (e.g., of FIGS. **36** and **37** and/or including an epoxy group). Unlike free-radical polymerization, cationic polymerization (e.g., including UV-initiated cationic ROP) is not inhibited by oxygen. However, polymer chain growth and cross-linking reactions may be inhibited by trace (e.g., less than 0.1 wt. %) of water and or chemicals (e.g., amines and/or urethanes). Nevertheless, initiating moieties in UV-initiated cationic ROP are relatively chemically stable for extended durations (e.g., more than 24 hrs.) In this way, polymerization in UV-initiated cationic ROP may continue even in the absence of visible light. In

addition, some monomers (e.g., of FIGS. **36** and **37**) and/or oligomers may be cured with lower UV dosages, depending on the amount of the protective layer **2860** sought for preparation.

(257) FIG. **35** shows several example onium salts **3500** suitable for usage as cationic photo-initiators for the protective layer **2860** of the battery of FIG. **28**, according to some implementations. The onium salts **3500** may be one example of the inorganic and/ionic conductor **3018** and may thereby be used to initiate cross-linking of the first polymeric chain **3010** with the second polymeric chain **3020** to form the protective layer **2860**. In some aspects, the onium salts **3500** may include diphenyliodonium salts and/or triphenylsulfonium salts (both shown in FIG. **35**), as well as diazonium salts, diaryliodonium salts, ferrocenium salts and/or various other metallocene compounds (not shown in FIG. **35** for simplicity). Efficiency of the onium salts **3500** as cationic photo-initiators may at least in part depend on their respective solubility in the protective layer **2860** (e.g., when formed as a resin), and/or their respective polarity and/or surface charge. Generally, solubility increases with increasing size of anions in respective onium salts **3500** because charge is dissipated over a relatively larger surface area of the anion, which lowers hydrophilicity of the respective onium salt **3500**. In this way, the solubility and reactivity of at least some of the onium salts **3500** in non-ionic resins may increase along the order of tetrafluoroborate (BF.sub.4−)<hexafluorophosphate (PF.sub.6−)<hexafluoroarsenate (AsF.sub.6−)<fluoronium (SbF.sub.6−). In some aspects, antimony salts (e.g., including the fluoronium anion) may be selected most often for use as cationic photo-initiators, due to their relatively higher solubility and reactivity, over the other listed salts. In addition, besides solubility alone, spectroscopic properties such as range of light absorption and/or bond cleavage efficiency may affect the rate of initiation of polymerization of monomers in the first polymeric chain **3010** and/or the second polymeric chain **3020**.

(258) FIG. **36** shows a several example monomers **3600** of various cationic photo-polymerizable compositions suitable for forming the protective layer of the battery of FIG. **28**, according to some implementations. The monomers **3600** may each be one example of repeating monomer unit “C” and/or “D” of FIG. **30** and may thereby be used to initiate cross-linking of the first polymeric chain **3010** with the second polymeric chain **3020** to form the protective layer **2860**. That is, multiple of instances of repeating monomer unit “C” may attach to exposed carbon and/or other atoms to form the first polymeric chain **3010** of multiple units, e.g., “C”-“C”-“C”- . . . etc., in the manner shown in FIGS. **30** and **36**. Repeating monomer unit “D” may be identical or dissimilar to repeating monomer unit “C,” and thereby form the second polymeric chain in a similar manner to the first polymeric chain by attaching to additional instances of repeating monomer unit “D,” e.g., “D”-“D”-“D”- . . . etc., in the manner shown in FIGS. **30** and **36**.

(259) In some aspects, cationic UV resin formulations formed of at least some of the monomers **3600** may include examples of one or more cycloaliphatic epoxies, any one of which may be used as an epoxide group. Cycloaliphatic epoxies tend to be the relatively more reactive compared to linear aliphatic groups or aromatic epoxy molecules, such that when any one or more of the monomers **3600** are used to produce the protective layer **2860**, a tight network of polar groups may form to yield a relatively brittle polymer-based final product. To reduce undesirable brittleness, some UV cationic resin formulations may be prepared to include polyols (not shown in FIG. **36** for simplicity) as reactive diluents and performance modifiers. In some aspects, the polyols may act as monomeric materials, reacting into formed epoxy-based networks. In comparison to cycloaliphatic epoxy groups alone, many different polyols are available ranging from di and tri-functional glycols, polycaprolactone oligomers, and even high order dendritic polyols. Selecting between the relatively higher number of available polyols may assist in fine-tuning of end-usage properties of the protective layer **2860**.

(260) FIG. **37** shows ultraviolet (UV) curable monomers **3700** suitable for forming the protective layer **2860** of the battery **2800** of FIG. **28**, according to some implementations. Monomers **3700** may be one example of monomer “B” and/or monomer “C” of FIG. **30** and may include diglycidyl

1,2-cyclohexanedicarboxylate (DG-CHDC), 3,4-epoxycyclohexylmethyl-3′,4′-epoxycyclohexane carboxylate (ECC), triphenylsulfonium triflate (TPS-TF), diphenyliodonium triflate (DPI-TF), diphenyliodonium hexafluorophosphate (DPI-HFP), poly[(phenyl glycidyl ether)-co-formaldehyde] (PPGEF), and/or glycidyl 2,2,3,3-tetrafluoropropyl ether (GTFEP).

(261) In some aspects, the protective layer **2860** may be formed from multiple polymers (e.g., formed from the monomers **3700**) uniformly mixed together and/or cross-linked (e.g., via ultraviolet (UV curing) with a UV-curing wavelength of 254 nanometers (nm)) to form a three-dimensional (3D) lattice disposed on the anode **2820**. The protective layer (e.g., when formed as the 3D lattice) may be formulated according to one or more recipes disclosed in Examples 24-28 below:

Example 24: UV-Curable Recipe

(262) TABLE-US-00005 Function Components Wt. % Soft Oligomeric Epoxy (e.g., used as a Polyoxyethylene bis(glycidyl ether) 16.64 flexible spacer to decrease brittleness (PEG-DEG-500) having an average M.sub.n of the protective layer 2860) of 500 Rigid Polymeric Epoxy (e.g., used as a Undiluted clear difunctional bisphenol 3.62 flexible spacer to decrease brittleness A/epichlorohydrin derived liquid epoxy of the protective layer 2860) resin (e.g., EPON ™ Resin 828) Fast Curing Epoxy Monomer Diglycidyl 1,2-cyclohexanedicarboxylate 8.04 (DG-CHDC) Photo-initiator Triphenylsulfonium triflate (TPS-TF) 5.00 Inorganic and/or Ionic Conductor Lithium 16.7 bis(trifluoromethanesulfonyl)imide (LiTFSI) Inorganic and/or Ionic Conductor (e.g., Lithium lanthanum zirconium oxide 50.0 used as a UV-screen to minimize acid (LLZO) (500 nm particle size) formation on exposed surfaces of the protective layer 2860)

Example 25: UV-Curable Recipe

(263) TABLE-US-00006 Function Components Wt. % Soft Oligomeric Epoxy (e.g., used as a Polyoxyethylene bis(glycidyl ether) 19.64 flexible spacer to decrease brittleness of (PEG-DEG-500) having an average M.sub.n the protective layer 2860) of 500 Rigid Polymeric Epoxy (e.g., used as a Undiluted clear difunctional bisphenol 3.62 flexible spacer to decrease brittleness of A/epichlorohydrin derived liquid epoxy the protective layer 2860) resin (e.g., EPON ™ Resin 828) Fast Curing Epoxy Monomer 3,4-epoxycyclohexylmethyl-3′,4′- 8.04 epoxycyclohexane carboxylate (ECC) Photo-initiator Diphenyliodonium 2.00 hexafluorophosphate (DPI-HFP) Inorganic and/or Ionic Conductor Lithium 16.7 bis(trifluoromethanesulfonyl)imide (LiTFSI) Inorganic and/or Ionic Conductor Lithium lanthanum zirconium oxide 50.0 (LLZO) (500 nm particle size)

Example 26: UV-Curable Recipe

(264) TABLE-US-00007 Function Components Wt. % Soft Oligomeric Epoxy Polyoxyethylene bis(glycidyl ether) (PEG- 17.8 DEG-500) having an average M.sub.n of 500 Rigid Polymeric Epoxy PPGEF (Number Average Molecular 3.5 Weight (Mn) = 345) Fast Curing Epoxy Monomer Diglycidyl 1,2-cyclohexanedicarboxylate 7.0 (DG-CHDC) Photo-initiator Diphenyliodonium triflate (DPI-TF) 5.0 Inorganic and/or Ionic Conductor Lithium bis(trifluoromethanesulfonyl)imide 16.7 (LiTFSI) Inorganic and/or Ionic Conductor Lithium lanthanum zirconium oxide 50.0 (LLZO) (500 nm particle size)

Example 27: UV-Curable Recipe

(265) TABLE-US-00008 Function Components Wt. % Soft Oligomeric Epoxy Polyoxyethylene bis(glycidyl ether) (PEG- 19.8 DEG-500) having an average M.sub.n of 500 Rigid Polymeric Epoxy PPGEF (Number Average Molecular Weight 4.5 (Mn) = 570) Fast Curing Epoxy Monomer 3,4-epoxycyclohexylmethyl-3′,4′- 7.0 epoxycyclohexane carboxylate (ECC) Photo-initiator Diphenyliodonium hexafluorophosphate (DPI- 2.0 HFP) Inorganic and/or Ionic Conductor Lithium bis(trifluoromethanesulfonyl)imide 16.7 (LiTFSI) Inorganic and/or Ionic Conductor Lithium lanthanum zirconium oxide (LLZO) 50.0 (500 nm particle size)

(266) In some aspects, polyethylene glycol (PEG), other polyethers and/or other polyols may be used and/or substitutes for any of the components listed in Examples 23-27 depending on the

performance requirements for the protective layer **2860**. In addition, variations of Examples 24-27 are possible where component loading levels are adjusted +/−3% from that listed. In addition, all listed components, except for polyoxypropylenediamine (e.g., JEFFAMINE® D-230), are compatible with UV-catalyzed cationic ROP. In this way, the first polymeric chain **3010** may initiate cross-linking with the second polymeric chain **3020** to form the carbon-carbon bonds **3026** and produce a 3D lattice disposed on the anode **2820**. The 3D lattice may be formed of the various components listed in each Example, where the components are uniformly mixed together, interconnected with each other, and dispersed throughout the protective layer **2860**. That is, the first polymeric chain **3010** and the second polymeric chain **3020** are exemplary and additional polymeric chains are possible, depending on the Example. Some Examples may include additional polymeric chains cross-linked to each other as well as one or more of the first polymeric chain **3010** or the second polymeric chain **3020**. In addition, each component listed in any one or more of the Examples 21-22 and/or 24-28 may be formed of a corresponding polymeric chain.
(267) That is, in Example 26, which is representative of the other Examples, polyoxyethylene bis(glycidyl ether) (PEG-DEG-500) having an average M.sub.n of 500 may be denoted as monomer “B” in the first polymeric chain **3010** and diglycidyl 1,2-cyclohexanedicarboxylate (DG-CHDC) may be denoted as monomer “C” in the second polymeric chain **3010**. Monomer “B” may bond with additional instances of monomer “B” and also bond with one or more instance of monomer “C” in 3D to form the 3D lattice. Additional components (not shown in FIG. **30** for simplicity), may be denoted as monomer “D” and so forth. For example, in Example 26, PPGEF (Number Average Molecular Weight (Mn)=345) may be denoted as monomer “D” and bond to additional instances of itself as well as monomer “B” and/or monomer “C” in 3D to form the 3D lattice, where lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) and/or lithium lanthanum zirconium oxide (LLZO) (500 nm particle size) may be depicted as the inorganic and/or ionic conductor **3018** and dispersed throughout the protective layer **2860**.
(268) Certain fast-curing cycloaliphatic epoxy monomers (e.g., 3,4-epoxycyclohexylmethyl-3′,4′-epoxycyclohexane carboxylate (ECC)), such as included in Examples 25 and 27 above, may be incorporated into the protective layer **2860**, to rapidly cross-link together at higher cross-linking density levels such as the cross-linking density **3300**B of FIG. **33**B via cationic polymerization processes to produce a network of polar groups to at least partially trap the TFSI.sup.− anions **2826** within the protective layer **2860**. In addition, in some aspects, various UV initiators (e.g., onium sales of hexafluorophosphate (PF.sub.6.sup.−), which may be beneficial for formation of the solid-electrolyte interphase **2840**) may be substituted for the photo-initiators disclosed in Examples 24-27. In some other aspects, the anion for fluoroantimonic acid (SbF.sub.6.sup.−) may be substituted for the photo-initiators disclosed in Examples 24-27. The anion for fluoroantimonic acid (SbF.sub.6.sup.−) is soluble in the electrolyte **2830** and may facilitate alloying of at least some regions of the protective layer **2860** with at least some of the lithium cations (Li.sup.+) **2825**. In this way, alloying of some regions of the protective layer **2860** may consume at least some of the lithium cations (Li.sup.+) **2825**, thereby removing the consumed lithium cations from participating in lithium-lithium metallic bonding to form undesirable dendrites from the anode **2820** towards the cathode **2810**.
(269) In addition, in one implementation, lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) may act as a co-initiator once UV curing has been initiated (e.g., by any of the photo-initiators listed in Examples 24-27), thereby increasing curing rates. For example, activation of the ROP reaction of an example monomer, poly(ethylene glycol) diglycidyl ether (DGEPEG) may be achieved using lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) with a subsequent propagation step. In the activation step, the lithium cation (e.g., provided by LiTFSI) attacks the carbon-oxygen bond on one or more epoxides of DGEPEG. As this reaction occurs under acidic conditions, the epoxide is converted to a hydroxyl ion. During the initiation and propagation steps, the hydroxyl ions react with epoxides and other hydroxyl ions via a nucleophilic reaction yielding chain extension via the

formation of C—O—C bonds.

(270) In alternative to Examples 24-27 presented earlier, the protective layer **2860** may be formulated to include fluorinated materials (e.g., fluoropolymers, such as glycidyl 2,2,3,3 tetrafluoropropyl ether (GTFEP)) to be at least partially grafted onto the carbon atoms **3014**. In this way, GTFEP may be used as a source of the fluorine ions (F^-) **3012**, which may later dissociate from their respective carbon atoms to combine with the lithium cations (Li^+) **2825** to produce lithium fluoride (LiF) via the Wurtz reaction, as discussed elsewhere in the present disclosure. Example 28 may be prepared according to the procedure provided by Example 23 to include the following components:

Example 28: Fluorinated Polymer

(271) TABLE-US-00009 Weight Percent Components (wt. %) of Total Polyoxyethylene bis(glycidyl ether) (PEG- 14.7 DEG-500) having an average M_n of 500 Glycidyl 2,2,3,3- 10.5 tetrafluoropropyl ether (GTFEP) Undiluted clear difunctional bisphenol 2.7 A/epichlorohydrin derived liquid epoxy resin (e.g., EPON ™ Resin 828) Polyoxypropylenediamine (e.g., JEFFAMINE ® 7.5 D-230) Lithium bis(trifluoromethanesulfonyl)imide 14.3 (LiTFSI) One or more surfactants (e.g., cationic 1.0 surfactants such as alkyl trimethyl ammonium, R—$N(CH_3)_3^+$, dissolved in seawater (SW)) Lithium lanthanum zirconium oxide (LLZO) 49.3 (500 nm particle size)

(272) Example 28 as presented above may not be initiated by UV curing, as polyoxypropylenediamine (e.g., JEFFAMINE® D-230) may not be compatible with UV-catalyzed cationic ROP. However, in some aspects, polyoxypropylenediamine (e.g., JEFFAMINE® D-230) may be replaced by 3,4-epoxycyclohexylmethyl-3′,4′-epoxycyclohexane carboxylate (ECC) to thereby render Example 28 UV-curable, similar to Examples 23-27.

(273) FIG. **38** shows several example non-reactive diluents **3800** suitable for usage as additives to adjust dilution levels in UV-curable formulations prepared with the UV curable monomers of FIG. **37**, according to some implementations. In one implementation, the non-reactive diluents may include 1,2-Dimethoxyethane (DME) and/or triethylene glycol dimethyl ether. In some aspects, the non-reactive diluents **3800** may be removed from the protective layer **2860** after cross-linking by baking at 100° C. In some other aspects, the non-reactive diluents **3800** may be retained in the protective layer **2860** to enhance diffusion of the lithium cations (Li^+) **2825**.

(274) FIG. **39** shows several example reactive diluents **3900** suitable for usage as additives to adjust dilution levels in UV-curable formulations prepared with the UV curable monomers of FIG. **37**, according to some implementations. In some aspects, reactive diluents may function as flexibilizers to release stress of at least some of the carbon atoms **3014** (e.g., which may be cross-linked to each other) within the protective layer **2860** to thereby minimize pinhole formation. In addition, in some aspects, the non-reactive diluents **3800** and/or the reactive diluents **3900** may be or include materials used as additives for UV-curable formulations in the amount between 1 wt. % and 50 wt. % per formulation weight to adjust the viscosity depending on deposition method (e.g., low viscosity for spray coating compared to high viscosity for slot die or draw-down coating methods). In this way, the remaining components are diminished in proportion to the amount of diluent added.

(275) FIG. **40**A shows a graph **4000**A of capacity (% of initial) against cycle number, according to some implementations. The graph **4000**A depicts performance of the battery **2800** when prepared according to Example 22 against Example 21 and an unprotected 40 μm lithium anode (e.g., provided as a reference, “Ref.”). Example 22 shows consistently higher capacity retention (e.g., in % of original) per operational discharge-charge cycle number than both Example 21 and the unprotected 40 μm lithium anode.

(276) FIG. **40**B shows a graph of capacity (mAh/g) against cycle number, according to some implementations. The graph **4000**A depicts performance of the battery **2800** when prepared according to Example 22 against Example 21 and an unprotected 40 μm lithium anode (e.g.,

provided as a reference, “Ref.”). Example 22 shows consistently higher capacity retention (e.g., in milli-amp hours per gram, mAh/g) per operational discharge-charge cycle number than both Example 21 and the unprotected 40 μm lithium anode. The graph **4000**A depicts performance of the battery **2800** when prepared according to Example 22 against Example 21 and an unprotected 40 μm lithium anode (e.g., provided as a reference, “Ref.”). Example 22 shows consistently higher capacity retention (e.g., in % of original) per operational discharge-charge cycle number than both Example 21 and the unprotected 40 μm lithium anode.

(277) FIG. **41**A shows another graph of capacity (% of initial) against cycle number, according to some implementations. The graph **4100**A depicts performance of the battery **2800** when prepared according to Example 28 against Example 21 and an unprotected 40 μm lithium anode (e.g., provided as a reference, “Ref.”). Example 22 shows consistently higher capacity retention (e.g., in % of original) per operational discharge-charge cycle number than both Example 21 and the unprotected 40 μm lithium anode.

(278) FIG. **41**B shows another graph of capacity (mAh/g) against cycle number, according to some implementations. The graph **4100**B depicts performance of the battery **2800** when prepared according to Example 28 against Example 21 and an unprotected 40 μm lithium anode (e.g., provided as a reference, “Ref.”). Example 22 shows consistently higher capacity retention (e.g., mAh/g) per operational discharge-charge cycle number than both Example 21 and the unprotected 40 μm lithium anode.

(279) FIG. **42**A shows another example battery **4200**A, according to some other implementations. The battery **4200**A may be an example of other battery configurations disclosed herein. In one implementation, the battery **4200**A may be implemented as a lithium-sulfur battery, and may include a cathode **4210**, an anode structure **4222** including an anode **4220** positioned opposite to the cathode, a separator **4250** positioned between the anode **4220** and the cathode **4210**, and an electrolyte **4230**. The anode **4220** may be disposed on and/or coupled with a substrate **4201**, such as a metal current collector formed from nickel (Ni) or Aluminum (Al), etc. The cathode **4210** may be disposed on and/or coupled with a substrate **4202**, such as a metal current collector formed from nickel (Ni) or Aluminum (Al), etc. In some aspects, the electrolyte **4230** may be formulated by mixing at least two or more solvents, such as those disclosed in Examples 1-20 presented earlier. The electrolyte **4230** may be dispersed throughout the cathode **4210** and in contact with the anode **4220**. In some aspects, the anode **4220** may be a single foil of solid metallic lithium. In this way, at least some lithium cations (Li.sup.+) **2825** output by the anode **4220** may participate in dissociation reactions and/or combination reactions during operational discharge-charge cycling of the battery **4200**A. That is, lithium cations (Li.sup.+) **4225** output from the anode **4220** may be transported through the electrolyte **4230** and retained in their electrochemically favored positions (not shown in FIG. **42**A for simplicity) within the cathode **4210** during discharge cycles of the battery **4200**A. Then, during charge cycles of the battery **4200**A, the lithium cations (Li.sup.+) **4225** may be forced to return to the anode **4220** upon exposure to an outside current source.

(280) In addition, a solid-electrolyte interphase (SEI) layer **4240** may be formed on the anode **4220**. In some aspects, a protective layer **4260** may be formed at least partially within and/or on the SEI layer **4240** and face the cathode **4210**. In some aspects, the SEI layer **4240** may be formed from one or more compounds on the anode **4220** responsive to on one or more oxidation-reduction reactions involving lithium cations (Li.sup.+) and one or more solvents of the electrolyte **4230**. In some implementations, the protective layer **4260** may be at least partially formed from carbonaceous materials including one or more of flat graphene, wrinkled graphene, carbon nano-tubes (CNTs), carbon nano-onions (CNOs), or non-hollow carbon spherical particles (NHCS), one or more of which may be one example of the carbonaceous structure **956** of FIG. **9**B.

(281) In one implementation, the anode **4220** of the anode structure **4222** may be formed as a single layer of solid lithium, which may output lithium cations (Li.sup.+) during operational discharge cycling of the lithium-sulfur battery. The SEI layer **4240** may be formed on the single

layer of solid lithium, and the protective layer **4260** may be formed on and at least partially disposed within the SEI layer **4240** responsive to operational discharge-charge cycling of the lithium-sulfur battery **4200**A. The protective layer **4260** may be an example of other protective layer configurations disclosed herein, including the protective layer **2860** of FIG. **28**.

(282) In addition, or the alternative, the protective layer **4260** may include a polymeric backbone chain **4262** formed of interconnected carbon atoms **4263**. In this way, at least some of the interconnected carbon atoms **4263** may move during operational discharge-charge cycling of the battery **4200**A and define a cooperative segmental mobility (also referred to as "segmental motion") of the protective layer **4260**. Additional polymeric chains **4264** may be cross-linked to one another and to at least some of the interconnected carbon atoms **4263** of the polymeric backbone chain **4262**. Each of the additional polymeric chains **4264** may be formed of interconnected monomer units **4266**. In some aspects, a plasticizer **4268** may be dispersed throughout the protective layer **4260** without covalently bonding to at least some of the interconnected carbon atoms **4263** of the polymeric backbone chain **4262**. In some aspects, the plasticizer **4268** may be formed of and/or may include one or more of a polyethylene glycol (PEG or PEO) based oligomer, a nitrile, such as succinonitrile, glutaronitrile, adiponitrile. In addition, or the alternative, the plasticizer **4268** may be formed from and/or may include a solvent, including dimethoxyethane (DME), tetrahydrofuran (THF), diethyl ether, dioxolane (DOL), tetraethylene glycol dimethyl ether (TEGDME), toluene, bis(2,2-trifluoroethyl ether) (TEE), fluoroethylene carbonate (FEC), diethyl carbonate (DEC), dimethyl carbonate (DMC), propylene carbonate (PC), and/or ethylene carbonate (EC). The plasticizer **4268** may separate adjacent monomer units (e.g., by a spacing **4267** of FIG. **42**B) of the interconnected monomer units **4266** of at least some of the additional polymeric chains **4264**. Increasing the separation between adjacent monomer units may increase the cooperative segmental mobility of at least some of the additional polymeric chains **4264**, thereby increasing an ionic conductivity of the protective layer **4260**. In addition, in some aspects, increasing the concentration levels of the plasticizer **4268** in the protective layer **4260** may increase lithium cation (Li.sup.+) conductivity through the protective layer **4260**. In some other aspects, linear polymeric and/or oligomeric chains (not shown in FIG. **42**A for simplicity) may be covalently bonded, grafted, and/or cross-linked by cross-linking **4265**) to at least some of the interconnected carbon atoms **4263** of the polymeric backbone chain **4262**. In this way, the linear polymeric and/or oligomeric chains may increase the cooperative segmental mobility (e.g., the maximum cooperative segmental mobility) of the protective layer **4260**, which may increase lithium cation (Li.sup.+) conductivity through the protective layer **4260**. For example, oligomeric substances such as polyoxypropylenediamine (e.g., JEFFAMINE® M-600) may increase the cooperative segmental mobility (e.g., the maximum cooperative segmental mobility) of the protective layer **4260**.

(283) In some instances, the protective layer **4260** may be configured to melt at a glass transition temperature (T.sub.g), such that increasing the glass transition temperature causes a reduction in the cooperative segmental mobility (e.g., the maximum cooperative segmental mobility) of the polymeric chains. In addition, reducing the cooperative segmental mobility (e.g., the maximum cooperative segmental mobility) of polymeric chains may decrease lithium cation (Li.sup.+) conductivity through the protective layer **4260**. In this way, aspects of the present disclosure may maximize lithium cation (Li.sup.+) conductivity through the protective layer **4260** by configuring the glass transition temperature to be less than room temperature (e.g., 18° C.-22° C.).

(284) In some aspects, the protective layer **4260** may be prepared by Example 23 according to the formulation recipe provided by Example 29 disclosed below:

Example 29: Plasticizer-Inclusive Recipe

(285) TABLE-US-00010 Function Components Wt. % Wt. % Range Monomer "C" of Poly(ethylene glycol) diglycidyl 17.5 10.0-50.0 FIG. 30 ether (PEGDGE) Monomer "D" of Undiluted clear difunctional 3.2 1.0-30.0 FIG. 30 bisphenol A/epichlorohydrin derived liquid

epoxy resin (e.g., EPON ™ Resin 828) Cross-Linker Polyoxypropylenediamine (e.g., 9 5.0-25.0 JEFFAMINE ® D-230) Plasticizer Tetraethylene glycol dimethyl ether 3.6 1.0-40.0 (TEGDME or tetraglyme) Inorganic and/or Ionic Lithium 16.7 5.0-50.0 Conductor (e.g., salt) bis(trifluoromethanesulfonyl)imide (LiTFSI) Inorganic and/or Ionic Lithium lanthanum zirconium oxide 50.0 25.0-90.0 Conductor (LLZO) (500 nm particle size)

(286) In some other implementations, the protective layer **4260** may be formed on the anode as a three-dimensional (3D) polymeric lattice (not shown in FIG. **42**A for simplicity) that includes a first polymeric chain and a second polymeric chain positioned opposite one another. In some aspects, the first and second polymeric chains may be examples of the first and second polymeric chain **3010** and **3020**, respectively of region “A” shown in the diagram **3000** of FIG. **30**. In some implementations, each of the first and second polymeric chains may include carbon atoms at least temporarily chemically bonded to oxide ions (O.sup.2−), fluorine anions (F.sup.−), and/or nitrate anions (NO.sub.3.sup.−). Lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) (not shown in FIG. **42**A for simplicity) may be dispersed throughout the 3D polymeric lattice to dissociate into lithium cations (Li.sup.+) **4225** and TFSI− anions **4226**. In this way, the first and second polymeric chains may form the 3D polymeric lattice with a cross-linking density sufficient to trap the TFSI− anions **4226** by cross-linking **4265** with each other. For example, the cross-linking **4265** may be initiated upon exposure to an energetic environment including ultraviolet (UV) energy and LiTFSI, where LiTFSI may serve as a polymerization co-initiator compound.

(287) In other implementations, the first and second polymeric chains may form the 3D polymeric lattice through cross-linking polymerization reactions, which may include a ultraviolet (UV) curing that may progress at a curing rate. In some aspects, the polymerization co-initiator compound may increase the curing rate. In some instances, additives are dispersed uniformly throughout the 3D polymeric lattice, and may include lithium nitrate (LiNO.sub.3), inorganic ionically-conductive ceramics including lithium lanthanum zirconium oxide (LLZO), NASICON-type oxide Li.sub.1+xAl.sub.xTi.sub.2-x(PO.sub.4).sub.3 (LATP) or lithium tin phosphorus sulfide (LSPS), or nitrogen-oxygen containing additives. In this way, inorganic ionically-conductive ceramics may be uniformly embedded in the 3D polymeric lattice and/or uniformly distributed in the protective layer **4260**. In some aspects, the protective layer **4260** may include desiccated solvents.

(288) In some additional or alternative implementations, the protective layer **4260** may trap various types of anions (not shown in FIG. **42**A for simplicity). For example, the protective layer **4260** may be formed of multiple ingredients including relatively pliable oligomeric epoxy and/or polyol based compounds, a relatively rigid polymeric epoxy based compound, and/or photo-initiator molecules. In some aspects, at least some of the relatively pliable oligomeric epoxy and/or polyol based compounds may prevent formation of pinholes in the protective layer **4260**. Lithium-containing salts dispersed uniformly throughout the protective layer **4260** may dissociate into lithium (Li+) cations and various types of anions.

(289) In addition, in some instances, the protective layer **4260** may be formed on the anode responsive to exposure to an ultraviolet (UV) energetic source that facilitates a UV curing of at least some of the ingredients of the protective layer **4260**. In addition, in some aspects, the protective layer **4260** may include non-reactive diluents including 1,2-Dimethoxyethane (DME), tetrahydrofuran (THF), triethylene glycol dimethyl ether, or 2-Methyl-2-oxazoline (MOZ). In some other aspects, the protective layer **4260** may include reactive diluents including 1,3-Dioxolane (DOL), 3,3-Dimethyloxetane (DMO), 2-Ethyl-2-oxazoline (EOZ), or ε-Caprolactone (CL). In this way, a per-unit formulation weight of the protective layer **4260** may be based on a concentration level of non-reactive diluents or reactive diluents relative to ingredients of the protective layer **4260**. In some aspects, the reactive diluents may reduce mechanical stress of at least some cross-linking units within the protective layer **4260**.

(290) In some aspects, reactive diluents may be removed from the protective layer **4260**. In some other aspects, reactive diluents may remain in the protective layer **4260** after cross-linking **4265** of

at least some of the multiple ingredients (e.g., relatively pliable oligomeric epoxy and/or polyol based compounds, the relatively rigid polymeric epoxy based compound, and/or photo-initiator molecules) with one another. The retention of reactive diluents in the protective layer **4260** after cross-linking of two or more ingredients may increase lithium cation (Li.sup.+) **4225** diffusion through the electrolyte **4230**. In one implementation, the relatively rigid polymeric epoxy based compound may be formed from several repeating epoxy monomer units. For example, each repeating epoxy monomer unit is 3,4-epoxycyclohexylmethyl 3,4-epoxycyclohexanecarboxylate (ECC), which may cross-link with additional ECC monomer units and produce a network of polar groups (not shown in FIG. **42**A for simplicity). The network of polar groups may trap at least some anions produced upon dissociation of lithium-containing salts.

(291) FIG. **42**B shows a diagram **4200**B of an enlarged section “E” of the battery **4200**A of FIG. **42**A, according to some implementations. In some aspects, the plasticizer **4268** may cross-link (e.g., by the cross-linking **3265**) with additional monomer units and is not chemically (e.g., covalently) bonded with the interconnected carbon atoms **4263** of the polymeric backbone chain **4262**. At least some of the cross-linking **4265** of the plasticizer **4268** may volumetrically expand and/or contract to impart flexibility to the protective layer **4260**. In this way, the protective layer **4260** may expand and/or contract as needed to accommodate volumetric expansion of the anode **4220** resulting from operational discharge-charge cycling of the battery **4200**A.

(292) In one implementation, the plasticizer **4268** may affect the cooperative segmental mobility (e.g., the maximum cooperative segmental mobility) of the protective layer **4260**. For example, in the absence of the plasticizer **4268**, the 3D lattice of the protective layer **4260** may be relatively rigid due to higher degrees of cross-linking **4265**, which in turn may minimize the spacing **4267** between adjacent monomer units of the interconnected monomer units **4266**. Introduction of the plasticizer **4268** between adjacent monomer units may increase the spacing **4267**, which provides the additional polymeric chains more volume in which to move, thereby resulting in additional segmental motion of the protective layer **4260**. For example, since the plasticizer **4268** is not covalently bonded to the interconnected carbon atoms **4263** of the polymeric backbone chain **4262**, the plasticizer may be able to retain a relatively higher freedom of mobility within the 3D lattice of the protective layer. This relatively higher freedom of mobility of the plasticizer **4268** may expand the spacing **4267** between adjacent monomer units, thereby increasing the flexibility of the protective layer **4260** as may be necessary to accommodate volumetric expansion of the anode **4220** associated with operational discharge-charge cycling of the battery **4200**A.

(293) As used herein, a phrase referring to “at least one of” or “one or more of” a list of items refers to any combination of those items, including single members. For example, “at least one of: a, b, or c” is intended to cover the possibilities of: a only, b only, c only, a combination of a and b, a combination of a and c, a combination of b and c, and a combination of a and b and c.

(294) The various illustrative components, logic, logical blocks, modules, circuits, operations, and algorithm processes described in connection with the implementations disclosed herein may be implemented as electronic hardware, firmware, software, or combinations of hardware, firmware, or software, including the structures disclosed in this specification and the structural equivalents thereof. The interchangeability of hardware, firmware and software has been described generally, in terms of functionality, and illustrated in the various illustrative components, blocks, modules, circuits and processes described above. Whether such functionality is implemented in hardware, firmware or software depends upon the application and design constraints imposed on the overall system.

(295) Various modifications to the implementations described in this disclosure may be readily apparent to persons having ordinary skill in the art, and the generic principles defined herein may be applied to other implementations without departing from the spirit or scope of this disclosure. Thus, the claims are not intended to be limited to the implementations shown herein but are to be accorded the widest scope consistent with this disclosure, the principles and the novel features

disclosed herein.
(296) Additionally, various features that are described in this specification in the context of separate implementations also can be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also can be implemented in multiple implementations separately or in any suitable subcombination. As such, although features may be described above in combination with one another, and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.
(297) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Further, the drawings may schematically depict one more example processes in the form of a flowchart or flow diagram. However, other operations that are not depicted can be incorporated in the example processes that are schematically illustrated. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the illustrated operations. In some circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems can generally be integrated together in a single product or packaged into multiple products.

Claims

1. A lithium-sulfur battery comprising: a cathode; an anode structure positioned opposite to the cathode, the anode structure comprising: a single layer of solid lithium; a solid-electrolyte interphase layer formed on the single layer of solid lithium; and a protective layer formed on and at least partially disposed within the solid-electrolyte interphase layer responsive to operational discharge-charge cycling of the lithium-sulfur battery, the protective layer comprising: a polymeric backbone chain formed of interconnected carbon atoms collectively defining a cooperative segmental mobility of the protective layer; a plurality of additional polymeric chains cross-linked to one another and to at least some carbon atoms of the polymeric backbone chain; and a plasticizer dispersed throughout the protective layer and configured to separate at least some of the plurality of additional polymeric chains and increase the cooperative segmental mobility of the protective layer; a separator positioned between the anode structure and the cathode; and an electrolyte dispersed throughout the cathode and in contact with the anode structure, wherein the protective layer is characterized by a glass transition temperature of between 60° C. and 81° C. and forms a cross-linked three-dimensional (3D) polymeric lattice characterized by chemical resistance to dissolution in the electrolyte.
2. The lithium-sulfur battery of claim 1, wherein the protective layer is configured to melt at the glass transition temperature.
3. The lithium-sulfur battery of claim 1, wherein an increase in the glass transition temperature is associated with a decrease in the cooperative segmental mobility of the protective layer.
4. The lithium-sulfur battery of claim 3, wherein the decrease in the cooperative segmental mobility of the protective layer is associated with a decrease in a lithium ion (Li.sup.+) conductivity associated with the protective layer.
5. The lithium-sulfur battery of claim 1, wherein an increase in an amount of the plasticizer is associated with an increase in a lithium ion (Li.sup.+) conductivity associated with the protective layer.